The Incidence of Diversity: Labor Market Effects of Corporate Diversity Pledges*

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Abstract

Do diversity initiatives create new positions for minorities, or move them between firms? I build a comprehensive dataset of the numerical, dated workforce diversity targets of American companies, and match them to administrative and LinkedIn data on their employees. Relative to never-pledged firms, those with a numerical race pledge raised their share of Black workers by 5.4 percent on average over the four years after the pledge. These Black hires were no more often placed in diversity or staff roles, and fell no further behind the white men hired alongside them in college rank, experience, promotion or retention. Although the pledgers largely fell short of what they promised, they diversified their workforces by tilting the flow of outside hires toward Black candidates. For every 100 positions the pledgers added for Black workers at director and above, 45 went to Black workers who moved from similar jobs at other firms.

Keywords: Racial Inequality, Corporate Diversity Pledges, Labor Market Incidence, Vacancy Chains, Voluntary Affirmative Action, Internal Labor Markets, Corporate Governance
JEL Codes: J15, J23, J63, J71, G34, M14, M51

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1 Introduction

Racial gaps in economic outcomes in the United States are large and persistent. Black Americans earn less, hold less wealth, and move up the income distribution less often than white Americans (Bayer and Charles, 2018; Derenoncourt et al., 2024; Chetty et al., 2020). Corporate America plays a role in this. For instance, Black applicants are less likely to be called back than white applicants with the same résumé (Bertrand and Mullainathan, 2004; Kline et al., 2022). In the aftermath of the murder of George Floyd in 2020 and the protests that followed, consumers, employees, and institutional investors pressed companies to disclose their demographics and set targets for minority representation (Balakrishnan et al., 2023; Bourveau et al., 2026). Many of the largest U.S. firms responded with public promises about the racial and female make-up of their workforces. The median race target implied an increase in the share of the targeted group of one percentage point per year for three to five years.

Companies promised fast and visible change, but those pressing them wanted a lasting increase in opportunities for minorities. A firm can raise minority representation at the top in three ways. First, it can move minority workers up from below the director level, by promoting its own employees or by hiring from lower positions at other firms. This adds minority directors to the economy, although a firm's own pipeline takes years to build. Second, a firm in a hurry can lower its screening standards for minority candidates or create window-dressing roles. A lower standard can backfire, because it weakens the targeted group's incentive to invest in skills (Lundberg and Startz, 1983; Coate and Loury, 1993). Third, it can hire minorities who already hold similar positions at other firms. However, such a hire merely moves a director from one firm to another, adding none to the economy. Welch (1976) predicted this for employment quotas that bind only some employers: when minority workers are scarce, these employers draw them away from the others. The benefits then go to minorities who are already employed rather than to a larger group of workers.

I study the effects of these pledges to answer three questions. Did the firms that set numerical targets actually change the composition of their workforce? Did they lower the bar for the minorities they hired, and were the new positions window dressing? Finally, did the pledges create new opportunities for minorities, or reallocate existing ones between firms?

To answer these questions, I construct a dataset of the pledges of every S&P 1500 firm, coded from the firms' own filings, reports and press releases from 2015 to 2025. It shows

that 186 firms made numerical commitments to diversify their workforce. For effects at the company level, I collect the EEO-1 reports that firms file with the federal government each year on the composition of their workforce. For effects at each level of the company, and to follow workers between jobs, I use LinkedIn data on 23.5 million U.S. workers from 2014 to 2024. LinkedIn does not record race, ethnicity or gender, so I predict them from each profile's name, location and college. I validate this name-based measure against self-reported race in voter files and against profile photographs.

Since the pledges mostly addressed the Black population, I compare the Black share at the pledgers with that at never-pledged S&P 1500 firms in an event study design. While pledger and non-pledger firms are not identical, two factors speak for the validity of this design. First, the Black share followed the same trend at pledgers and controls before the pledges. Second, the name-based estimates hold when I reweight the never-pledged firms to match the pledgers' size, sector and 2019 Black share. They also hold with sector-by-year fixed effects and with other staggered-adoption estimators. Furthermore, to study worker-level outcomes within firms, I use triple-difference designs comparing Black new hires with the white men hired by the same firm in the same year. This allows me to compare the two groups' observable characteristics, such as experience, college ranking and the rank of the previous employer, and to test whether the gaps between them widened after the pledges.

I build a simple framework in which a firm meeting a diversity target can promote from within, hire from below the director level, lower its bar or hire sitting directors away from other firms. The pools of candidates that the firm can reach determine which of these margins it uses. To see which margins the pledgers used, I split each pledger's yearly change in the Black share of a seniority band into hires from outside, promotions from below, separations and promotions out of the band. To see what the outside hires did to other employers, I trace the people the pledgers hired back to the firms they left, which my data allow because they also cover firms outside the S&P 1500. For each Black director who left a director job elsewhere for a pledger after its pledge, I then check who, if anyone, filled her old seat within a year.

I document four findings. First, after a numerical race pledge, the Black share of the pledgers' workforces rose relative to never-pledged firms. In the official EEO-1 reports, the rise averages 0.57 percentage points over the four years after the pledge, which is 5.4 percent of the pledgers' 2019 share and corresponds to about 24,000 more Black employees. In the same reports, firms that made general diversity statements without a number show no detectable increase. Using the aggregate-level data of Onorato et al. (2026), who

infer race from both photographs and surnames, I estimate a firm-wide rise of 0.38 percentage points at the pledgers, and with my name-based measure a rise of 0.21 points. Split by seniority, the name-based measure shows increases in every band, including 0.20 points at director and above. On the firm-years that both sources cover, my firm-wide estimate is still smaller than in the EEO-1 reports, mostly because predicting race from names misses some Black workers. Even the largest estimate is far below the one percentage point a year that the median target promised.

Second, firms accomplished this compositional shift mostly through external hiring rather than internal promotions. Specifically, of every 100 Black workers added at director and above, 79 were hired from outside and 21 were promoted from lower bands. This was not accompanied by an expansion of total headcount, which, if anything, grew more slowly at pledgers than at controls.

Third, the pledges did not widen the gaps in observable characteristics between Black hires and the white men hired alongside them. Among hires into director and above, this holds for college ranking, experience, predicted pay, retention and promotion. The gap in the rank of their previous employers narrowed, so the pledgers did not recruit their Black hires from lower-ranked firms. The new positions also do not look like window dressing: classifying each hire's title by function and role, I find that the new Black hires were placed no more often in diversity, human-resources or staff roles, or in roles new to the firm. The stable gaps have two readings. Either firms had been passing over qualified Black candidates, or the supply of qualified Black candidates was thin and the pledgers met their targets partly by hiring sitting directors away from other firms.

Fourth, about three in five of the added Black directors were new to the economy. Of the 79 outside hires in every 100 added seats, 45 came straight from a director-level job at another firm. Those firms refilled the vacated seats with a Black worker only 7.4 percent of the time, no more often than they usually hire Black directors. Each such move therefore mostly reallocated a Black director from one firm to another. Counting the replacements, the net number of Black directors in the U.S. labor market rose by about 58 for every 100 the pledgers added. Never-pledged index firms, which form the comparison group, supplied only 5 percent of the increase in Black hires, so poaching from them biases the estimate at director and above by at most a sixth.

Taken together, the findings imply that looking at firm-level effects of corporate diversity initiatives overstates their aggregate effect. I find no sign that the increase came from a lower bar, the route by which a target can backfire in Coate and Loury (1993). The incidence of a target therefore depends on the margin used to meet it. A target on a firm's

stock of directors credits a director hired away from another firm the same as one promoted from within. A target on promotions, and on hires from below the director level, would credit only the margins that add Black directors one for one, which in this episode produced 55 of the 58.

I contribute to three literatures. The first studies racial inequality and labor market discrimination (see, for instance, Becker, 1957; Phelps, 1972; Arrow, 1973; Coate and Loury, 1993; Giuliano et al., 2009; Lang and Lehmann, 2012; Hsieh et al., 2019; Kline et al., 2022; Onorato et al., 2026), a literature reviewed by Bayer et al. (2025). Work on affirmative action, reviewed by Holzer and Neumark (2000a), has mostly studied mandates and quotas, such as federal contractor obligations (Leonard, 1990; Miller, 2017; Kurtulus, 2016; Holzer and Neumark, 1999, 2000b), bans on race-conscious admissions (Hinrichs, 2012; Bleemer, 2022), caste reservations (Bertrand et al., 2010) or board quotas (Bertrand et al., 2019). To my knowledge, this is the first paper in labor economics to study corporate diversity pledges. Two characteristics set them apart from policies that have been studied in this literature. First, unlike mandates and quotas, they are voluntary commitments by firms. Second, unlike most policies shown to narrow racial gaps, such as minimum wages (Derenoncourt and Montialoux, 2021; Wursten and Reich, 2023), they target the upper levels of the corporate ladder.

The closest paper in this literature is Bertrand et al. (2019), who find that Norway's board quota raised the representation and pay of the women appointed, with little spillover to the women below them. In contrast, the pledges studied here reach below the board. I contribute to this literature by following workers over a decade across large pledging and non-pledging firms. This lets me separate growth from reallocation and outside hiring from promotion, and compare the Black hires with the white men hired alongside them.

The second literature studies the displacement and general-equilibrium effects of labor market policies. In Crépon et al. (2013), job-search assistance raised the employment of treated job seekers in France partly at the expense of untreated job seekers in the same labor markets. Related studies find spillovers on non-participants from job-search assistance in Denmark (Gautier et al., 2018), longer unemployment insurance in Austria (Lalive et al., 2015), training in France (Ferracci et al., 2014), and a public employment program in India (Muralidharan et al., 2023). Most of these papers study supply-side interventions that change the search behavior or skills of workers, and they measure displacement by comparing local labor markets with more and fewer treated workers. In contrast, the pledges raised the demand for Black directors at a subset of firms that com-

pete for a thin pool of senior talent. Comparing local markets would miss these firm-to-firm spillovers, because the pledgers hired most of their Black directors from other industries or occupations, and markets narrow enough to compare do not contain such hires. I therefore measure the general-equilibrium incidence directly. For each Black director whom a pledger hired from a director job at another firm, I go back to the firm she left and check who, if anyone, filled her seat. Applying the vacancy-chain accounting of White (1970) and Mercan and Schoefer (2020) to matched worker-firm data, I find that firm-level estimates overstate the increase in Black directors in the economy by about 72 percent.

The third literature studies corporate commitments and their consequences (see, for instance, Flammer, 2015; Bhagwat et al., 2020; Edmans et al., 2023; Balakrishnan et al., 2023; Gormley et al., 2023; Raghunandan and Rajgopal, 2024; Cai et al., 2024; Bolton and Kacperczyk, 2026; Chen et al., 2026; Law and Tan, 2026). A central question in this literature is whether voluntary diversity commitments change what firms do or only what they say, which Baker et al. (2024) call diversity washing. I contribute a dataset that codes ten years of filings, reports and press releases of every S&P 1500 firm and separates dated numerical race targets from general statements. In the official EEO-1 reports, the firms that made only general statements show no detectable change in their Black share relative to never-pledged firms, while the firms that set a dated numerical target raised theirs. The Black workers these firms hired were not placed more often in diversity, human-resources or staff roles. I also bring general-equilibrium incidence into the study of corporate governance. Studies that evaluate firms one at a time miss that, when more than a hundred large employers pursue similar targets at once, the same Black directors move between them: of every 100 Black directors the pledgers added, 45 came straight from a director-level job at another firm.

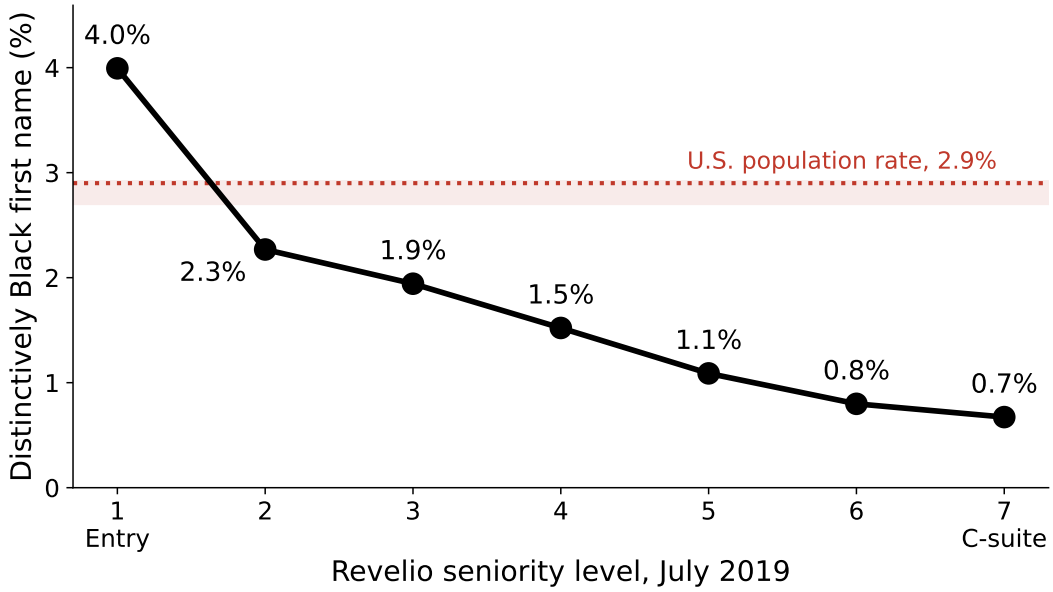
The rest of the paper is structured as follows: Section 2 describes the institutional setting, Section 3 the data, and Section 4 a simple framework and the empirical strategies. Section 5 reports the effects on the workforce, Section 6 reports the worker-level comparisons, Section 7 discusses the aggregate-level effects and Section 8 concludes.

2 Institutional Setting

Minorities face steep barriers to advancing up the corporate ladder in the United States (Stainback and Tomaskovic-Devey, 2009; Bourveau et al., 2026). Economy-wide, Black

workers in 2018 held 15 percent of all jobs at private employers with 100 or more employees, but only 7.5 percent of first- and mid-level management roles and 3.3 percent of executive positions (U.S. Equal Employment Opportunity Commission, 2018). Within the S&P 1500 firms in my sample, LinkedIn data reveals a monotonic decline across seven seniority rungs. Figure 1 plots the share of workers in July 2019 with a distinctively Black first name, which represents 2.9 percent of the U.S. population (Rosenman et al., 2023) but falls from 4.0 percent at entry level to 1.1 percent among directors and 0.7 percent in the C-suite.

FIGURE 1: BLACK REPRESENTATION ACROSS THE CORPORATE HIERARCHY



Notes: The figure plots the share of workers with a distinctively Black first name, by the Revelio seniority level held on July 1, 2019. A first name is distinctively Black if at least 80 percent of the people with that name are Black in the voter-file tables of Rosenman et al. (2023). The sample is the 10.4 million workers whose first name appears in the tables, 95 percent of all 11.0 million at the 1,448 S&P 1500 firms on that date (all 1,449 index firms except Old Dominion Freight Line, which is missing from the panel used for this figure). The dotted line is the U.S. population rate, 2.9 percent: the share of each racial group with a distinctively Black first name in the six-state voter files behind the name tables, weighted by the national racial mix. Sources: Rosenman et al. (2023), Revelio Labs.

Discriminatory practices on the basis of race have been illegal in the U.S. since the Civil Rights Act of 1964, which bars race-based hiring and promotion. It only allows racial preferences in temporary voluntary plans that correct a manifest imbalance (Holzer and Neumark, 2000a). However, there is no law that requires private employers to meet racial targets in their workforces. The closest was the mandate for federal contractors to keep an affirmative action plan. This mandate (Executive Order 11246) forced federal contractors with at least 50 employees and a $50,000 contract to keep written affirmative action plans

with placement goals (Miller, 2017; U.S. Department of Labor, 2024).

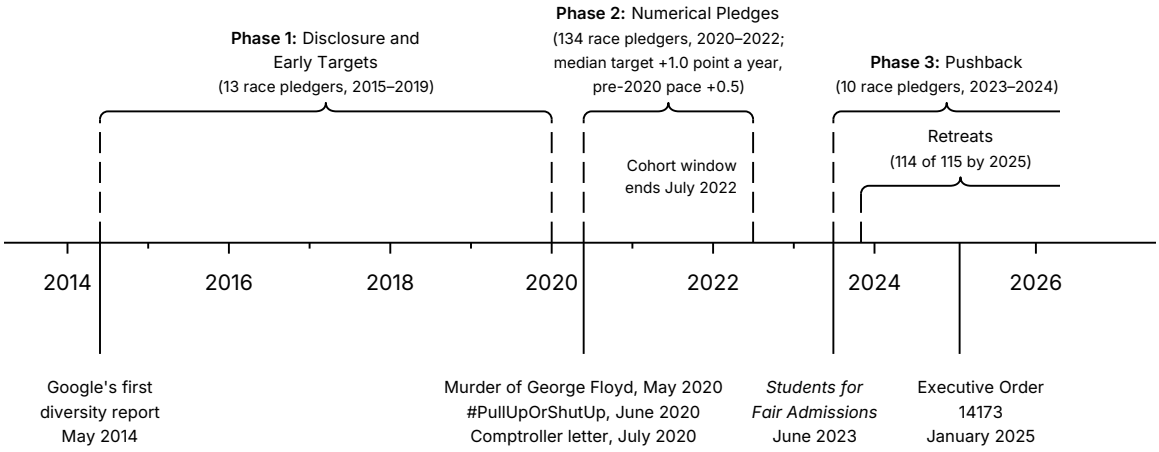
Within these limits, large firms have run diversity programs for decades: training for managers, diversity criteria in managers' evaluations, and committees and staff responsible for diversity (Kalev et al., 2006). The evidence on these programs is mixed. Training changes attitudes briefly and behavior little (Kalinoski et al., 2013; Bezrukova et al., 2016; Chang et al., 2019), evaluation criteria do little for the share of women and Black workers in management, and committees and staff, which assign responsibility, come with the broadest increases (Kalev et al., 2006). None of these commits the firm to a number that outsiders can check.

The murder of George Floyd on May 25, 2020, led to pressure on corporations by consumers, employees, and institutional investors (Balakrishnan et al., 2023; Pajuste et al., 2024). On social media, the #PullUpOrShutUp campaign asked brands that had posted statements of support to publish, within 72 hours, how many Black employees they had in their corporate offices and in leadership roles, and it asked consumers to stop buying from these brands until they did (Callahan, 2020). Employees made similar demands inside their firms. For instance, a group of Amazon employees asked the company to release data on race and promotions and to set diversity targets for its management (Weise, 2020). Institutional investors then turned these demands into governance campaigns. On July 1, 2020, the New York City Comptroller wrote, on behalf of three of the city's pension funds, to 67 S&P 100 companies that had issued statements in support of racial equality, asking them to commit to publishing their EEO-1 reports (Office of the New York City Comptroller, 2020). Over the following months, BlackRock, State Street and Vanguard asked their portfolio firms to disclose the same data.

Many of the largest publicly traded companies answered with publicly disclosed, numerical targets for the composition of their U.S. workforce. In all, 157 S&P 1500 firms set a numerical race target between 2015 and 2024. About two thirds of the firms with such targets set them in either 2020 or 2021. In contrast to the contractor plans, which were confidential, pledges were public and they were accompanied by proxy statements, sustainability reports and voluntary disclosures of workforce data. Figure 2 depicts the timeline of events. While a few firms set numerical targets before 2020, the majority of the pledges came in a wave after the events following the death of George Floyd in 2020. By 2025, most pledgers have either softened or withdrawn their targets.

Table 1 quotes the pledges of Cisco, McDonald's and State Street. To give an example, Cisco aimed to increase Black representation especially at the top of the company, with a goal of a 75% increase in representation from Director through Vice President levels

FIGURE 2: TIMELINE OF CORPORATE DIVERSITY PLEDGES, 2014 TO 2026



Notes: The phase labels count the 157 numerical race pledgers by calendar year of their first target. The median target and the pre-2020 pace use the 72 race targets that state both shares and both dates; the pace is the 2015 to 2019 trend at the targeted levels, measured on LinkedIn and scaled for misclassification. Retreats count the 115 main-sample pledgers by the first filing that weakens or ends their race target. Sources: pledge dataset; LinkedIn profiles via Revelio Labs; Forms 10-K and DEF 14A.

TABLE 1: EXAMPLES OF CORPORATE NUMERICAL RACE PLEDGES

Firm	Pledge
Cisco	"achieve a 25% increase in representation of all employees who self-identify as AA/Black from entry-level through manager level and a 75% increase in representation of employees who self-identify as AA/Black from Director through VP+ level" by 2023
McDonald's	"increase representation of historically underrepresented groups in leadership roles, Senior Director and above in the U.S., from 29% to 35% by the end of 2025."
State Street	"triple our Black and Latinx leadership (senior vice presidents+) and double our percentage of Black and Latinx populations over the next three years."

Notes: The table reports the numerical race pledges of Cisco, McDonald's and State Street. The pledges are quoted from the firms' own documents: Cisco's statement "Social Justice Beliefs and Actions" (September 2020), McDonald's announcement of February 19, 2021, and State Street's statement "10 Actions to Address Racism and Inequality" (July 2020). Source: pledge dataset (Section 3.1).

by 2023. McDonald's aimed to raise the share of historically underrepresented groups in U.S. leadership roles, senior director and above, from 29 to 35 percent by the end of 2025, which represents about 1.2 percentage points a year. It also linked executive pay to meeting its diversity goals. State Street, like about one fifth of race pledgers did, mentioned both Black and Latin employees in its pledge.

When firms explained why they set targets, they mostly pointed to their customers or to their own employees. Part of the pressure, however, came from shareholders. The Comptroller's letters, for instance, asked for a written commitment by August 30, 2020, and warned of shareholder proposals and opposition to the re-election of directors at firms that did not commit (Office of the New York City Comptroller, 2020). In the 2021 proxy season, shareholders filed 42 proposals asking S&P 1500 firms to publish their EEO-1 reports, against 15 a year earlier (Sullivan & Cromwell LLP, 2021). Public targets may also have served recruiting: BNY Mellon set its goals in part to "help position our firm as a competitive choice with Black and Latinx professionals" (The Bank of New York Mellon Corporation, 2021).

Some firms may also have set targets as part of a broader business strategy, for instance to serve a diverse customer base or to compete for talent. Dell, for example, called closing the employment gap of underrepresented minorities in its industry "critical to meeting future business needs and ensuring that diverse perspectives reflect our global customer base" (Dell Technologies Inc., 2024). About three in ten pledgers in my main sample tied executive pay to their diversity goals. From 2022, Macy's included its ethnic representation goal at director and above in its annual incentive goals (Macy's, Inc., 2022), and from 2021 McDonald's based 15 percent of its executive vice presidents' annual bonus on human-capital metrics (CNBC, 2021).

The ambition of the pledges created doubt about whether and through which means the targets could be achieved. At least one chief executive said in writing where the difficulty lay. In the June 2020 memo that committed Wells Fargo to doubling its Black leaders within five years, its chief executive wrote that "the unfortunate reality is that there is a very limited pool of black talent to recruit from with this specific experience as our industry does not have enough diversity in most senior roles" (Bloomberg News, 2020). When Reuters reported the remark in September, he apologized within a day (Moise et al., 2020; Associated Press, 2020). The pledgers' own reports name the ways around this constraint: Uber planned to double Black representation in its leadership "through pipeline development and hiring" (Uber Technologies, Inc., 2021), and Amazon reported that it had doubled its Black directors and vice presidents "through external hiring as well as inter-

nal promotions" (Amazon.com, Inc., 2021).

Since 2023, there have been notable cases of pushback against these initiatives. Examples include the Supreme Court's decision in *Students for Fair Admissions*, which raised the legal risk of race-conscious decisions, and an executive order in January 2025 that revoked the contractor mandate to have affirmative action plans.

3 Data and Measurement

3.1 Corporate Diversity Pledges Dataset

To identify firms that made numerical diversity pledges, I construct a new dataset of corporate commitments that covers every firm in the S&P 1500 from 2015 to 2025, based on publicly available company documents. I code the treatment variable from the firms' proxy statements, annual and current reports, sustainability and diversity reports and press releases.[1]

I start by defining a dictionary of keywords related to racial diversity pledges (e.g., "African American", "people of color", "BIPOC" and "underrepresented minorities"; Appendix A.1). For each firm and with the help of an LLM, I go through 10 years of their documents looking for those keywords and extracting the sections that contain them. I record the verbatim quote, and extract the source, the group, the number, the deadline, the positions covered and the stated announcement date. Every record that could make a firm a pledger is checked by hand against its primary text. Three independent readers then re-read the firms whose status or date is hardest to establish, together with a random sample of the rest. Finally, in a blinded double-coding of 100 random pledges from their quotes alone, the codes agree with the stored group, target and deadline in 89 to 98 percent of cases (Appendix A.4).

I define a numerical race pledge as a public target that the firm itself sets for its U.S. workforce or part of it. The target must name a racial or ethnic group that includes Black or Hispanic workers, and it must set a numerical goal for that group's share or headcount, to be reached by a deadline. I classify vague language, a goal without a number or a date, membership in a coalition and a donation as general statements. To give an example, Walmart's 2021 annual report promises "a diverse, equitable and inclusive company" and "a

[1]Proxy statements are forms DEF 14A and DEFA14A, annual reports form 10-K, and current reports form 8-K with their exhibits; sustainability reports include ESG reports.

diverse associate base" but names no group and gives no number or date. Appendix A.2 details the specific definition of numerical race pledgers, gives the specific conditions and explains how I resolve the hard cases.

Of the 386 firms in the S&P 1500 that made some public diversity commitment, 186 made a numerical pledge between 2015 and 2024, and the other 200 made only general statements or set targets outside the definition.[2] Among these numerical pledgers, 157 set targets for race or ethnicity (of those, 49 pledged exclusively that) and 137 set gender targets (of those, 29 pledged gender only). The main analysis uses the 115 race pledgers of the main sample, which pledged between 2018 and 2022 (Section 4.2).

Race pledgers came from all eleven GICS sectors, and no sector accounts for more than a fifth of them (Appendix Table A.2): consumer discretionary, industrials, financials, information technology and health care each have between 17 and 31 race pledgers. However, relative to each sector's size in the index, consumer-facing firms pledged most often. About a fifth of consumer staples and communication services firms and a sixth of consumer discretionary firms set a numerical race target, against about 2 percent of energy, utilities and real estate firms.

I code a pledger's date as the day of the first public announcement of its earliest qualifying race target, and its cohort as the year of the first July 1 on or after that date. Panel A of Figure 3 depicts the number of firms making numerical commitments by the year of their first numerical race pledge, or of their first gender pledge if they set no race target. Eighteen firms pledged between 2015 and 2019. Following the pressure from the Black Lives Matter movement, 66 percent of the workforce commitments came in 2020 and 2021, and the wave tailed off through 2024.[3] The dataset also records what each firm later did with its target. By their 2025 filings, 114 of the 115 pledgers in the main sample had softened or withdrawn their targets, which I call a retreat.

To see which positions the targets cover, I map the specific wording of each pledge to the levels of the corporate ladder. For instance, vice presidents map to Revelio level 6, executive, and above. Panel B of Figure 3 counts the targets that cover each level, and almost every pledger sets a target for the senior levels. Specifically, 89 percent of race pledgers and 86 percent of gender pledgers have a target that covers director-level

[2]The general pledgers that serve as a comparison group are the 193 firms with general statements and no numerical race target: the 186 firms that made only general statements and seven firms that made general statements about race but set a numerical target only for women.

[3]Among the firms with both race and gender targets, 18 had set their gender target in an earlier year than their race target. While gender targets continued commitments that many firms had already made, race targets were the new commitment of 2020. The race goals were thus the larger shock to what firms had promised.

positions. Forty-four percent of race pledgers have a race target for the whole workforce, against 25 percent of gender pledgers, and 29 percent limit their race targets to director and above, against 47 percent for gender.[4] Seventy-two race targets of 42 pledgers state the target share, the starting share and both dates. Among these, the median asks for a rise of 1.0 percentage point a year, and a quarter ask for more than 1.5 points.[5]

Since the vast majority of race pledgers cover Black workers, I concentrate the estimation on the Black share. In the main sample, 42 percent of the race pledgers name Black workers explicitly, and the rest use an umbrella group that includes them. About a fifth of race pledgers name Hispanic workers, always alongside Black workers (Table A.4). I therefore use the Black and Hispanic share as the secondary outcome.

3.2 Official and Photo-Based Race Data

EEO-1 reports are documents collected by the government each year that count the firm's U.S. employees by job category, sex and race or ethnicity. This data is, in most cases, self-reported. Every private employer with at least 100 employees and every federal contractor with at least 50 employees and a $50,000 contract files this report. Hispanic workers of any race form one group, so the Black group excludes them. Because the reports are confidential, I combine three sources. For 2016 to 2020, I use the consolidated reports of federal contractors that the Labor Department's Office of Federal Contract Compliance Programs released under the Freedom of Information Act. From 2021 on, I use the reports that firms published themselves and the reports that corporations attach to their annual filings with the Illinois Secretary of State. Each report also prints the previous year's totals, which I use when that year's report is missing and the two head counts differ by less than 0.3 log points. A firm's count jumps when it acquires or sells a business. To deal with this, I cut a firm's series at those events and at known mergers and give each stretch its own fixed effect (Appendix E.3). In the event study of Section 5.1, I keep the 105 pledgers and 708 controls with at least one report.

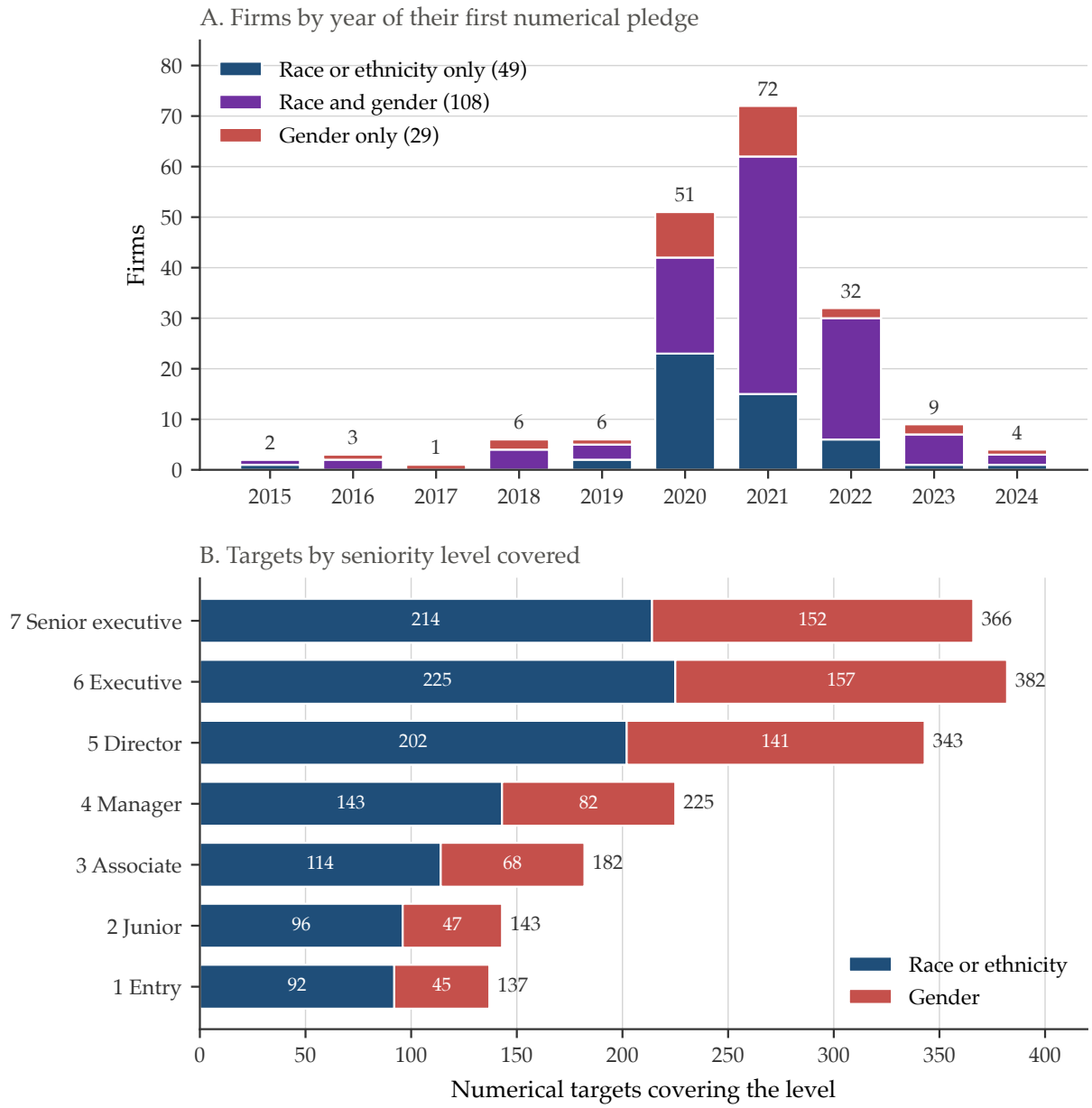
Since the job categories of the reports do not line up with the seniority bands of Section 3.3, I use them only for the Black share of all employees. After 2020, I only have the reports of firms that published them or filed them in Illinois.

The second measure comes from Onorato et al. (2026), who predict the race of

[4]The race targets name the United States in 82 percent of cases, while seven in ten gender targets are global.

[5]Table A.3 reports the distribution by the lowest level a target covers and by the group it names.

FIGURE 3: ADOPTION AND SCOPE OF NUMERICAL DIVERSITY PLEDGES



Notes: Panel A counts the 186 S&P 1500 firms with a numerical pledge covering U.S. workers by calendar year of their first numerical race pledge, or of their first gender pledge if they have no numerical race target. Panel B counts their targets whose scope covers each Revelio level. A target that spans several levels counts at each of them, and the 11 joint targets count once as race and once as gender, so the 401 targets give 412 entries. Levels 1 and 2, 3 and 4, and 5 to 7 are the entry, middle and director-and-above bands. Source: pledge dataset.

LinkedIn users at Fortune 1000 firms from profile photographs and surnames, and report each firm's Black share by calendar year. I observe 92 of my pledgers and 491 of my controls in their panel.

I also use three other sources. I get revenue, net income and total assets from the SEC's XBRL company facts to compare pledgers and controls in Section 4.4. The race predictions of Section 3.4 use IPEDS counts of graduates by race, and I validate them against self-reported race in the North Carolina voter file (Appendix C.3).

3.3 The Workforce Panel

The workforce data come mainly from a November 2025 snapshot of LinkedIn profiles (Revelio Labs, 2025). LinkedIn data has been widely used to measure labor-market outcomes at the level of firms and workers (see, for instance, Berry et al., 2026; Jeffers, 2024; Baker et al., 2024; Cai et al., 2024; Bloom et al., 2026). A profile typically lists a worker's history of job positions, each with a title, a location, a start date, and an end date. I measure each firm's workforce on July 1 of each year. This gives each firm a single headcount per year and places the 2020 snapshot before the pledges of 95 of the 115 pledgers. A worker counts at firm f on July 1 of year t if the most recent position she started on or before that date is at f and her profile shows no end to it before that date. The panel runs from 2014 to 2025 and covers 1,504 firms, including the 1,449 index firms. From 2014 to 2024, it holds 23.5 million workers and 126.5 million worker-years. The estimation sample keeps 1,177 of the index firms, the 115 pledgers of the main sample and 1,062 never-pledged firms (Section 4.2), with 6.4 million workers on July 1, 2019.

There are two concerns about using LinkedIn data in this context. The first is the selection into LinkedIn. The second is differential coverage rates by race, especially if minorities face discrimination on the platform (Evsyukova et al., 2025). To address these concerns, Table 2 compares the panel with the firms' official EEO-1 reports and with a representative sample of private-sector employees from the Current Population Survey. At the median of the 586 firms with a 2019 EEO-1 report, LinkedIn shows 67 workers for every 100 employees the firm reports to the government. By industry, coverage is highest in information and finance and lowest in utilities and transportation (Appendix B.3). On these firms, the panel matches the reports on the share of women (41.8 against 41.9 percent) and comes close on the Black share (13.3 against 14.2 percent), but it finds fewer Hispanic workers (10.9 against 14.7 percent) and more Asian workers (14.7 against 10.7 percent). Weighted by each firm's EEO-1 employment, the panel's shares are 14.3 per-

TABLE 2: WORKERS IN THE LINKEDIN PANEL, EEO-1 REPORTS AND THE CPS, 2019

	LinkedIn panel		EEO-1	CPS	EEO-1
	All firms (1)	Firms with EEO-1 report (2)	Same firms (3)	Employers of 1,000+ (4)	All private filers (5)
Race and ethnicity (percent of the four groups)					
Black	13.7	13.3	14.2	14.4	15.8
Hispanic	11.7	10.9	14.7	15.8	16.3
Asian or Pacific Islander	12.9	14.7	10.7	7.9	7.7
White	61.6	61.1	60.5	61.9	60.2
Black, unweighted mean across firms	12.2	11.7	9.1		
Industry (percent of workers)					
Goods-producing	30.5	34.1	33.1	16.5	21.5
Trade, transportation and utilities	22.3	18.0	25.1	28.0	23.5
Information, finance, professional services	37.7	41.8	29.1	22.7	25.7
Education and health	4.2	4.1	7.7	20.3	20.1
Leisure, hospitality, other services	5.3	2.0	5.0	12.5	9.2
Workers and firms					
Female (percent)	43.7	41.8	41.9	48.2	48.1
Under 30 (percent)	37.4			28.5	
Bachelor's degree or more (percent)	80.0 [53.1]			39.2	
South (percent)	35.2			36.8	36.1
West (percent)	27.9			22.7	22.0
Median firm (thousand U.S. workers)	1.9	2.8	4.2		
Workers at firms of 10,000 or more (percent)	61.5	66.7	80.0		
LinkedIn workers per 100 EEO-1 employees		67			
Firms (pledgers)	1,177 (115)	586 (83)	586 (83)		
Workers (millions)	6.4	4.3	7.4	53.4	57.1

Notes: Columns 1 and 2 are the workers in the LinkedIn panel on July 1, 2019, at all estimation firms and at the firms with a 2019 EEO-1 report, and column 3 is those firms' FOIA EEO-1 reports for October to December 2019. Column 4 is private wage and salary workers at employers of 1,000 or more (March 2019 CPS); column 5 is the 281,650 establishments filing EEO-1 reports. Size counts LinkedIn workers in columns 1 and 2. Age comes from degree dates; brackets give a lower bound for degrees. Sources: Revelio Labs; OFCCP; EEOC; CPS.

cent Black, 11.4 percent Hispanic and 11.7 percent Asian, so the Asian gap comes from which firms LinkedIn covers best, and the Hispanic gap arises within firms. However, since the outcome is a share, low coverage matters only if it differs by race or between pledgers and controls. By race the difference is small: in North Carolina, 16.7 percent of Black and 15.5 percent of white registered voters have a matching profile (Appendix B.3). Across groups of firms, the ratio of LinkedIn workers to reported employees has similar means at pledgers, general pledgers and controls, but its median is lower at controls (Appendix B.3).

Compared to the private-sector employees of employers with 1,000 or more workers in

the Current Population Survey, the panel is close on the Black share but has fewer women, and it over-represents college graduates, younger workers and information, finance and professional services, with few workers in education and health. These are the jobs from which firms fill director positions, the level that most race targets cover. Across firms, the panel compresses the Black share: at the same 586 firms, the average firm's Black share is 11.7 percent on LinkedIn and 9.1 percent in the EEO-1 reports. Because the classifier gives every worker some probability of being Black, firms with few Black workers measure higher than they are and firms with many measure lower (Section 3.4). I use the following definitions throughout the paper.

1. Firm. Revelio links each employer to its ultimate parent. I make minor changes to this mapping, which I describe in Appendix B.2. I define firms by their current ownership to prevent acquisitions from suddenly moving a firm's measured workforce from one year to the next. For instance, a company acquired in 2022 counts as part of its acquirer in 2016 as well. I only keep workers with at least one U.S.-based position at the companies in my sample.

2. Level and band. Revelio maps every job title to one of seven corporate seniority levels, from entry level (level 1) to senior executive (level 7). For the purpose of simplicity, I group the levels into three bands: entry (levels 1 and 2), middle (3 and 4), and director and above (5 to 7).[6]

3. Hire. I count a position start as an outside hire if the worker's previous job was at another employer or she had none. Moves within the firm are left out.

4. Promotion. A promotion is a move into a higher band at the same firm, and a clear title promotion counts as a promotion (Appendix B.5). In the hire-level comparisons of Section 6, a hire is promoted within 24 months if she starts a later position at the same firm within two years at a higher level.

5. Lateral director move. A lateral director move is an outside hire into director and above of a worker whose last employee position before the entry was at director level or above at another U.S. employer, the origin.

6. Origin seat and vacancy chain. The seat at the origin counts as refilled when a different person starts there within 12 months in the same role, one of Revelio's 150 role groups,

[6] In 2019, 77 percent of workers in the top band sat at level 5 (director), 21 percent at level 6, and 2 percent at level 7. The levels come from Revelio's model of job titles, which can place the same title at different levels in different firms. Appendix B.5 describes a correction for titles whose level falls when the worker is promoted. After I drop cells with fewer than five workers, each band has about 12,500 firm-year cells in the estimation panel, and the median cell at director and above holds about 300 workers.

and in the same band; the first such start is the refill. Each move is thus the first link of a vacancy chain (White, 1970; Mercan and Schoefer, 2020), which Section 7 follows to the refill.

7. Employer characteristics. I rank employers by the value that workers' voluntary moves between them reveal (Sorkin, 2018), estimated on moves between U.S. employers from 2014 to 2019. The previous employer's rank is its percentile in this ranking, defined for hires who come from another employer. A firm's size is its LinkedIn headcount on July 1, 2019, and its sector is its GICS sector.

3.4 Measuring Race

Since LinkedIn profiles do not report race or ethnicity, I predict them using other pieces of information available in the profiles, as other studies do (e.g., Berry et al., 2026; Onorato et al., 2026). I classify people into one of four categories, white, Black, Hispanic of any race, and Asian or Pacific Islander, using Bayesian Improved First-name Surname Geocoding (BIFSG; Voicu, 2018). BIFSG adds first names to the surname-and-place method of Elliott et al. (2009) and Imai and Khanna (2016). BIFSG has been used to measure racial disparities in tax audits (Elzayn et al., 2025).[7] For group r, first name f, surname s, and county g,

$$P(r \mid f, s, g) \;\propto\; P(r \mid g)\, P(f \mid r)\, P(s \mid r),$$

which treats the terms as independent within a group. I take the name terms from the tables that Rosenman et al. (2023) estimate from the voter files of six states that record race. For locations, I use the Census Bureau's estimate of each group's share of the population in the county where a worker held most of her U.S. positions. I keep this county fixed over her career, so that a move does not change her predicted race (Appendix C.1). For graduates of colleges where most graduates are Black, I multiply the Black odds by the college's likelihood ratio from IPEDS graduates by race, with a weight fitted on North Carolina voters (education-augmented BIFSG; Appendix C). For the share outcomes, I average the classifier's probabilities over the workers in a cell, and in the hire-level designs I use the most likely group.

Given that the classifier is probabilistic, the dependent variable has non-classical mea-

[7] Among North Carolina voters, BIFSG labels 71 percent of Black voters as Black, while Revelio's own race classifier labels them correctly only 55 percent of the time. Therefore, I use Revelio's race probabilities only as a robustness check (Appendix C.3).

surement error, which attenuates the estimated effects on the shares. Following Mahajan (2006) and Hu (2008), the measured Black share of a cell, or its Black and Hispanic share, is

$$m = \pi_0 + (\pi_1 - \pi_0)\, s,$$

where s is the true share and π_1 and π_0 are the classifier's average probabilities for the group among its members and among non-members. If π_1 and π_0 are the same at pledgers and controls, before and after the pledges, a change in s translates into a change in m that is $\pi_1 - \pi_0$ times as large. Because the classifier is more likely to misclassify a Black worker as something else than the other way around, $\pi_1 - \pi_0$ is below one. I report the estimates below as measured. Dividing them by $\pi_1 - \pi_0$ scales them into true units: for the Black share, the factor is 1.30 on the EEO-1 reports of federal contractors and 1.62 on the North Carolina voter file, which records self-reported race (Appendix Table C.1). Scored by its most likely label instead, the classifier labels 76 percent of Black voters and 6 percent of the others as Black.

4 Empirical Design

I estimate the effect of a numerical race pledge by comparing the workforces of pledging firms with those of never-pledged S&P 1500 firms, before and after each pledge. Since firms announced their pledges at different dates, mostly between 2020 and 2022, I run this comparison as a staggered event study of the Black share at the company level and in each seniority band (Section 4.3). For outcomes at a worker-level, a triple difference adds a second comparison within each firm, between white men and Black workers hired in the same year (Section 4.5).[8]

4.1 Conceptual Framework

To define the objects I estimate, I build a simple conceptual framework in which the two classic concerns about targets, a lowered bar (Coate and Loury, 1993) and reallocation under thin supply (Welch, 1976), are two margins of one problem. The framework is in partial equilibrium.

Setup. I consider a firm f with N_f seats at director and above, some of which it fills through turnover over the horizon of its target. A seat is filled through one of three

[8]The flow decomposition of Section 5.4 is described in Appendix E.2.

channels $m \in \{P, U, L\}$: promotion from the firm's own middle band, an outside hire who was not a director, or an outside hire who was a director at another employer. Channel m reaches a pool of Black candidates of size Z_m, thinner at the top. I assume that hiring x_m Black candidates through m costs $x_m^2/(2\phi_m Z_m)$, because each further hire requires searching further into the pool. I call ϕ_m the channel's speed. The speed of promotion, ϕ_P, rises with the horizon, because internal candidates take time to develop, and the speed of lateral hiring, ϕ_L, is the largest, because a sitting director is already vetted and can be hired at once. On channel L, the convexity comes from a ladder of origins: inducing a director to move costs more the higher her employer ranks relative to the firm (Sorkin, 2018), and the firm takes the cheapest origins first. I also allow the firm to fill seats below its usual standard, $x_Q \geq 0$, at a productivity cost $\kappa_f x_Q + x_Q^2/(2\psi_f)$, which is the lowered bar of Coate and Loury (1993). Adding a seat costs a salary, whereas redirecting turnover costs only search. With induced hires at a few percent of turnover, the firm therefore adds no seats.

Without a pledge, the firm values a Black director at $v_f \geq 0$ per head. With a pledge, its target names an increase Γ_f by the deadline, counted over the channels in a set M_f (all three for a target on the stock of Black directors). I model the cost of falling short as $\frac{\eta_f}{2}(\Gamma_f - C_f)^2$, where C_f is the counted increase and η_f is the bite. The bite is zero without a target and small for a general statement. It is large for a public number, and larger still when pay is tied to it.

The firm's problem. The firm solves

$$\max_{x_P, x_U, x_L, x_Q \geq 0} \; v_f\Big(\sum_m x_m + x_Q\Big) - \sum_m \frac{x_m^2}{2\phi_m Z_m} - \kappa_f x_Q - \frac{x_Q^2}{2\psi_f} - \frac{\eta_f}{2}\big(\Gamma_f - C_f\big)^2. \tag{1}$$

Let $\lambda_f \equiv \eta_f(\Gamma_f - C_f)$ be the shadow value of a counted Black director. The first-order conditions give

$$x_m = \big(v_f + \lambda_f \mathbf{1}\{m \in M_f\}\big)\,\phi_m Z_m, \qquad x_Q = \psi_f\,(v_f + \lambda_f - \kappa_f)^+, \tag{2}$$

so a pledge adds $\lambda_f \phi_m Z_m$ on each counted channel. Writing $\Sigma_M \equiv \sum_{m \in M_f} \phi_m Z_m$ and leaving the bar aside, I obtain

$$\lambda_f = \frac{\eta_f \Gamma_f}{1 + \eta_f \Sigma_M}, \qquad \Delta_f = \Gamma_f\,\frac{\eta_f \Sigma_M}{1 + \eta_f \Sigma_M}, \tag{3}$$

where Δ_f is the number of Black directors the pledge adds. I derive four implications.

Level and bite. The Black share rises by Δ_f / N_f, which increases with bite, ambition and pools. The fraction of the target delivered, $\eta_f \Sigma_M / (1 + \eta_f \Sigma_M)$, identifies the bite relative to the pools. The pledgers delivered a tenth to an eighth of the promised pace, which puts $\eta\Sigma$ near 0.1 and means that their targets had weak bite. I compare numerical pledgers (large η_f) with general pledgers (small η_f).

Channels. The increase on channel m is proportional to $\phi_m Z_m$, so outside hiring dominates promotion where the firm's own pipeline is thin relative to the outside pools and the horizon is short. With the quadratic cost, the added hires keep the firm's usual mix. If the marginal cost rises more slowly on the fast channel, they tilt toward it instead. Either way, the mix of all Black hires moves little, because the added hires are a small share of them.

The bar and the ladder. The bar falls only if $v_f + \lambda_f > \kappa_f$. A lower bar would widen the qualification gap between Black and white-male hires in proportion to x_Q relative to all Black hires. A gap that does not widen is consistent with a shadow value below the cost of an under-qualified director. On the ladder, as x_L rises, the marginal lateral hire comes from a higher-ranked origin. The origin rank of Black lateral hires then rises with the pledge relative to white-male hires.

Incidence. As in Welch (1976), the target binds at the pledgers and not at the firms they hire from. A promotion or a hire from below the director level adds a Black director to the economy. However, a lateral hire merely moves a Black director from one firm to another, and the seat she leaves at origin o is refilled with probability ρ, by a Black worker with probability r_o, the origin's own rate. For every Black director a pledger adds, the net number of Black directors in the labor market rises by

$$\mathcal{I} = 1 - \sigma_L \, (1 - \rho r), \tag{4}$$

where σ_L is the lateral share of the added seats and r is the average of r_o. With $\sigma_L = 0.45$ and $\rho r = 0.074$, $\mathcal{I} = 0.58$. The pledges lower r_o only by depleting the pool, which they cannot do at five percent of director seats. The framework therefore predicts reallocation without depletion.

Target design. A target on flows that counts promotions and hires from below has two effects. First, it makes the firm's count match the market's, so that $\mathcal{I} = 1$. Second, it raises the shadow value from $\eta\Gamma/(1 + \eta\Sigma)$ to $\eta\Gamma/(1 + \eta\Sigma_{PU})$, with $\Sigma_{PU} = (1 - \sigma_L)\Sigma$, which increases hiring on the slow channels. Such a target adds more net new Black directors

than a stock target if and only if

$$\eta \, \Sigma_{PU} > \frac{\rho r}{1 - \rho r}, \tag{5}$$

where the right-hand side is about 0.08 at the values above. With $\eta\Sigma$ near 0.1 and $\sigma_L = 0.45$, the two targets are within a percent of each other. At the bite observed here, a flow target would have produced about as many net new Black directors as the stock targets did, and it would have made the pledgers' count match the market's. The flow target's additions rise relative to the stock target's as bite grows.

4.2 Sample and Timing

In my main specification, the treated firms are the 115 firms that made a numerical race pledge between 2018 and 2022, and the controls are the 1,063 firms with no pledge signal in any source (see Table 3).[9] I define a firm's cohort G_f as the year of the first July 1 on or after its earliest numerical race pledge. A firm that announced its pledge in September 2020, for instance, belongs to the 2021 cohort. I keep a pledger in its cohort even if it later softened or withdrew its target. To separate the effect of a pledge from shocks that were common to all firms in 2020, I compare the change in a pledger's workforce after its pledge with the change at the controls over the same years. The four snapshots before each pledge let me see whether the two groups were already moving apart.

4.3 Event-Study Design

The main outcome of this section is the Black share of a firm's workforce both at the company level and in each of the three seniority bands defined in Section 3.3. For firm f, band b, and year t, $y_{f,b,t}$ is that share on July 1, in percent, so coefficients are in percentage points. In my main specification, I estimate race probabilistically (Section 3.4). Therefore, each worker enters the share with the classifier's probability of being Black rather than the most likely category. I perform robustness checks with the most likely category. I drop cells with fewer than five workers.

[9] I exclude four other groups of firms from the main comparison (Table 3), since they faced the same public pressure and may also have changed whom they hired: the general pledgers, which I use instead as an alternative comparison group, most of the firms that pledged after 1 July 2022 (Appendix A.3), the firms with targets for women only, and the firms whose only targets concern the board. Board members are not employees, so they do not show up in the positions recorded on LinkedIn. Because one control firm, Old Dominion Freight Line, has no firm-band-year cell with at least five workers, every share estimate uses 1,062 controls. The exhibits that count all the firms in the sample definition list 1,063 controls.

TABLE 3: SAMPLE CONSTRUCTION AND CLASSIFICATION OF S&P 1500 FIRMS

	Firms	Role in the analysis
A. Classification of the index		
Numerical race pledge, main estimation sample	115	Treated
Other numerical race pledges	28	Dropped
Other numerical pledges after 1 July 2022	17	Held out
Targets for the board only	11	Held out
Targets for women only	22	Held out
General pledgers (no numerical race target)	193	Alternative comparison group
No commitment	1,063	Controls
All firms in the index	1,449	
B. Pledge data: firms with a numerical pledge covering U.S. workers, any date		
Race and gender	108	
Race only	49	
Gender only	29	
All numerical pledgers (401 targets)	186	Description (Section 3.1)
C. Workforce panel, 2014 to 2024		
All firms in the panel	1,504	23.5 million workers, 126.5 million worker-years
Main estimation sample	1,177	14.3 million workers, 68.5 million worker-years

Notes: The table classifies the S&P 1500 firms by pledge status (Panel A), counts the numerical pledgers (Panel B) and gives the size of the workforce panel (Panel C). Panel B includes every firm with a numerical pledge covering U.S. workers, at any date; seven of the 193 general pledgers in Panel A set a numerical gender target and are counted under Gender only. The 1,504 panel firms of Panel C include the 1,449 index firms. Sources: pledge dataset; LinkedIn profiles via Revelio Labs (November 2025 release).

I estimate the following interaction weighted event study estimator of Sun and Abraham (2021):

$$y_{f,b,t} = \alpha_f^b + \delta_t^b + \sum_{g=2018}^{2022} \sum_{\substack{e=-4 \\ e\neq -1}}^{3} \beta_{g,e}^b \, \mathbf{1}\{G_f = g\} \, \mathbf{1}\{t - g = e\} + \varepsilon_{f,b,t}. \tag{6}$$

I use this estimator because, with pledges at different dates, a regression with a single post-pledge indicator would also use firms that had already pledged as controls for later pledgers. Such a regression can put negative weights on some of the effects if they differ across cohorts or grow over time (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfœuille, 2020). In Equation (6), I instead compare each cohort only with the never-pledged firms. The estimates are similar with other staggered-adoption estimators (Appendix D.4; Callaway and Sant'Anna, 2021; Borusyak et al., 2024). To summarize the path, I first average the cohort coefficients at each event time. Each cohort's weight is its share of the treated firms observed at that event time. The headline number is the mean

of these averages over event times 0 to 3, the first four snapshots after the pledge:

$$\tau_b = \frac{1}{4}\sum_{e=0}^{3}\sum_{g\in\mathcal{G}_e}\omega_{g,e}\,\beta^b_{g,e}, \qquad \omega_{g,e} = \frac{N_g}{\sum_{g'\in\mathcal{G}_e} N_{g'}}. \tag{7}$$

Here, $\mathcal{G}_e$ is the set of cohorts observed at event time e, and N_g is the number of treated firms in cohort g. Every cohort is observed in all four pre-pledge years and at event times 0 to 2; the panel ends in 2024, so event time 3 is observed only for the 2018 to 2021 cohorts (83 firms). Cells are unweighted, so the estimand is the average effect on a pledging firm, not the effect on the average worker at a pledging firm. I do the worker-weighted version, which weights each cell by its number of workers, in a robustness check (Appendix G; Solon et al., 2015).

Standard errors are clustered at the ultimate parent company level. I also report randomization inference, which reassigns pledge status and cohort years across firms within size and sector strata and compares the observed t-statistic with the resulting distribution.[10]

4.4 Identification and Threats

These estimates represent the average treatment effect on the treated (ATT) under three assumptions. First, parallel trends requires that, without their pledges, the measured Black share would have moved as it did at the controls. The closest empirical sign for this would be a violation of pre-treatment trends. Second, no anticipation requires that pledgers did not change whom they hired or promoted before event time 0, the first snapshot after their announcement. If they had, then the Black share would have already risen between event times -1 and 0. Third, the stable unit treatment value assumption (SUTVA) requires that a firm's share not depend on whether other firms pledged. In particular, if pledging firms poach minority managers from the controls, that would inflate the estimate. It is also feasible that control firms could quietly follow the pledgers in their attempts. That would shrink the coefficients. Section 5.4 empirically tests the poaching concern by looking at the company-to-company flows by treatment status, and poaching from the controls turns out to matter little for the estimate. For the economy as a whole, however, the movement of Black workers from other employers to the pledgers is what Section 7 measures: there, I follow these workers back to the employers they left and ask

[10]The test is exact under the sharp null that no pledge moved any firm's share; because the statistic is studentized, it remains approximately valid for a zero average effect.

who filled the seats they left behind.

Pledgers and controls are not equal in observables. Table 4 compares the race pledgers of the main estimation sample and the general pledgers with the controls; its workforce rows are for 2019 and its financial rows for fiscal 2019.[11] The pledgers are much larger: the average pledger has about six times the revenue, eight times the assets and six times the U.S. LinkedIn workforce of the average control. Pledgers were also more exposed to public pressure. Eighteen percent of them own one of the world's 100 most valuable brands, against almost none of the controls. More of the pledgers are consumer-facing firms and federal contractors, and a smaller share of their executives who give to political campaigns lean Republican. In the LinkedIn data, the Black share at director and above in 2019 is nearly the same at pledgers, general pledgers and controls (10.60, 10.99 and 10.26 percent). The Black and Hispanic share differs by less than a point (Table 4). On the EEO-1 reports that federal contractors file, pledgers were already ahead among officials and managers, with 5.6 percent Black against 4.1 percent at the controls.

Size, sector and exposure to public pressure predict which firms pledged. In a logit of the pledge on 31 firm characteristics, only firm size and one sector are significant at the 5 percent level, and the model ranks a random pledger above a random control 91 percent of the time (87 percent out of sample). Neither the 2019 Black share, in the LinkedIn data or in the EEO-1 reports, nor its change from 2015 to 2019 predicts the pledges. To measure how exposed each firm was to the pressure of 2020, I combine five measures into one index, their first principal component. These are advertising spending relative to revenue in 2017 to 2019, how consumer-facing the business is, ownership of a top-100 brand, the protests after the murder of George Floyd in the counties where the firm's employees live, and the share of the firm held by investors that signed an investor statement on racial equity in June 2020. In the highest decile of this index, 30 percent of the firms pledged, against 2.5 percent in the lowest decile (Figure 4, Panel A). When I hold sector, size, the 2019 Black share at director and above and its trend before 2020 fixed, one standard deviation of the index raises the probability of a pledge by 7.0 percentage points. However, neither the index nor any of its components significantly predicts, at the 5 percent level, how the Black share changed within sectors before 2020, although the confidence intervals are wide (Panel B).

Another concern is that large firms may have been hit harder by shocks in these years, such as the pandemic hiring freeze, the move to remote hiring or the later layoffs in tech-

[11] Fiscal 2019 is the fiscal year ending between June 2019 and May 2020, as first reported in an annual report.

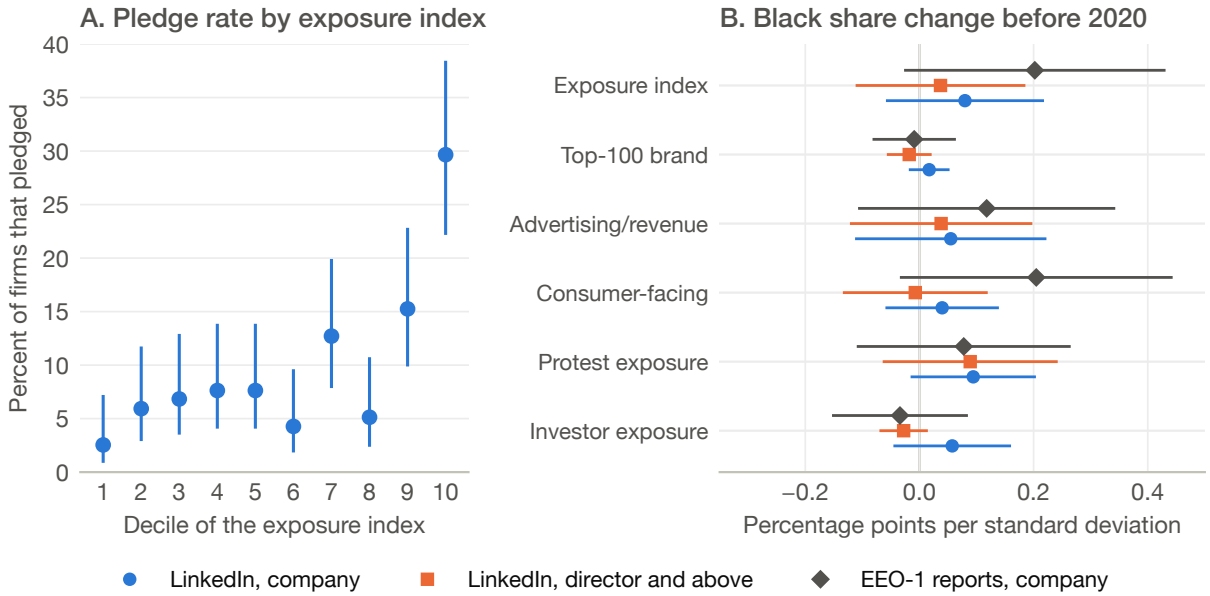
TABLE 4: BASELINE FIRM CHARACTERISTICS BY DIVERSITY PLEDGE STATUS (2019)

	Race pledgers ($N = 115$)	General pledgers ($N = 193$)	Never-pledged controls ($N = 1{,}062$)	Race minus control: Difference	Race minus control: p-value
Panel A: Size and workforce					
Revenue ($ billions)	26.7	30.4	4.6	22.1	<0.001
Net income ($ billions)	3.19	3.02	0.30	2.89	<0.001
Total assets ($ billions)	95.0	90.7	11.5	83.5	0.001
LinkedIn workers, 2019 (thousands)	21.7	18.7	3.7	18.1	<0.001
EEO-1 U.S. employees, 2019 (thousands)	31.1	31.0	8.7	22.4	<0.001
Director and above, 2019 (percent)					
Black	10.60	10.99	10.26	0.34	0.343
Black and Hispanic	19.16	19.25	18.69	0.47	0.315
Female	36.77	35.15	32.40	4.36	<0.001
Panel B: Exposure to pressure (percent)					
Interbrand top-100 brand, 2020	18.3	8.8	0.2	18.1	<0.001
Consumer-facing sector	35.7	21.2	19.7	16.0	<0.001
Federal contractor, 2016 to 2020	77.4	62.7	52.5	24.8	<0.001
Right-leaning executive donors	33.7	43.5	47.3	−13.6	<0.001
Democratic vote in headquarters state	55.9	53.4	52.6	3.3	<0.001

Notes: The table reports means of baseline firm characteristics for the race pledgers, the general pledgers and the never-pledged controls. The last two columns give the difference between race pledgers and controls and the p-value of a Welch t-test. LinkedIn workers and shares at director and above come from my panel on July 1, 2019. Revenue is available for 115, 190 and 1,012 firms (race pledgers, general pledgers, controls), a 2019 EEO-1 report for 78, 105 and 455, donor data for 107, 158 and 729, and a headquarters state for 104, 187 and 1,032. A firm is a federal contractor if it has an EEO-1 report for any year from 2016 to 2020 released under the Freedom of Information Act. Consumer-facing sectors are retail, restaurants and hotels, airlines, media, telecommunications, retail banking and consumer goods. Right-leaning executive donors are the share of a firm's executives giving to campaigns in 2012 to 2020 whose donations lean Republican (at least five donors). Democratic vote is the 2016 Democratic share of the two-party vote in the headquarters state. Sources: pledge dataset; SEC XBRL company facts; EEO-1 reports; LinkedIn profiles via Revelio Labs; Interbrand; DIME; 2016 election returns.

nology and finance. To address this, I estimate a version with sector-by-year fixed effects, which absorb the part of these shocks that firms in the same sector share (Section 5.2). Because these fixed effects do not absorb shocks that hit large firms harder within a sector, I also reweight the controls to match the pledgers' size, sector and 2019 Black share at director and above. Finally, the race classifier makes errors, which only rescale the estimates if their rates are the same at pledgers and controls and stay the same after 2020. To check this, I use the North Carolina voters matched to the panel. Among these voters, the classifier's error rates are close at pledgers and controls and did not change differently after 2020 (Appendix C).

FIGURE 4: EXPOSURE TO PUBLIC PRESSURE, PLEDGING AND THE BLACK SHARE BEFORE 2020



Notes: Panel A plots the percent of firms with a numerical race pledge by decile of the exposure index, with 95 percent Wilson intervals (115 pledgers, 1,062 never-pledged firms). Panel B plots coefficients per standard deviation of the index and its components from regressions of the pre-2020 change in the Black share on the measure and sector fixed effects, with 95 percent robust confidence intervals. LinkedIn changes run from 2015 to 2019, EEO-1 changes from 2016 to 2019 (446 firms). Sources: Revelio Labs; EEO-1 reports; SEC filings; Interbrand; Crowd Counting Consortium; pledge dataset.

4.5 Triple Differences

The second question of this paper is whether the pledgers lowered the bar for the Black workers they hired, and whether the new positions were window dressing. The event study compares a demographic share across firms. Other outcomes are measured on hires, such as a new hire's university, previous employer, starting title or later promotion. For these outcomes, I add a second comparison group inside each firm, made up of the white men hired in the same year. The triple difference compares Black hires in the main specification, and Black and Hispanic hires in the secondary one, with these white-male hires, after the pledge relative to before, at pledgers relative to the controls. The superscript BH below stands for whichever group is compared.

The sample is every U.S. hire from outside the firm at the 115 pledgers and the 1,063 controls between 2016 and 2024. I count a position start as an outside hire if the worker's previous job was at another employer or she had none. Moves within the firm are left out, as in the decomposition of Section 5.4. Each hire gets its most likely ethnicity label

(Section 3.4).

For an outcome y, firm f, band b, and hire year t, let $\bar{y}^{BH}_{f,b,t}$ and $\bar{y}^{WM}_{f,b,t}$ be the mean outcome among the compared group's hires and among white-male hires, and let $n^{BH}_{f,b,t}$ and $n^{WM}_{f,b,t}$ be their numbers. The regression mirrors Equation (6), with the gap between the two groups as the outcome:

$$\bar{y}^{BH}_{f,b,t} - \bar{y}^{WM}_{f,b,t} = \alpha^b_f + \delta^b_t + \sum_{g=2018}^{2022} \sum_{\substack{e=-4 \\ e\neq -1}}^{3} \theta^b_{g,e} \, \mathbf{1}\{G_f = g\} \, \mathbf{1}\{t - g = e\} + u_{f,b,t}, \tag{8}$$

estimated with weights $h_{f,b,t} = n^{BH}_{f,b,t} \, n^{WM}_{f,b,t} / (n^{BH}_{f,b,t} + n^{WM}_{f,b,t})$. With these weights, the regression on the gap is equivalent to the full triple difference with firm-by-year, firm-by-group and group-by-year fixed effects, and a gap built from a handful of hires gets little weight. I drop a cell if the firm hired no worker from one of the two groups in that band and year. Since a pledge raises the number of Black hires, it changes both the weights and the set of cells. The post-pledge coefficients therefore lean toward the firms that responded most.

I estimate the equation as a stacked regression (Cengiz et al., 2019): each pledge cohort forms a stack with all controls, and the fixed effects are specific to each stack. The window, reference year and clustering follow the event study, and θ_b averages the coefficients over event times 0 to 3. Since retention and promotion are measured 24 months after the hire, their window starts at -6 and their reference year is -3.

The identifying assumption of this triple differences in differences estimate is that, absent the pledge, the gap in observable variables between incoming Black hires and incoming white-male hires would have moved at pledgers as it did at the control firms. Because a fraction of Black hires would have been hired without the pledge, θ averages over all of them. Appendix H.3 derives the bounds on the added hires and the size of the label shift.

4.6 Counting New Black Directors

The event studies count the Black directors that the pledges added at the pledgers, including those who were already directors at other firms. Without the pledge, such a director would have stayed at her firm, and her seat would not have opened. I therefore count an added Black director as new to the economy if the pledger promoted her or hired her from below the director level. A Black director hired straight from a director job at another firm counts only if a Black worker took the seat she left, which happened for 7.4

percent of these seats. I compare this rate with the rate at which the same employers usually hire Black directors. Following each vacated seat further, until someone new to the director level fills it, changes the count by less than one director per 100 (Section 7).

5 Effects on Workforce Composition

5.1 The Black Share

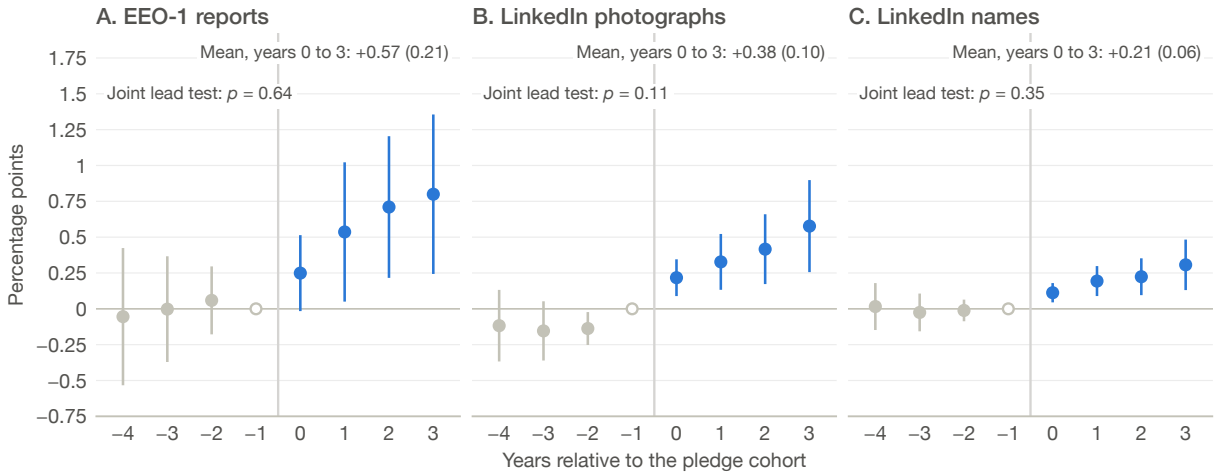
I document that the Black share of all employees at the pledging firms rose after a numerical race pledge regardless of whether I use official firm demographics, the photo-based measure of Onorato et al. (2026) or my measure predicting race from names, location and education. Figure 5 plots the coefficients of Equation (6) for three sources, each on every pledger and control it covers. The three sources are the firms' own EEO-1 reports, which count workers by self-reported race; race predicted from LinkedIn profile photographs and surnames by Onorato et al. (2026); and my name-based measure. Averaged over event years 0 to 3, the Black share rose by 0.57 percentage points in the EEO-1 reports (wild-bootstrap 95 percent interval 0.21 to 0.94), by 0.38 points in the photographs and by 0.21 points in the name-based measure. None of the joint tests rejects flat pre-trends at the 10 percent level (p = 0.64, 0.11 and 0.35). The differences in effect sizes across measures are also consistent with the amount of measurement error one would expect from official data (little measurement error), photographs (more) or names, location and education (the most).

On the firm-years covered by both sources, the name-based estimate is 0.23, against 0.57 in the EEO-1 reports on those same firm-years. Comparing the Black shares of the two sources across firms before 2020 gives a company-level correction factor of 1.8, which turns the 0.23 into 0.42, inside the EEO-1 interval. For the rest of the paper I use the name-based measure, since it is the only one of the three that covers almost every firm in every year and that I can split into the seniority bands of Section 3.3.

After a numerical race pledge the Black share rises in every band and keeps rising. Figure 6 plots the coefficients of Equation (6) for each band, and Table 5 reports their average over event years 0 to 3.[12] At director and above the average effect is 0.20 percentage points, significant at the 1 percent level. The coefficient grows from 0.07 points in the first snapshot after the pledge to 0.38 three years later, 3.5 percent of the pledgers' 2019 share

[12]The tables report standard errors clustered by firm and the figures 95 percent intervals. In the text I give significance levels, and intervals where a null or a bound is the result.

FIGURE 5: EFFECT OF NUMERICAL RACE PLEDGES ON THE FIRM-WIDE BLACK SHARE ACROSS RACE MEASURES



Notes: The figure plots event-study coefficients for the Black share of all employees in three sources. The estimator is the interaction-weighted event study of Sun and Abraham (2021), Equation (6), with firm and year fixed effects, reference year −1 and 95 percent intervals from firm-clustered standard errors. Labels give the average over years 0 to 3 (standard error). Panel A uses the EEO-1 reports of 105 pledgers and 708 never-pledged controls, each dated to the first July 1 after its count, and its lead test also covers year −5. Panel B uses the photo-based panel of Onorato et al. (2026) (92 pledgers, 491 controls, calendar years), and Panel C my name-based measure (115 pledgers, 1,062 controls). Sources: EEO-1 reports, Onorato et al. (2026), Revelio Labs.

of 10.6 percent.[13] The factors of Appendix Table C.1 put the average effect at director and above between 0.26 and 0.54 percentage points. Panel A of the table gives the averages for the other bands. Multiplied by the pledgers' LinkedIn headcounts in 2023, the averages amount to about 1,200 more Black workers at director and above and 5,800 across the firms. The Black and Hispanic share rises by about as much as the Black share, and the Hispanic share alone does not move in any band (Table 5, Panel A). This is congruent with the pledges, since only a fifth of them mention Hispanic workers directly, and always alongside Black workers.

Before the pledges the paths are flat outside the middle band. The joint test that the three pre-pledge coefficients are zero does not reject at director and above, at entry or for the whole firm (Table 5, Panel B). The exception is the middle band, whose coefficient three years before the pledge is −0.18, significant at the 5 percent level, with a joint p of 0.08.

[13] Event year 3 is observed only for the 83 firms of the 2018 to 2021 cohorts, so part of the jump between years 2 and 3 reflects the change in which firms are observed. For those 83 firms the path at director and above rises steadily, from 0.11 to 0.19, 0.30 and 0.38.

The effect had not faded by the end of the window, when the backlash began.[14] In every band the coefficient at event year 3 is the largest of the path. The July 2025 snapshot adds event years 4 and 5 (shaded in Figure 6). Event year 4, which rests on 83 pledgers, is significant at the 1 percent level in every band: 0.34 points at director and above, 0.38 in the middle band, 0.43 at entry and 0.31 for the whole firm. Event year 5 rests on 20 pledgers and is imprecise (0.28 at director and above).[15] Aligned on each firm's first retreat, the firm-wide Black share at the first July 1 after it did not fall (Appendix Table E.1, Panel B). The 95 percent interval excludes falls larger than about two fifths of the pledge effect. The number of Black hires per 100 positions did not fall in the quarters after the retreat either (Panel A of that table; Appendix E.1). Half of the retreating firms are observed for only one or two quarters after their retreat.

5.2 Placebo, General Pledgers and Robustness

The placebo in the second row of Table 6 shows no rise in a share that the pledgers' targets do not name: the 39 pledgers without a numerical gender target show no rise in the white-women share.[16] If anything the white-women share fell, significantly only in the middle band and at the 10 percent level, and the placebo has no significant lead. The third row replaces the never-pledged controls with the 193 general pledgers, firms that made public diversity commitments under the same public pressure but set no numerical race target. Against them, the estimate at director and above is unchanged (0.22, significant at the 5 percent level), but below the top it falls by about half or more and loses significance. Relative to the never-pledged firms, the general pledgers' Black share thus rose somewhat below the top, through their own statements or a trend shared with the pledgers, but not at the top. The part of the effect specific to a numerical target is largest at the level most targets named (Figure 3). The difference across bands is imprecise.

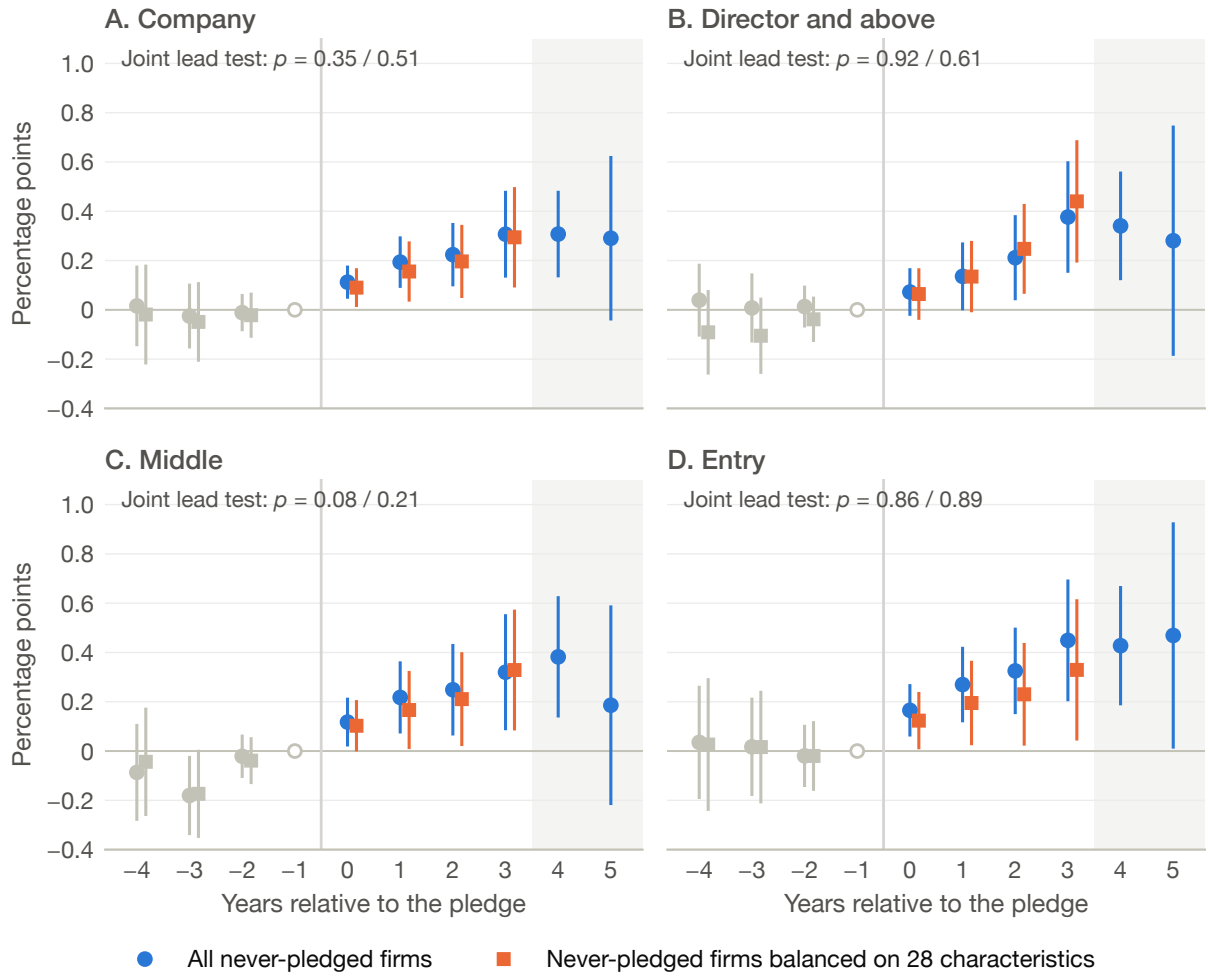
Changes to the panel or the fixed effects leave the estimates positive and of similar size, although some leads then fail. The fourth row keeps only the firms observed at every event year, which raises the estimates outside the middle band and leaves the middle band unchanged. The leads then fail in the middle band and for the whole firm ($p = 0.01$).

[14] The firm-wide increase did not reverse at the first July snapshot after firms retreated, although half of the firms are observed for only one or two quarters after retreat, too briefly to rule out a slow reversal.

[15] Some job changes of 2025 had not yet reached the profiles when the data were extracted in October 2025. Reweighting the cells for incomplete recording moves the year-4 estimates by -0.05 to $+0.08$ points. In the run that adds 2025, years -4 to 2 are the same to the third decimal. Year 3, however, also includes the 2022 cohort, whose effect at director and above is near zero, and its coefficient there falls to 0.26.

[16] Gender is predicted from first-name probabilities (Revelio Labs, 2025).

FIGURE 6: EFFECT OF NUMERICAL RACE PLEDGES ON THE BLACK SHARE BY SENIORITY LEVEL



Notes: The figure plots event-study coefficients for the Black share of the company and of each band. The estimator is the interaction-weighted event study of Sun and Abraham (2021), Equation (6), with firm and year fixed effects, reference year −1 and 95 percent intervals from firm-clustered standard errors. Circles compare the 115 numerical race pledgers with 1,062 never-pledged controls, and the shaded years 4 and 5 rest on 83 and 20 pledgers. Squares reweight the controls by entropy balancing on 28 characteristics, on common support (110 pledgers and 886 controls, effective number of controls 65). Sources: Revelio Labs; pledge dataset.

TABLE 5: EFFECT OF NUMERICAL RACE PLEDGES ON BLACK AND HISPANIC WORKFORCE SHARES

	Name-based measure				Company, other sources	
	Director and above	Middle	Entry	Company	EEO-1 reports	Photographs
Panel A: ATT, mean of event years 0 to 3 (percentage points)						
Black	0.199***	0.226***	0.302***	0.209***	0.574***	0.384***
	(0.072)	(0.077)	(0.080)	(0.056)	(0.212)	(0.104)
Black and Hispanic	0.229**	0.248**	0.319***	0.225***	0.353	0.436***
	(0.098)	(0.102)	(0.114)	(0.081)	(0.287)	(0.127)
Hispanic	0.030	0.022	0.016	0.015	−0.209	0.052
	(0.071)	(0.077)	(0.079)	(0.055)	(0.281)	(0.071)
Panel B: Joint test of the leads (p-value)						
Black	0.92	0.08	0.86	0.35	0.64	0.11
Black and Hispanic	0.68	0.14	0.76	0.48	0.25	0.39
Hispanic	0.42	0.67	0.64	0.65	0.08	0.55
Black share in 2019 at the pledgers (percent)	10.6	11.7	13.8	12.7	10.6	9.2
Numerical race pledgers	115	115	115	115	105	92
Never-pledged controls	1,062	1,062	1,062	1,062	708	491

Notes: The table reports the effect of numerical race pledges on workforce shares. The estimator is the interaction-weighted event study of Sun and Abraham (2021), Equation (6), with firm and year fixed effects, event years −4 to 3 and reference year −1. Panel A gives the ATT of Equation (7), the mean of the coefficients in event years 0 to 3, with standard errors clustered by firm. Panel B gives the p-value of the joint test that the leads, −4 to −2 (−5 to −2 for the EEO-1 reports, whose event years start at −6), are zero. The first four columns compare the pledgers with the never-pledged controls in unweighted firm-band-year cells, with shares measured as in Section 3.4. The last two columns are the whole-firm estimates of Figure 5. The 2019 row averages the pledgers' shares in 2019. In the last two columns it covers the 86 and 91 pledgers observed that year. In the EEO-1 column, the rows with Hispanic workers use the 3,968 of 3,999 firm-years that have a Hispanic count. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Sources: pledge dataset; LinkedIn profiles via Revelio Labs; EEO-1 reports; Onorato et al. (2026).

The last row drops the 2024 snapshot, which profiles may not yet reflect, and the estimates change little. Sector-by-year fixed effects, which absorb shocks common to an industry, give 0.24 at director and above, significant at the 1 percent level.[17] Reweighting the controls to the pledgers' size, sector and 2019 Black share gives 0.200 at director and above and 0.226 at the company level (Appendix Figure G.2). With them the middle band's leads fail ($p = 0.07$). Adding the four firms that pledged before 2018 does not change the results quantitatively (Appendix G). I perform randomization inference randomly assigning treatment status to companies and run 5,000 draws. The true assignment sits at

[17] With sector-by-year fixed effects the estimates are 0.20 in the middle band, 0.28 at entry and 0.20 across the firm. Against the general pledgers, they are 0.27 at director and above, significant at the 5 percent level, and, below the top, larger than without these fixed effects, 0.17 to 0.21, significant at the 5 or 10 percent level.

98.9 percent of the distribution at director and above, 99.9 percent at entry and across the firm, and 100 percent in the middle band, which is equivalent to p-values of 0.012, 0.002, 0.001 and 0.0002.

The pledgers also differ from the controls in their visibility and their exposure to public pressure (Table 4), so I also reweight the controls by entropy balancing on 28 characteristics: size in workers and in market value, sector, region, the 2019 Black share at director and above and across the company, advertising spending, how consumer-facing the business is, headquarters in a state that voted Democratic in 2016, the political leaning of executive donors, federal contracting, institutional ownership, having signed the 2019 Business Roundtable statement, membership in the S&P 500 and an index of public pressure, with indicators for missing values. Five of the largest pledgers (Microsoft, Alphabet, PepsiCo, Accenture and Amazon) have no comparable controls, and 176 small controls have no comparable pledgers. Dropping them leaves 110 pledgers and 886 controls. Against these reweighted firms, the effect is 0.22 points at director and above and 0.18 points for the whole company, both significant at the 1 percent level, and no band fails the lead test (squares in Figure 6). The pressure index includes diversity proposals that shareholders filed through 2022, and without it the effects are 0.20 and 0.18. Ownership of a top-100 brand, which almost no control has, stays unbalanced (14.5 against 6.8 percent after reweighting).

5.3 Headcount and Other Groups' Shares

One hypothesis is that firms on an expansion path would find it easier to accommodate a growing minority share, since a high number of openings would give space to several people from various demographics. Appendix Figure E.1 applies the same estimator to the logarithm of each band's LinkedIn headcount, and Appendix Table E.4 reports the averages. At the typical pledger, headcount over the four years after the pledge was about 5 percent lower relative to the controls than in the year before (-4.8 log points; first row, Company column).[18] Relative headcount had already fallen by about 3.4 percent between event years -4 and -1, and it fell by about 7.5 percent between years -1 and 3. Removing that earlier decline as a linear trend shrinks the average drop to 1.6 log points, which is not significant. The firms' own EEO-1 reports, filed by 105 pledgers and 708 controls, are too noisy to confirm the decline, but they rule out a relative expansion of more than about 9 percent (Appendix Table E.3). The top band's share of the workforce

[18]The cells are unweighted, so these are statements about the average pledger. Summed over firms, the pledgers' LinkedIn headcount grew faster than the controls' from 2019 to 2023, only because of Amazon.

TABLE 6: EFFECT OF NUMERICAL RACE PLEDGES ON THE BLACK SHARE: PLACEBO AND ROBUSTNESS CHECKS

	Director and above	Middle	Entry	Company	Firms: Treated	Firms: Control
Main estimate	0.199***	0.226***	0.302***	0.209***	115	1,062
	(0.072)	(0.077)	(0.080)	(0.056)		
Placebo: no gender target, white-women share	−0.219	−0.428*	−0.221	−0.239	39	1,062
	(0.176)	(0.238)	(0.169)	(0.169)		
General pledgers as controls	0.215**	0.071	0.140	0.108	115	193
	(0.097)	(0.090)	(0.091)	(0.066)		
Balanced cohorts	0.245***	0.234***	0.347***	0.236***	83	1,062
	(0.080)	(0.087)	(0.094)	(0.066)		
No 2024 snapshot	0.226**	0.202**	0.312***	0.221***	115	1,062
	(0.088)	(0.085)	(0.103)	(0.067)		

Notes: The table compares the main estimate of the effect of numerical race pledges with a placebo and robustness checks. The estimator is the interaction-weighted event study of Sun and Abraham (2021), Equation (6), with firm and year fixed effects, event years −4 to 3 and reference year −1. Estimates are ATTs of Equation (7), the mean of the coefficients in event years 0 to 3, in percentage points, on the LinkedIn firm-band-year cells of Section 4.3, with firm-clustered standard errors in parentheses. Unless the row says otherwise, the comparison is between the numerical race pledgers of the main sample and the never-pledged controls, and the outcome is the Black share. The placebo uses the 39 main-sample pledgers without a numerical gender target. The balanced-cohorts row keeps the 2018 to 2021 cohorts, each firm observed at every event time from −4 to 3. The last row ends the panel in 2023. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Sources: pledge dataset; LinkedIn profiles via Revelio Labs (November 2025 release); education-augmented BIFSG race probabilities, first-name gender probabilities.

rose (third row of Appendix Table E.4) only because entry-level headcount fell fastest (first row, Entry column). That share was already rising before the pledge. At the typical pledger, then, the Black share rose without a relative expansion of positions, so some other group's share fell.

I next ask which groups' shares fell as the Black share rose. In raw terms, the white-male share fell by 0.4 to 0.5 points in every band, more than twice the rise in the Black share at director and above and across the firm (Panel A of Table 7). The five groups' shares (Black, Hispanic, Asian, white women and white men) sum to 100 percent in every cell, so their effects sum to zero. The Asian share rose, and the Hispanic share did not move. The white-women share moved little, significantly only in the middle band and at the 10 percent level.

Both movements had begun before the pledges and kept their pace afterwards. The Black share, in contrast, breaks from its trend. Its pre-pledge coefficients have almost no slope, and the trend-adjusted increase is 0.23 points at director and above, significant at

the 5 percent level, and stays significant at entry and across the firm (Panel B of Table 7 and Appendix E.4).

5.4 Flows and Origins of the Hires

There are four different ways in which a firm could change its Black share at any particular step of its corporate ladder. It could hire Black workers from outside the firm, promote them into those positions from below, promote them out of those positions, or lose workers of a given group when they leave the firm. To disentangle these four channels, I split each pledger's yearly change in the Black share into the four flows and compare, cohort by cohort, how each flow changed after the pledge relative to the controls (Appendix E.2). The results are depicted in Figure 7. At director and above, hiring contributes 83 percent of the net yearly change in the Black share. Promotion into the band adds a fifth. Separations are not significant, which suggests that white incumbents did not leave faster after the pledge. In the middle band, hiring also drives the increase, promotion from the entry band adds a smaller amount, and separations offset about half of the total.

The fact that hiring contributes the majority of the effect raises the concern that untreated firms were poached by the pledgers, violating SUTVA and inflating the event study estimates. The fact that my worker panel includes the origin firm for every moving worker allows me to categorize Black inflows into pledging firms in different bins and see how those particular flows changed after the pledge (Table 8). I find that poaching accounts for little of the increase. At director and above, the increase came in similar parts from other numerical race pledgers, general pledgers, private firms with at least 1,000 U.S. workers, and smaller private firms. By contrast, never-pledged index firms sent 9 percent of the entrants and supplied 5 percent of the increase. Even at the upper 95 percent limit of the Black hires from controls, the bias is at most 0.033 points, a sixth of the estimate, and at the point estimate it is zero.

I also find that the pledges did not widen the set of companies from which a company hires. For the 95 pledgers of the 2021 and 2022 cohorts, the gap between Black and white-male hires in the share coming from employers new to the firm did not change, and the quarterly Black hire rate shows no step at the announcement either.[19] Total and white-male hiring fell around the pledges, while the Black hire rate rose by an insignificant 0.05

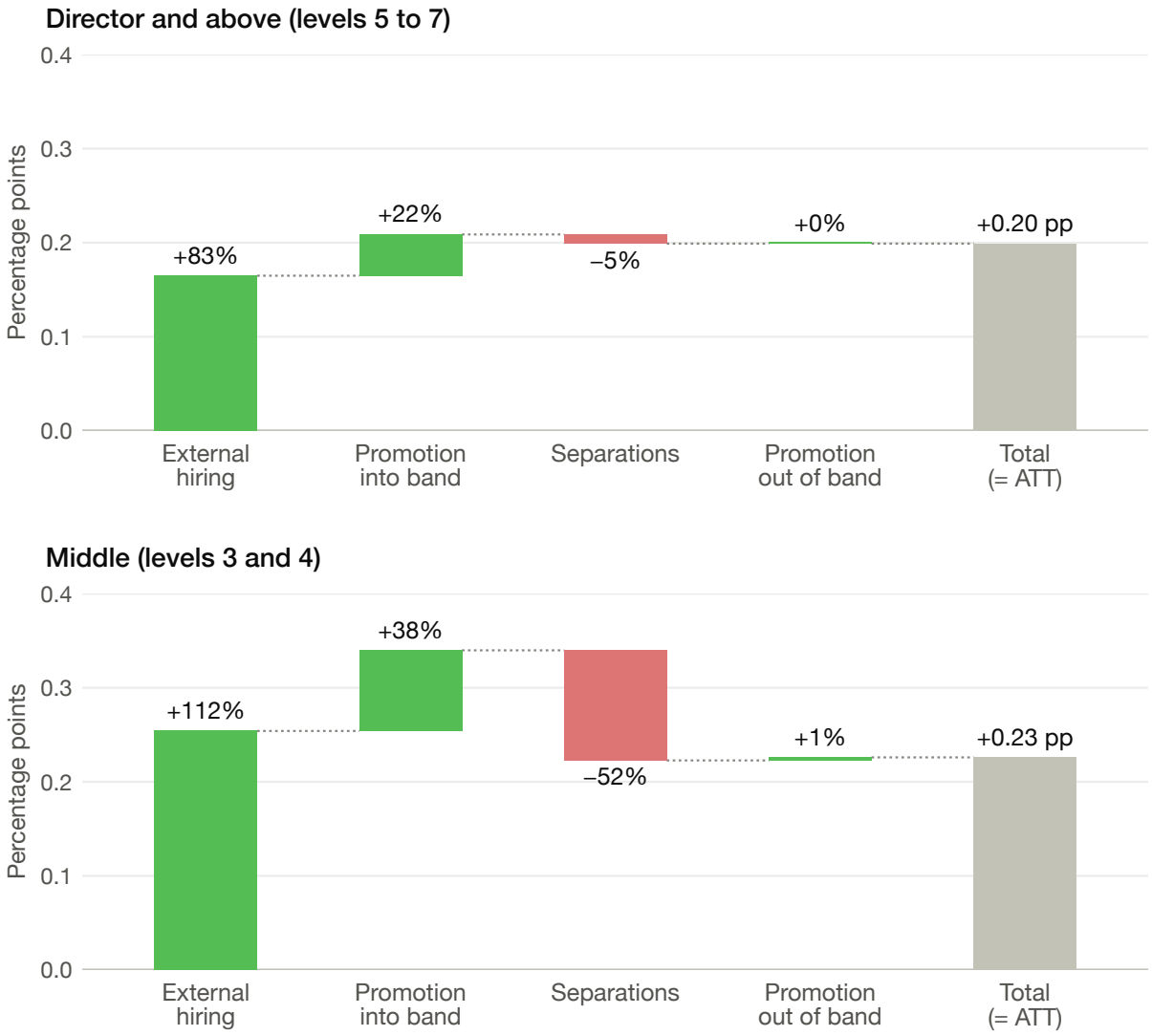
[19]The step is +0.002 per 100 positions a quarter, on a level of about 0.32. The 95 percent interval rules out a step larger than about 0.03 in either direction, under a tenth of the level.

TABLE 7: EFFECT OF NUMERICAL RACE PLEDGES ACROSS RACIAL AND GENDER GROUPS

	Director and above	Middle	Entry	Company
Panel A: Raw estimates				
Black	0.199***	0.226***	0.302***	0.209***
	(0.072)	(0.077)	(0.080)	(0.056)
	[0.921]	[0.082]	[0.864]	[0.351]
Hispanic	0.030	0.022	0.016	0.015
	(0.071)	(0.077)	(0.079)	(0.055)
	[0.416]	[0.669]	[0.637]	[0.655]
Asian	0.245***	0.420***	0.178	0.309***
	(0.094)	(0.139)	(0.114)	(0.085)
	[0.008]	[0.004]	[0.043]	[0.001]
White women	0.028	−0.262*	−0.060	−0.049
	(0.121)	(0.137)	(0.108)	(0.094)
	[0.736]	[0.098]	[0.594]	[0.117]
White men	−0.503***	−0.406**	−0.436***	−0.485***
	(0.134)	(0.158)	(0.114)	(0.103)
	[0.049]	[0.001]	[0.464]	[0.043]
Panel B: Adjusted for each group's pre-pledge trend				
Black	0.233**	0.109	0.330***	0.264***
	(0.096)	(0.114)	(0.121)	(0.083)
Hispanic	−0.073	−0.038	−0.107	−0.018
	(0.099)	(0.130)	(0.123)	(0.078)
Asian	−0.161	−0.036	0.044	0.034
	(0.152)	(0.172)	(0.148)	(0.097)
White women	0.065	−0.242	−0.074	−0.063
	(0.185)	(0.170)	(0.149)	(0.103)
White men	−0.056	0.212	−0.237	−0.182
	(0.198)	(0.206)	(0.183)	(0.131)
Pre-pledge slope, Asian (pp a year)	0.163***	0.182***	0.054	0.110***
	(0.050)	(0.055)	(0.046)	(0.035)
Pre-pledge slope, white men (pp a year)	−0.179***	−0.247***	−0.080	−0.121***
	(0.065)	(0.062)	(0.060)	(0.045)

Notes: The table reports the effect of a numerical race pledge on each group's share of a band and of the firm (Company column), in percentage points. Estimates are τ_b (Equation (7)), the mean of the event-year coefficients 0 to 3 of the interaction-weighted event study of Sun and Abraham (2021), Equation (6), with firm and year fixed effects, event years -4 to 3 and reference year -1. The sample is 115 numerical race pledgers and 1,062 never-pledged controls. Firm-clustered standard errors are in parentheses and p-values of the joint lead test in brackets. Shares average education-augmented BIFSG probabilities, and the white probability is split by the first name's probability of being female. Panel B subtracts a line fitted by generalized least squares through the pre-pledge coefficients, anchored at zero in year -1; the last two rows report its slope. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Sources: LinkedIn profiles via Revelio Labs (November 2025 release); pledge dataset.

FIGURE 7: DECOMPOSITION OF THE POST-PLEDGE INCREASE IN THE BLACK SHARE: HIRING, PROMOTIONS AND SEPARATIONS



Notes: The figure decomposes the event-study estimate of Table 5 into four flows, each labeled with its share of the net yearly change in the Black share at pledgers relative to controls. The flows come from a stacked difference in differences that compares the change after the pledge (event years −1 to 3) with the change before (−4 to −1), cohort by cohort. Workers are pooled across firms and demotions are excluded. Appendix Table E.2 gives the flows with 95 percent bootstrap intervals. Sources: LinkedIn profiles via Revelio Labs; pledge dataset.

per 100 positions a year (Table 8, total row).[20] Black hires thus ran 0.11 per 100 positions a year above what the fall in total hiring would imply at the earlier Black share, a gap significant at the 1 percent level. Therefore, the pledges tilted a shrinking flow of hires toward Black candidates.

TABLE 8: EXTERNAL HIRING CONTRIBUTIONS TO THE BLACK SHARE AT DIRECTOR AND ABOVE BY ORIGIN EMPLOYER

Origin (percent of entrants)	Black hires per 100 positions	White-male hires per 100 positions	Contribution to the Black share (pp)	Share of the increase (percent)
Other numerical race pledgers (11.1)	0.0026	−0.077	0.0165***	21.0
	(0.0131)†	(0.050)	(0.0048)	(7.5)
General pledgers (10.1)	0.0038	−0.043	0.0108**	13.7
	(0.0108)	(0.039)	(0.0043)†	(4.8)
Never-pledged S&P 1500 firms (9.3)	0.0024	−0.005	0.0038	4.8
	(0.0068)	(0.025)	(0.0039)	(4.8)
Private, 1,000+ U.S. workers (21.0)	0.0128	−0.056	0.0200**	25.4
	(0.0160)	(0.051)	(0.0079)	(7.4)
Smaller private, self-employed (21.1)	0.0047	−0.060	0.0187***	23.7
	(0.0133)	(0.060)	(0.0072)	(7.3)
Government (1.7)	0.0022	−0.002	0.0017	2.2
	(0.0031)	(0.006)	(0.0025)	(3.4)
Education, nonprofits, health (4.6)	0.0087**	0.015	0.0022	2.7
	(0.0044)	(0.011)†	(0.0027)	(3.4)
No prior job or a 2+ year gap (12.0)	0.0028	−0.002	0.0006	0.8
	(0.0132)†	(0.028)†	(0.0077)†	(10.6)
Firm abroad, other index, unknown (9.1)	0.0090**	0.012	0.0045	5.7
	(0.0043)	(0.017)†	(0.0032)	(4.3)
Total	0.0490	−0.217	0.0787***	100
	(0.0639)	(0.246)	(0.0237)	

Notes: The table reports the change in external hiring into director and above, and its contribution to the Black share, by the hire's previous employer (the origin). Each cell is the change at pledgers minus the change at controls, event years 0 to 3 against −3 to −1, with cohorts stacked and workers pooled across firms, so the origin rows add up to the total. Black hires are weighted by the education-augmented BIFSG probability of being Black and white-male hires by the probability of being white and male. Hires are per 100 positions a year (before the pledge, 1.34 Black and 3.97 white-male hires), and contributions are in percentage points a year. Bootstrap standard errors (1,000 firm-cluster draws) are in parentheses. A dagger (†) after a standard error marks an estimate whose joint lead test gives $p < 0.10$. The sample is 115 numerical race pledgers and 1,063 never-pledged controls. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Sources: LinkedIn profiles via Revelio Labs (November 2025 release); pledge dataset.

[20] Dated by start month, total hiring fell by 0.21 hires per 100 positions a quarter and white-male hiring by 0.07, both significant at the 5 percent level.

6 Black Hires Compared with White-Male Hires

Since the majority of the compositional shift came through hiring, I now turn to the characterization of those workers. From the LinkedIn profiles, I build the hire-level outcomes defined in Appendix Table H.1. The estimates are the triple differences of Section 4.5, which compare Black and white-male hires in the same firm and hire year, before and after the firm's pledge, at the pledgers relative to the controls. The reported effect is the average over event years 0 to 3.

6.1 Placement in DEI, Human-Resources and Staff Roles

A firm under pressure to show progress could place Black hires in visible roles that carry little authority, such as diversity officer or a human-resources post, or in staff functions away from the line jobs that lead to the top. Panel A of Table 9 tests this for hires into director and above. None of these placements rises. DEI and human-resources titles fell relative to white men. The share in staff rather than line functions, the path-to-power score of the function, the chance of a role new to the firm and the chance of managing people did not move.

Relative to white men, the Black hires' chance of holding a DEI title fell by 0.42 points. The decline is relative: after 2020 DEI titles became more common among Black hires at the pledgers and at the controls, but much more so at the controls, where they had been rare. DEI titles are rare in any case: after the pledges, 0.9 percent of the Black hires into director and above at the pledgers held one. Relative to white men, the Black hires' chance of a human-resources title fell by 0.87 points.

6.2 Qualifications, Pay and Career Outcomes

I find that the pledges did not change how the Black hires into director and above compare with the white men hired alongside them. Panel B of Table 9 reports the triple difference for six outcomes, which compares Black and white-male hires in the same firm and hire year, before and after each firm's pledge, at the pledgers relative to the controls. Specifically, the gaps in predicted pay, schooling and retention show no change distinguishable from zero. The gap in the rank of the previous employer, in contrast, narrowed by 1.8 percentile points, significant at the 1 percent level and with flat leads (Figure 8, panel C). Pledging firms did not dip into lower-ranked firms to recruit Black workers. The picture

TABLE 9: JOB PLACEMENT, QUALIFICATIONS AND CAREER OUTCOMES OF EXTERNAL BLACK HIRES (DIRECTOR AND ABOVE)

	Raw gap, Black minus white men				Triple	
	Pledgers		Controls		difference	Lead
	Before	After	Before	After	θ	p-value
Panel A. Placement and role						
DEI title (pp)	0.58	0.73	0.28	0.77	-0.42^{***}	0.46
					(0.15)	
HR title (pp)	1.11	1.20	1.47	1.88	-0.87^{***}	0.70
					(0.27)	
Staff rather than line function (pp)	6.12	9.67	7.03	8.89	0.63	0.13
					(1.11)	
Path-to-power score of the function (SD)	-0.069	-0.019	-0.128	-0.063	-0.038	0.24
					(0.030)	
Role new to the firm, rolling definition (pp)	-0.16	-0.21	-1.27	-1.05	0.04	0.56
					(0.44)	
Manages people (pp)	-4.59	-4.01	-4.50	-4.88	0.28	0.77
					(1.04)	
Panel B. Qualifications, pay and careers						
Top-100 university (pp)	-4.25	-4.07	-6.86	-5.89	-0.88	1.00
					(1.09)	
Years since bachelor's degree	-3.34	-2.85	-3.55	-3.00	0.61^{**}	$0.04^{\dagger}$
					(0.28)	
Previous employer's rank (percentile, movers only)	-4.19	-3.14	-2.69	-2.46	1.79^{***}	0.84
					(0.57)	
Predicted salary (dollars)	$-14{,}217$	$-13{,}204$	$-17{,}362$	$-16{,}282$	391	0.48
					(1,352)	
Still at the firm after 24 months (pp)	-2.42	-2.54	-2.49	-2.39	-0.36	0.22
					(1.00)	
Promoted within 24 months (pp)	-0.08	-0.15	-0.32	-0.38	-0.66	0.22
					(0.41)	
External Black hires	19,601	31,589	26,468	34,085		
External white-male hires	89,907	104,275	136,564	153,548		

Notes: The table reports the gap between external Black and white-male hires into director and above. The sample is U.S. external hires, 2016 to 2024, at the 115 pledgers and 1,063 never-pledged controls, with race given by the most likely label of Section 3.4. Panel A leaves out non-employee titles, which lowers its hire counts by 1 to 2 percent. Raw gaps are Black minus white-male means for 2016 to 2019 and 2020 to 2024 (to 2022 for 24-month outcomes). θ is the mean over event years 0 to 3 of the change in that gap within firm and July 1 hire year, pledgers relative to controls. It comes from the stacked triple difference of Equation (8), with firm-clustered standard errors and reference year -1 (-3 for 24-month outcomes). Lead p-values test the three leads jointly, and † marks a test that rejects at 5 percent. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Sources: LinkedIn profiles via Revelio Labs; pledge dataset.

is the same when I count every position start, moves within the firm included, and under the alternative designs of Appendix Table H.2.

These estimates reflect changes in gaps that already existed. Before and after 2020, Black hires into director and above entered lower-paid positions than the white men hired alongside them and had less experience, at the pledgers and at the controls alike. For example, after 2020 Black hires at the pledgers entered positions predicted to pay $13,200 less and had 2.8 fewer years since their bachelor's degree.[21]

The pledges also slightly changed how far up the ladder the Black hires had already climbed. Among hires with an earlier job on record, the share of Black hires taking their first director-level job fell by 4.0 percentage points relative to white men, significant at the 1 percent level.[22] The hires were not bid up: relative to white men, they did not climb more levels on the way in, and the predicted pay of their new position did not rise more relative to their previous one (Appendix Table F.3).[23]

The two remaining coefficients cannot be read as effects of the pledges. The experience gap narrowed relative to the controls, but it had already been narrowing before the pledges and fails the lead test (Figure 8, panel B). More than half of the insignificant drop of 0.66 points in promotion within 24 months comes from event year 3, which rests on three pledgers (panel F); over event years 0 to 2 it is an insignificant -0.32.

These comparisons average over all Black hires, most of whom would have been hired without the pledges. On average, the pledges moved the predicted pay of Black hires at the top, relative to white men, by between $-$\$2,260 and \$3,040 (95 percent). Relative to white men, I find no change distinguishable from zero in their retention and promotion over 36 months or in the share entering positions in the bottom fifth of the firm's pay. On a base of 22.8 percent, the bottom-fifth test would detect a shift of about 2.1 points, a tenth of the base share, with 80 percent power (Appendix Table H.4).[24] This rules out a lowered

[21] The first four columns of Table 9 give the raw gaps, pooled over firms, before and after 2020. The gaps in experience, retention and promotion are about the same at the pledgers and at the controls, and the pay gap, about \$3,000 smaller at the pledgers, changed by the same amount at both after 2020. The schooling gap is smaller at the pledgers throughout, and the previous-employer gap was wider there before 2020 and narrowed afterwards.

[22] Among hires with an earlier job on record, 64 percent of Black hires into director and above had held a director-level job before, against 73 percent of white men. At the pledgers that share rose from 60 to 67 percent after the pledges, and at the controls from 62 to 66 percent. Among hires whose profiles cover at least three earlier years, the decline is 3.8 points.

[23] For all hires, the 95 percent intervals rule out average increases larger than 0.02 levels and 2.3 log points. Revelio predicts pay from the title, place and employer, so these estimates cannot show a premium within a title.

[24] The 36-month estimates rest on event years 0 to 2 of the 2020 to 2022 cohorts, the last year on 16 pledgers alone, and are no tighter than the 24-month ones. The 36-month retention estimate is at the margin of its

bar large enough to show in the average Black hire, but not a moderate one among the few hires the pledges added, about 7 percent of the Black hires into director and above after the pledges. At the 95 percent limits, these could still have entered positions predicted to pay about $45,000 less than the other Black hires, or been 13 points more likely to fall in the bottom fifth of pay. Yet they came from previous employers at least as well ranked as the white men hired alongside them (Appendix Tables H.5 and H.4). Below the top, three gaps changed within firms (Appendix H.2).

7 Net New Black Directors

Section 5.4 showed that the pledgers raised their Black share at director and above mostly by hiring from outside the firm, and that these hires came from many kinds of employers, with only a small part coming from the controls. A Black worker hired away from a director job at another firm leaves that seat behind, so an increase at the pledgers does not need to be an increase for the U.S. labor market as a whole. In this section, I count how many of the Black directors that the pledges added were new to the economy.

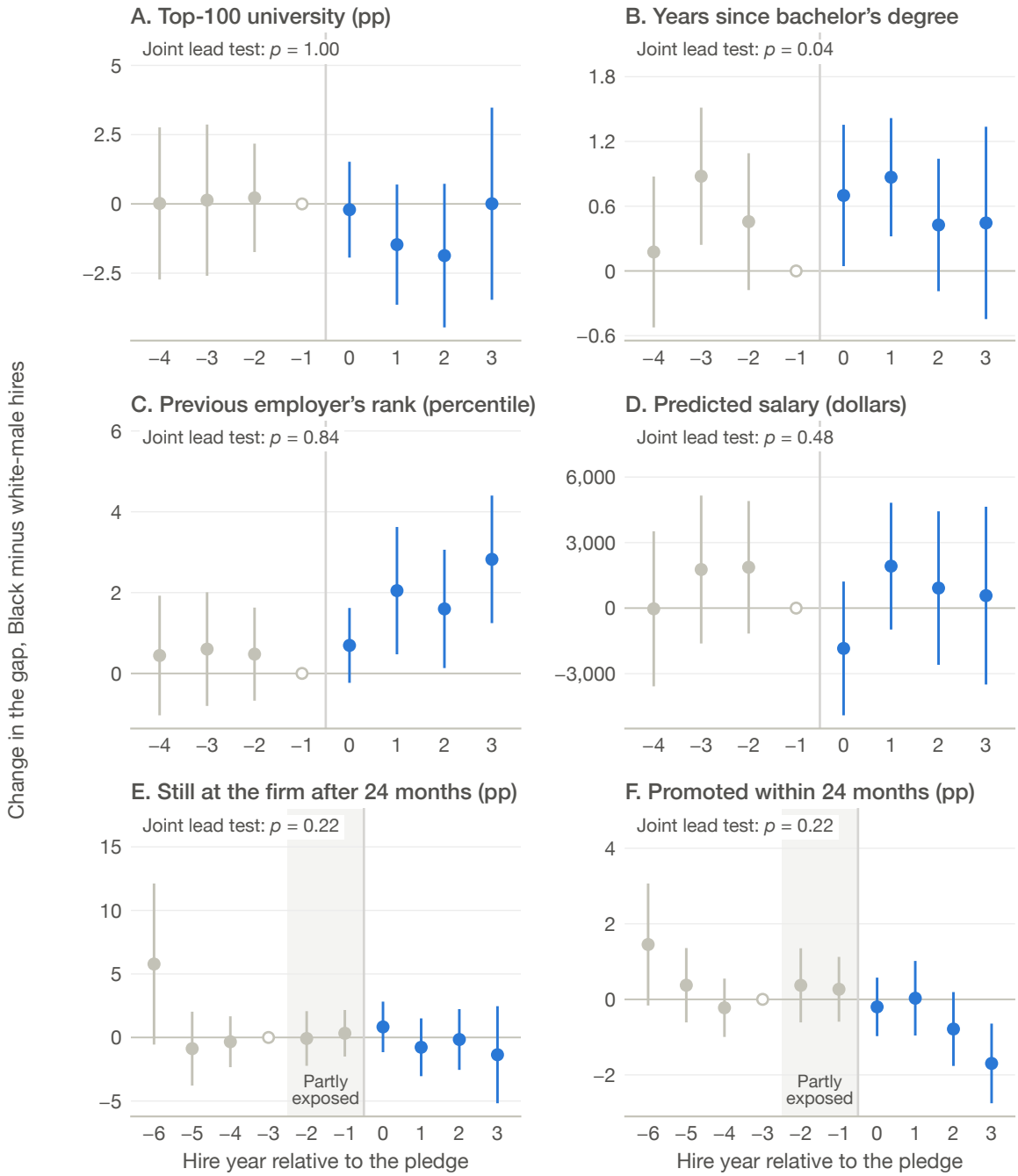
7.1 Where the Added Black Directors Came From

Table 10 splits the increase at director and above by the job that each new Black director held just before. Of every 100 Black directors added, 21 were promoted inside the firm and 79 were hired from outside. Since 34 of these outside hires were not directors in their previous job, about 55 in 100 were new to the director level. The other 45 came straight from a director-level job at another firm (95 percent interval 28 to 83), and few of them moved up a level on the way in. Such moves predate the pledges and are as common at the controls. At the pledgers, 39 percent of the Black outside hires into director and above came straight from a director-level job before the pledges and 42 percent after, against 40 and 43 percent at the controls. Counted by heads, about four in ten of the added outside hires also came from a director-level job.[25] Among the Black directors who moved from a director-level job elsewhere to a pledger after its pledge, 57 percent had started a director-level job at the firm they left within the previous two years, against 52 percent of those

lead test ($p = 0.05$) because of one early cell of a single cohort. The share of hires whose previous employer ranks in the bottom fifth of the firm's 2016 to 2019 hires has a joint lead test of $p = 0.06$ (Panel D of Appendix Table H.4) and fails a test for a linear pre-trend ($p = 0.009$).

[25] If I instead count anyone who had ever held a director-level job, outside hires who had been directors make up 70 of every 100 added (Appendix F.3).

FIGURE 8: CHARACTERISTICS AND CAREER OUTCOMES OF EXTERNAL BLACK VS. WHITE-MALE HIRES (DIRECTOR AND ABOVE)



Notes: The figure plots coefficients of the stacked triple difference, Equation (8), for six outcomes of external hires into director and above. Each coefficient is the change in the gap between Black and white-male hires in the same firm and July 1 hire year, at pledgers relative to controls, with 95 percent confidence intervals based on firm-clustered standard errors. The reference year is −1, and −3 for the two 24-month outcomes. Their shaded years are partly exposed to the pledge. Sources: LinkedIn profiles via Revelio Labs; pledge dataset.

who moved before the pledge and 53 percent of the white men who moved to a pledger after its pledge.

TABLE 10: MARKET-LEVEL DECOMPOSITION OF ADDED BLACK POSITIONS AT DIRECTOR AND ABOVE

	Per 100 added	95% interval	Black workers	
			Average	Year 3
A. Routes into the added seats				
Promoted inside the firm	21	[−4, 43]	250	480
Outside hire, not a director just before	34	[−4, 53]	400	770
Outside hire, already a director	45	[28, 83]	540	1,040
of which moving up a level	6	[−1, 13]	70	140
Total	100		1,200	2,300
B. Seats left by the movers				
Moved seats refilled by a Black worker (7.4%)	3		40	80
New Black directors	58	[23, 75]	700	1,330
C. New Black directors under other choices				
Following each vacated seat to its end	57	[22, 74]		
Without Amazon	63	[40, 79]		
Each pledger counted equally	69	[26, 102]		
Netting only seats at other U.S. employers	65	[39, 80]		
Refill in any role within 12 months	59	[25, 75]		
Refills within 24 months	58	[24, 75]		
Two-stage estimate on the entries	60	[40, 74]		
with the cohort weights of Panel A	62	[29, 83]		

Notes: The table splits the Black workers the pledges added at director and above by the job each held just before (Panel A) and counts the new Black directors (Panel B). Panel A estimates are stacked differences in differences of pledgers relative to controls, pooled across firms (Section 5.4; Appendix Table E.2). Shares are per 100 entries through inside promotions and outside hires, with 95 percent intervals from 1,000 firm-bootstrap draws, and leave out extra separations (about 5 percent of the net increase). Counts use the 1,200 added on average over event years 0 to 3 and the 2,300 in year 3. Panel B adds the moved seats that a Black worker refilled within 12 months in the same role (Appendix F.1). Panel C repeats the count of new Black directors under other choices: following each vacated seat until someone new to the director level fills it, dropping Amazon, counting each pledger equally, netting only the seats at other U.S. employers, two looser refill rules, and a two-stage estimate on the entries, first with its own cohort weights and then with those of Panel A (Appendix F.3). With equal weights the upper bound exceeds 100 because the share of sitting directors can be negative in a bootstrap draw. Sources: LinkedIn profiles via Revelio Labs (November 2025 release); pledge dataset.

7.2 The Seats the Movers Left Behind

The pledges moved Black directors between firms without thinning the pool they were drawn from: origins refilled their seats with Black workers at their usual rate, and the exceptions move the count by less than half a director per 100. Because my data covers

workers at every U.S. employer, I can follow the seats that the movers left behind (Appendix F). Table F.1 in Appendix F.1 follows the 9,348 Black directors who moved from a director-level job elsewhere to a pledger after its pledge. I count a seat as refilled when someone else starts at the origin in the same role and band within a year. The origin refilled the seat 78 percent of the time, and a Black worker filled it 7.4 percent of the time (95 percent interval 6.8 to 8.0).

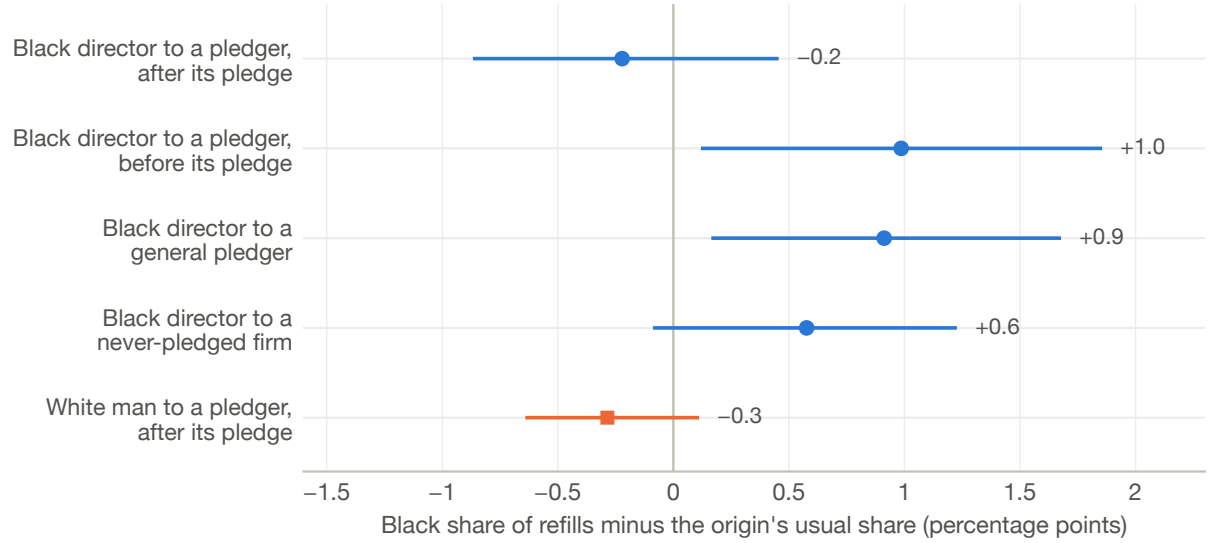
To test whether the pledges drained the pool of Black candidates at the origins, I compare the refills with the origin's usual share, the Black share of its other director-level starts in the two previous years, as the framework of Section 4.1 suggests. If the pool had been drained, the origins would have refilled the seats with Black workers less often than they usually hire Black directors. Figure 9 depicts this comparison. Black workers took 9.5 percent of the refills, both before and after the pledges. After the pledges, the refills fell short of the usual share by 0.2 points (95 percent interval -0.9 to 0.5), and seats that white men left for pledgers show the same shortfall (0.3 points, $p = 0.89$ for the difference). Before the pledges, and when Black directors left for general pledgers or never-pledged firms, the refills exceeded the usual share by 0.6 to 1.0 points. Relative to these moves, the shortfall after the pledges is 0.8 to 1.2 points ($p = 0.02$ against the seats left before the pledges). The refills were as often Black as before, but the movers now left origins whose other recent director-level starts were more often Black (9.8 against 8.5 percent). Even a shortfall of 1.2 points would lower the count of Section 7.3 by about 0.4 Black directors per 100 added. Employers that sent more of their departing directors to the pledgers before 2020 also show no detectable change in their Black director share (Appendix F.2).

Panel D of Table F.1 splits the origins. Employers with 82 or fewer director-level jobs make up half of the origins but a fifth of the moves. They refilled only 30 percent of the seats within a year in the same role and band, against 92 percent at larger employers, so a Black worker took 2.8 percent of their vacated seats against 8.8 percent. When I count any director-level start at the origin within a year as a refill, the small origins refilled 70 percent of the seats, and a Black worker took 9.0 percent of all their vacated seats, against 9.3 percent at larger origins (interval for the difference -1.3 to 1.9). Origins among the 115 pledgers refilled with Black workers at their usual rate. Where the origin was itself a numerical race pledger outside the main sample, the Black share of the refills was 3.3 points above the usual share (interval 0.0 to 5.6), although this difference shrinks when any director-level start counts as a refill.

A refill can itself be a move. About half of the Black workers who refilled these seats already held a director-level job, at another employer or in the same firm, so their own

seat opened in turn. When I follow each vacancy until someone new to the director level fills it, or until it stays empty, the refills add 6.4 new Black directors per 100 vacated seats instead of 7.4 (95 percent interval 5.6 to 7.1, Panel C of Table F.1). I use the simpler 7.4 below, since the difference moves the count by less than one director per 100 added (first row of Panel C in Table 10).

FIGURE 9: BLACK REFILLS OF THE SEATS LEFT BY DEPARTING DIRECTORS, RELATIVE TO THE ORIGIN'S USUAL SHARE



Notes: For each type of move from a director-level job at another U.S. employer, the figure plots the Black share of the workers who refilled the seats the movers left, minus the origin's Black share of director-level starts in the 24 months before, leaving out the mover's own start, in percentage points, with 95 percent intervals from 1,000 bootstrap draws clustered by origin. A seat is refilled when someone else starts at the origin in the same role and band within 12 months. Moves from 2015 to 2024. Sources: Revelio Labs, pledge dataset.

7.3 How Many Black Directors Were New

In Table 10, the inside promotions and the outside hires who were not directors just before count as new Black directors. A hire who moved from a director job elsewhere counts only if a Black worker took the seat she left. Since only a hire whose previous job was a director job leaves a director seat behind, I classify hires by that job. The added directors are entries, so a Black director who moves from one pledger to another counts at the pledger she joins, and I net the seat she leaves like any other. For every 100 Black directors the pledges added, the net number of Black directors in the U.S. labor market therefore rose by about 58 (95 percent interval 23 to 75). That is about 700 of the 1,200 added on average,

and 1,330 of the 2,300 three years after the pledge. Figure 10 shows this accounting. It starts from the 100 Black directors the pledges added, takes away the 45 director seats that the movers left at other firms, and adds back the 3 of those seats that a Black worker filled. The pledges were a small part of the economy-wide rise, and the rise elsewhere was not at their expense (Appendix F.4).

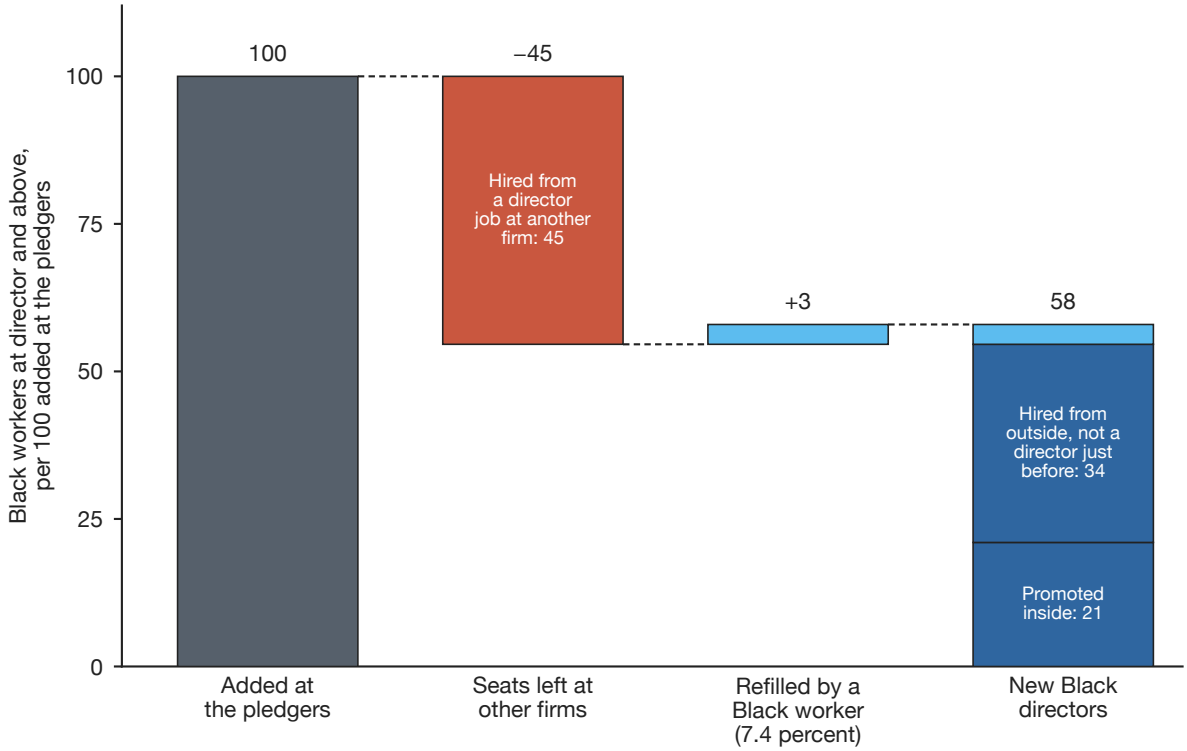
Almost all of the width of the interval comes from one number, the share of added hires who were already directors. If I hold that share at its point estimate, the uncertainty in the refill rate moves the count only between 57.7 and 58.2. My accounting choices move the count less than sampling does. These shares come from the pooled accounting of Section 5.4, in which larger pledgers weigh more. Without Amazon, 40 of every 100 added came straight from a director-level job and about 63 were new to the economy. Counting each pledger equally gives 34 and about 69, although the share of sitting directors then has an interval that includes zero (Panel C of Table 10).

The share of sitting directors also holds when I estimate it directly from the entries into director and above. I use the pledge as an instrument for whether an entrant is Black, and the outcome is whether the entrant is Black and came straight from a director-level job at another firm. Of every 100 Black entrants that the pledges added, 43 came from such a job (95 percent interval 28 to 65), which translates to 60 new Black directors (40 to 74). This interval is narrower than the accounting's because the regression weights each cohort by its entries. The 2022 cohort, in which the Black share of entrants did not move, then counts for less. With the cohort weights of Panel A, the share is 41 (18 to 76) (Panel C of Table 10; Appendix F.3).

8 Conclusion

Between 2018 and 2022, more than a hundred of the largest publicly traded U.S. companies promised to raise the share of minorities in their workforce. Using both official workforce data and LinkedIn profiles, I find that pledging firms indeed raised the share of Black people in their workforce. This increase came mostly through outside hiring and these new hires were not differentially placed in diversity or staff roles. However, following these hires back to the firms they left shows that the pledgers' own numbers overstate the change in the labor market as a whole. Of every 100 Black workers added at director and above, 45 came straight from a director-level job at another firm. Those firms refilled the vacated seats with a Black worker only 7.4 percent of the time, so the net number of Black directors in the U.S. labor market rose by about 58 for every 100 the pledgers added.

FIGURE 10: AGGREGATE INCIDENCE OF DIVERSITY PLEDGES: REALLOCATION VS. NET NEW BLACK DIRECTORS



Notes: The figure classifies every 100 Black workers the pledges added at director and above by the job each held just before and shows how many were new Black directors. Dark blue shows new Black directors and red shows hires from a director job at another firm. Light blue shows the 7.4 percent of those seats that a Black worker refilled within 12 months in the same role. Added workers are estimated as stacked differences in differences of pledgers relative to controls, pooled across firms (Section 5.4), averaged over event years 0 to 3. The 95 percent interval for the 58 new Black directors, 23 to 75, comes from 1,000 firm-bootstrap draws. Sources: Revelio Labs, pledge dataset.

These findings speak to two classic theories of affirmative action. First, I find no sign of a lowered bar as suggested by Coate and Loury (1993). After the pledges, the gaps between Black hires into director and above and the white men hired alongside them did not widen. Second, this reallocation of sitting directors is the type of sorting that Welch (1976) would have predicted. In this episode, it accounts for about 42 of every 100 Black workers the pledgers added at director and above.

My findings imply that firm-level estimates of corporate diversity initiatives overstate their aggregate effect. For the firm's purposes, a minority hired away from another firm counts the same as one promoted from within. In this episode, counting every added Black director as new would overstate the increase in the labor market by about 73 percent. Since an internal pipeline takes years to build and a sitting director can be hired

at once, firms end up hiring them from similar positions in similar firms. Targets and disclosures defined over promotions, and over hires from below the director level, would instead credit only the margins that add Black directors one for one, which in this episode produced 55 of the 58. In the framework of Section 4.1 a target on those margins would also raise the value firms place on them. Whether it adds Black directors depends on the target's bite: at the bite observed here it would have added about as many as the stock targets did, and it would have made the pledgers' count match the market's.

By their 2025 filings, nearly every pledger had softened or withdrawn its target. At the first snapshot after these retreats, the Black share at the pledgers had not fallen, although half of them are observed for only one or two quarters after retreating.

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A The Pledge Dataset

A.1 Sources and Candidate Firms

The universe is the S&P 1500 constituent snapshot, with each firm matched to its ultimate parent in the workforce data. Out of that sample, I decided to drop the following three firms: Sodexo, which is not an index member, Sylvamo, which has no pre-period workforce, and a mismatched LinkedIn record under Amazon's ticker. This leaves 1,449 firms.

I search every firm of the universe in four kinds of documents. The first is the proxy statements (DEF 14A): 9,104 filings of 1,399 firms from 2015 to 2021, and 8,462 proxy filings of the 965 control firms with an SEC identifier from 2022 to 2025. The others are the firm's annual and current reports with their exhibits, its sustainability, ESG and diversity reports, and its press releases and newsroom statements, found by web search. As a cross-check, I also collect the signatory lists of nine industry coalitions formed during the sample period:

- CEO Action for Diversity & Inclusion
- Paradigm for Parity
- White House Tech Inclusion Pledge
- OneTen
- Black Economic Alliance
- Catalyst CEO Champions for Change
- Just Capital Racial Equity Tracker
- As You Sow Racial Equity Audit campaign
- Bloomberg Gender-Equality Index

Signing one of the coalitions does not suffice to make the firm a pledger, and I use the lists only to flag firms for a second reading.

A language model (Claude, by Anthropic) reads the passages that the dictionary described below selects from these documents and returns one record per workforce target under a fixed extraction schema. A record holds the verbatim quote, its source document, the groups named, the number and its kind, the baseline and its year, the deadline, the positions covered, the first announcement date stated in the text, whether executive pay

depends on the target, and whether it comes with a commitment of money. Every record that could make a firm a pledger was then checked by hand against the primary text (Appendix A.4).

A dictionary of 99 terms, frozen on 9 September 2026, selects the passages that reach the extraction. It has 40 group terms (Black, African American, Hispanic, Latino, people of color, BIPOC, underrepresented minorities, ethnically diverse, and the like), 45 commitment terms (commit, pledge, aspire, target, goal, will, and the like) and 14 quantifiers (percentages, counts, dollar amounts, multipliers and year deadlines). A window of text is a hit when a group term and a commitment term fall within 300 characters of each other, and a strong hit when a quantifier accompanies them. Of 474 pledge quotes verified by hand, 98.3 percent are dictionary hits and 83.5 percent strong hits. I deduplicate the windows and queue them for extraction.

A.2 Classification Rules

A numerical race pledge is a commitment that meets the following five conditions.[26]

(i) The target is the firm's own and covers its U.S. workforce or part of it (the whole firm, management, officers, or hiring and promotion flows), explicitly or as a firm-wide or global goal. Goals of a coalition, a parent or a subsidiary, board-only targets, and targets for suppliers, customers, communities, philanthropy, lending or marketing do not count.

(ii) The target names a racial or ethnic group of U.S. workers that includes Black or Hispanic workers: a specific group (Black, African American, Hispanic, Latino) or an umbrella group (underrepresented minorities, people of color, BIPOC, racial or ethnic minorities, women of color). Targets for Asian workers only and for nationalities do not count. A joint target for women and minorities counts only if race carries a number of its own.

(iii) The target gives a number: a share, a percentage increase, a multiple such as doubling, or a count of people holding positions. Parity with no numeric level does not count, and neither does a count of hires or program places with no share attached.

(iv) The target has a deadline: a calendar or fiscal year, or a horizon measured from a dated document (e.g., "within five years").

[26] Each condition must be established in a primary document of the firm: a proxy statement, an annual or current report, a sustainability or diversity report, or a company press release.

(v) The target is a public goal for a future year. Three things that resemble one do not count: a metric in a single year's bonus scorecard that the firm never published as a multi-year goal, a report that a past goal was met, and a commitment to a process (training, surveys, diverse candidate slates) rather than to an outcome.

Rules (a) and (b) settle two kinds of hard cases, and rules (c) to (i) show how the conditions apply to the closest cases:

(a) A goal the firm calls aspirational or an ambition counts if it meets the five conditions. After 2020 the word served mostly as a legal hedge, so excluding such goals would remove many of the most specific targets.

(b) A target published only in a sustainability report or a press release counts, provided its date can be verified. A robustness check keeps only the pledges that appear in an SEC filing.

(c) Condition (iii): the parity goals of SAIC, Exxon and Carnival set no numerical level, so they are general.

(d) Condition (ii): Lumentum and CF Industries set only a joint number for "women and minorities", which is not a race target.

(e) Condition (ii): Philip Morris International's target of 20 percent Asian representation in senior roles covers no group that includes Black or Hispanic workers, so it is general.

(f) Condition (iv): the targets of Eli Lilly, General Mills and Morgan Stanley have no deadline.

(g) Condition (iii): a share of hires or promotions counts, but JPMorgan's commitment to hire 4,000 Black students is a count with no share attached.

(h) Condition (i): Astec, Inspire Medical, Acadia, Euronet, Harmonic, Azenta, Quanex and Dynatrace set race targets only for the board, so they are general. The 11 firms whose only targets concern women on the board (Thermo Fisher, Central Pacific Financial, Balchem, Repligen, Edgewell, ACI Worldwide, CoreCivic, HealthEquity, Ryman Hospitality, Interface and BJ's Restaurants, each in a proxy statement filed between 2022 and 2024) are not pledgers, and I hold them out of every estimate (Section 4). For Live Nation, Brinker and Ventas I leave out the board targets and count the others. Ball's only race target covers its South American packaging business, so it is general.

(i) Condition (v): all of American Airlines' numbers are retrospective.

The same conditions and rules, with women in place of the racial or ethnic group in condition (ii), define a numerical gender pledge. The 29 numerical pledgers without a race target include seven firms whose race commitment is a general statement but whose gender target meets the five conditions (TransUnion, TD SYNNEX, Yum! Brands, Eastman Chemical, Autodesk, Jacobs and Dow). They do not include three firms whose gender targets cover only workers outside the United States (DXC Technology, Aflac and Northern Trust). Of the race pledgers in the main estimation sample, 94 name the United States explicitly, 14 state no geography but use U.S. race categories, and 7 set global or multi-region targets that include the United States.

A.3 Dating and the Estimation Samples

I date a pledger by the day of the first public announcement of its earliest qualifying race target. A pledger with no race target is dated by its first gender target; for most of these only the year is known, and I set the date to 1 January of that year. The source can be a press release, a report, a proxy statement or a news article reporting the firm's announcement. A restatement does not reset the date, while an earlier qualifying target moves it back. Reports are dated by their release date, which can differ from the year on the cover. The cohort is the year of the first July 1 on or after the date. A firm that announced on or before 1 July 2022 is treated, however its pledge was found. A firm whose first announcement came later is held out as having pledged after the window. The exceptions are Domino's, WEX and IDEXX: their first announcements fall in the second half of 2022, and I code them as pledgers with the 2023 cohort. Neither group enters the estimates. For robustness checks, I keep a second date, that of the first SEC filing that mentions the pledge.

Two further rules make a firm's years before the pledge comparable to its years after it. I exclude firms created inside their event window, such as a spin-off whose earlier years in the panel belong to another workforce. The one such firm in the data is Sylvamo, spun off from International Paper in 2021. Mergers and divestitures inside the window are documented and handled with the same rules for treated and control firms (Appendix B.2).

Table A.1 reconciles this classification with the estimation samples and comparison groups of Sections 5 and 6. It does not list the hire-level subsamples or the robustness samples of Appendix G. Of the 157 race pledgers, 14 are held out as having pledged

after the window and 28 are not in the main sample, among them the seven with cohorts outside 2018 to 2022: Starbucks and Intel (2015), Iron Mountain and Airbnb (2017), and Domino's, WEX and IDEXX (2023). The other 115 form the main sample. The triple differences on hires and the retreat designs of Appendix E.1 use the same 115 pledgers and, where they need controls, the same 1,063.

The other samples follow from the classification of the index in Table 3. The 193 general pledgers are 186 firms whose commitments were all general statements and the seven firms of Appendix A.2 with a general race statement and a numerical gender target, which Section 3.1 counts among the 29 numerical pledgers without a race target. The 22 firms classified as having targets for women only are held out of the race comparisons. Three of them set gender targets only for workers outside the United States, so Section 3.1 does not count them as numerical gender pledgers (Appendix A.2). The placebo of Table 6 uses the 39 firms of the main sample that set no numerical gender target. The 17 firms held out as having pledged after the window are the 14 with a race target and three with a gender target only (Newmont, Chart Industries and Avery Dennison).

TABLE A.1: RECONCILIATION OF THE ANALYSIS SAMPLES

	Firms	Used in
A. From the race pledgers to the main sample		
Numerical race pledgers, any date	157	Section 3.1
Pledged after 1 July 2022, except Domino's, WEX and IDEXX	−14	Held out
Not in the main sample, including cohorts 2015, 2017 and 2023	−28	Dropped
Main sample	115	Sections 5 and 6
B. Subsets of the main sample		
Cohorts 2018 to 2021	83	Event years 3 and 4; Table 6
Cohorts 2018 to 2020	20	Event year 5, Figure 6
Cohorts 2021 and 2022	95	New employers, Section 5.4
No numerical gender target	39	Placebo, Table 6
C. Other groups of the index		
Never-pledged controls	1,063	Every comparison
General pledgers	193	Comparison group, Table 6
General statements only	186	
General race statement, numerical gender target	7	
Targets for women only	22	Held out
Numerical pledge after 1 July 2022, except Domino's, WEX and IDEXX	17	Held out
With a race target	14	
With a gender target only	3	
Targets for the board only	11	Held out

Notes: Panel A follows the race pledgers of Section 3.1 to the main sample. Panel B splits the main sample and Panel C gives the other groups of Table 3. Rows with a minus sign are subtracted to give the next row that is not indented; other indented rows detail the row above them. Source: pledge dataset.

A.4 Verification and Validation

Each pledger's classification, source and date were read against the primary text, and the quote was checked to belong to the firm. The firms whose status or date was hardest to establish, together with a random sample of the rest, were then read again by three independent readers. One checked the source, one the five conditions and the date, and one tried to refute the classification; a fourth resolved disagreements. One round of this check covered 51 firms: the three readers agreed on 36, and the fourth decided the other 15. The round included 12 randomly drawn firms that the first two passes had classified as pledgers. All 12 held, and 3 were re-dated.

Two exercises measure the quality of the stored fields. In a blinded double-coding, 100 randomly sampled pledges were coded again from their verbatim quotes alone, without access to the stored fields. Agreement is 89 percent on whether a pledge targets race ($\kappa = 0.78$) and 93 percent on gender ($\kappa = 0.86$). The numerical target matches exactly in 93 percent of cases (correlation 0.88), the deadline agrees within a year in 98 percent, and the classification as numerical agrees in 89 percent. The weakest field is the link to executive pay ($\kappa = 0.27$), which I use only to compare the pledgers that tied a target to pay with the others (Appendix A.7).

The second, a re-sourcing exercise, traced every stored quote back to its primary text. Of the 345 pledges checked, 48 appeared verbatim in SEC documents already on disk. The other 297 were traced to company pages, press releases, archived snapshots and a web search. In the end, 215 quotes (62 percent) match the source word for word, 87 differ by an elision or a few words, 25 are confirmed on group, number and deadline, and 18 (5 percent) could not be located. Eight rows had a substantive discrepancy with the source. Four assigned a percentage or a scope to the wrong level and were corrected, and two attributed another company's quote to the firm. A wider check of every race pledger confirmed these two attribution errors. It also found thirteen more firms whose race target rested on a commitment outside the workforce or on ambiguous text. I classify the fifteen firms concerned as general.

A last check compares the data with an independent catalog. The appendix of Cai et al. (2024) lists diversity targets found in firms' sustainability reports. It contains two pledgers that the proxy-based passes had missed, because their only target appeared in a standalone report. I added both and classified them under the rules above.

A.5 Screening the Controls

A control must show no commitment in any source. It appears on none of the nine coalition rosters, its proxy statements from 2015 to 2025 contain no qualifying language, and its annual reports, current reports and sustainability reports contain no target with a number and a deadline; every hit in these reports was read by hand. Every control flagged by any screen was also checked on the web. A board-diversity commitment alone does not disqualify a control.

I drew a random sample of 30 of the 362 controls whose documents use race language without a stated target. None had a numerical pledge, 13 had a general statement, and 17 had nothing that qualifies as a commitment. Their general statements escaped the screens, so these firms stayed among the controls. Misses of this kind make the controls resemble the pledgers and attenuate the estimates. Reading the 2022 to 2025 proxy statements turned up firms that pledged after the estimation window. These 17 firms are held out of every estimate. The same reading found the 11 firms whose only targets concern women on the board. They are not pledgers (rule (h) in Appendix A.2) and are held out as well.

A.6 Specificity and Ambition of the Targets

Targets also differ in how precisely they name positions. I classify the scope each target states as a named title (director, vice president, managing director, officer and similar), a generic scope ("leadership," "senior management") or the whole workforce. Just over a third of the 401 targets name a title, two fifths a generic scope and a quarter the whole workforce (Panel A of Table A.5). Race targets name a title more often than gender targets do (39.5 against 29.9 percent), and they cover the whole workforce more often (29.2 against 20.4 percent).

The targets in Table A.3 asked for a median of 1.0 point a year, as did the first race targets of 73 of the 115 main-sample pledgers that I can express as a change in points, 32 with a starting share from the firm's reports or EEO-1 filing. On LinkedIn, the targeted share had risen by a median 0.33 to 0.39 points a year in the five pre-pledge years (its Black part scaled by 1.62, or by 1.8 company-wide to 2.72 at director and above), and 84 to 86 percent asked for a faster rise, as did 14 of the 17 company-wide targets against the EEO-1 trend. With firms' own figures where they match the target and LinkedIn or EEO-1 otherwise, 26 to 29 percent of the 58 targets due by 2025 had been met by the deadline or, for the 40 still running, by the last year observed, and the median had covered about

TABLE A.2: PLEDGING FIRMS BY INDUSTRY

Sector (GICS)	Index firms	Numerical pledgers			Main sample	General pledgers	Controls	Race pledgers (%)
		All	Race	Gender only				
Consumer Discretionary	185	32	31	1	22	13	140	17
Industrials	246	38	29	9	17	26	182	12
Financials	237	22	21	1	16	27	185	9
Information Technology	171	24	21	3	16	14	133	12
Health Care	168	19	17	2	13	16	130	10
Consumer Staples	71	14	14	0	12	13	43	20
Materials	74	19	12	7	10	8	48	16
Communication Services	42	8	8	0	7	7	27	19
Real Estate	94	3	2	1	1	11	79	2
Energy	63	3	1	2	0	10	50	2
Utilities	56	3	1	2	1	7	46	2
Not classified	42	1	0	1	0	41	0	0
Total	1,449	186	157	29	115	193	1,063	11

Notes: Number of S&P 1500 firms in each sector. Numerical pledgers are the firms with a numerical pledge covering U.S. workers made between 2015 and 2024, including those that pledged after 1 July 2022; race pledgers set a race or ethnicity target (with or without a gender target). Main sample: the race pledgers of the main estimates. General pledgers and controls as in Table 3; the seven firms whose race commitment is a general statement but whose gender target is numerical appear both under Gender only and among the general pledgers, and the 11 firms whose only targets concern the board and three firms whose gender targets cover only workers outside the United States count only among the index firms. The last column gives race pledgers as a percentage of the sector's index firms. Sectors are the Global Industry Classification Standard sectors recorded in Compustat; 42 firms, 41 of them general pledgers, have no sector code in the extract.

TABLE A.3: ANNUAL INCREASE REQUIRED BY THE RACE TARGETS

	Targets	Firms	25th pct.	Median	75th pct.	Above 1.5 (percent)
All targets	72	42	0.67	1.00	1.56	26
By the lowest level covered						
From entry level	31	20	0.51	0.75	1.11	13
From levels 2 to 4	18	16	0.69	1.34	1.95	39
Director and above	23	17	0.95	1.13	1.75	35
By the group named						
Umbrella term	49	34	0.75	1.00	1.67	31
Named group	23	13	0.45	0.67	1.23	17

Notes: Race targets stated as a share that also state the starting share, its year and the deadline. The required annual increase is the target share minus the starting share, divided by the years between them, in percentage points. Umbrella terms are underrepresented minorities, people of color and similar; named groups are Black, Hispanic, or both.

TABLE A.4: MINORITY GROUPS MENTIONED IN THE RACE AND ETHNICITY PLEDGES

	Race only	Race and gender	All race pledgers	Main estimation sample
A. Groups named in any numerical race target (a firm can use several)				
Black or African American	28 (57)	31 (29)	59 (38)	48 (42)
Hispanic or Latino	14 (29)	19 (18)	33 (21)	26 (23)
Asian	2 (4)	4 (4)	6 (4)	6 (5)
Native American or Indigenous	2 (4)	4 (4)	6 (4)	5 (4)
Underrepresented minorities	9 (18)	29 (27)	38 (24)	29 (25)
People of color or BIPOC	16 (33)	30 (28)	46 (29)	29 (25)
Racial or ethnic minorities	2 (4)	22 (20)	24 (15)	14 (12)
Racially or ethnically diverse	9 (18)	23 (21)	32 (20)	22 (19)
B. Each firm once, in the first row that fits				
Black and Hispanic	14 (29)	19 (18)	33 (21)	26 (23)
Black, not Hispanic	14 (29)	12 (11)	26 (17)	22 (19)
Underrepresented minorities only	2 (4)	19 (18)	21 (13)	16 (14)
Umbrella that counts Asian workers	18 (37)	58 (54)	76 (48)	51 (44)
Historically Black colleges only	1 (2)	0 (0)	1 (1)	0 (0)
Firms	49	108	157	115

Notes: Number of firms (column percent in parentheses) whose numerical race targets use each wording, coded from the verbatim quote. Panel A counts every wording a firm uses; Panel B places each firm once, in the first row that fits. The first three columns follow Figure 3 and include the firms whose race pledges came after 1 July 2022; the last is the main estimation sample.

TABLE A.5: SENIORITY SCOPE AND FEATURES OF RACE AND GENDER TARGETS

A. Seniority scope of the target (percent of the row)

	Named title	Generic scope	Whole workforce	Targets
Race targets	39.5	31.3	29.2	233
Gender targets	29.9	49.7	20.4	157
Joint targets	27.3	45.5	27.3	11
All targets	35.4	38.9	25.7	401

B. Race and gender targets of the same 103 firms (percent of targets)

	Race targets	Gender targets
Names a title	38.0	34.1
Covers the whole workforce	29.3	21.1
States a baseline share	47.3	43.9
Tied to executive pay	22.0	24.4
Targets	150	123

Notes: Panel A: the 401 numerical targets of the 186 numerical pledgers, by whether the stated scope names a title (director, vice president, officer, partner, manager), a generic band ("leadership," "senior management") or the whole workforce; rows sum to 100 up to rounding. Panel B: the 103 firms with both a race and a gender target, and the share of their 150 race and 123 gender targets with each feature.

half of its promise; on LinkedIn and EEO-1 alone, 9 to 14 percent had. Summed over the 73, the event study's break relative to the controls amounts to a tenth to an eighth of the promised pace.

The 103 firms with separate race and gender targets write the two alike in most respects. Similar shares name a title (38.0 and 34.1 percent), state a baseline share (47.3 and 43.9 percent) and are tied to executive pay (22.0 and 24.4 percent). The race targets cover the whole workforce more often, 29.3 percent against 21.1 percent of the gender targets (Panel B of Table A.5).

Numerical and general pledgers also differ in what they put on record. Almost every numerical pledge promised some annual public reporting. Yet by text matching of their filings, only about half of the numerical pledgers disclosed the share they were starting from and reported progress in later years; among general pledgers, 4 and 16 percent did. Numerical pledgers are also ahead on pay links (30 against 3 percent of firms), published EEO-1 reports (23 against 2) and targets limited to the senior levels (40 against 12).

Table A.2 counts firms by sector, and Table A.4 groups the wording of the race targets. Of the 157 race pledgers, 59 name Black workers explicitly, 33 of them together with Hispanic workers. No firm names Hispanic workers without Black workers.

A.7 Pledges Tied to Pay or with a Disclosed Baseline

Two features make a pledge easier to check: a disclosed starting share, against which later reports can be compared, and a link to executive pay, which puts the target in the proxy statement each year. Among the 115 pledgers of the main sample, the 31 that tied a target to pay raised the Black share at director and above by 0.25 percentage points, against 0.16 for the others, but raised the firm-wide share by 0.18, against 0.20. The 38 that disclosed a baseline raised the firm-wide Black share by 0.28 points, against 0.15, but not the share at director and above (0.18 against 0.18). The evidence is mixed: with standard errors of about 0.1, none of these differences is statistically distinguishable from zero.[27]

[27] Main specification of Section 4.3, estimated separately on each group of pledgers against all controls. The standard error of each difference treats the two estimates as independent. Pay links and baselines are coded from the firm's own race targets.

A.8 Firm-Level Pledge Tables

Table A.6 lists the 115 race pledgers of the main estimation sample. For each firm it gives the cohort (the year of the first July 1 after the firm's first numerical race pledge), the month of that pledge, a short summary of its race targets that meet the five conditions, and the document that states them. Joint targets for women and minorities and parity goals without a level are left out, as the rules above require. When a firm later restated its target, that document can postdate the first announcement. A file with the quote, source, date, group, number, deadline and levels of every target, including the gender targets and the firms outside the main sample, is available from the author.

TABLE A.6: THE 115 RACE PLEDGERS OF THE MAIN ESTIMATION SAMPLE

Firm	Cohort	Date	Race target(s)	Source
Accenture Plc	2021	Sep 2020	By 2025: raise African American and Black share of the organization from 9% to 12% and of managing directors from 2.8% to 4.4%; raise U.S. Hispanic American and Latinx share from 9.5% to 13%.	Press release, Sep 10, 2020 (quotes CEO announcement Sep 1, 2020); DEF 14A 2021
Adobe, Inc.	2021	Sep 2020	Double underrepresented minorities in leadership positions by 2025; double Black representation as percentage of U.S. employees by 2025.	Proxy statement, 2021
Air Products & Chemicals, Inc.	2021	Oct 2020	At least 20% minority representation from a 17% base in U.S. professional and managerial population by 2025 (October 2020); raised to 30% by 2025 (November 2021).	Press release, November 2021; proxy statement, 2021
Alaska Air Group, Inc.	2021	Apr 2021	Increase racial diversity at all levels, including leadership, to at least 30% by 2025.	Press release, April 2021
Alphabet, Inc.	2020	Jun 2020	Improve leadership representation of underrepresented groups by 30% by 2025; more than double the number of Black+ Googlers at all other levels by 2025.	2021 DEF 14A proxy statement (CEO announcement Jun 17, 2020)
Altria Group, Inc.	2020	Jun 2020	Increase VPs and Directors who are Asian, Black, Hispanic or two or more races to at least 30% within five to ten years.	Corporate responsibility report, June 2020
Amazon.com, Inc.	2021	Nov 2020	Double number of Black directors and vice presidents again in 2021, after a goal to do so in 2020 was achieved; increase hiring of Black mid-level employees by 30% in 2021.	Proxy statement, Apr 2021

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Table A.6, continued

Firm	Cohort	Date	Race target(s)	Source
Arrow Electronics, Inc.	2022	Mar 2022	Grow U.S. leaders of underrepresented race and ethnicity (supervisors and above) by 0.5 percentage points in 2022 and by 2 percentage points by end of 2025.	Environmental, Social, and Governance Report, Mar 2022
AvalonBay Communities, Inc.	2021	Dec 2020	Increase under-represented minorities in leadership (officers and director-level associates and above) from 15% to 20% by 2025 and to 25% by 2030.	Press release, December 2020
Axalta Coating Systems Ltd.	2022	Apr 2022	Fill 30% of management positions in U.S. and Canada with underrepresented racial/ethnic groups by 2030.	Proxy statement, Apr 2022
Axcelis Technologies, Inc.	2022	Mar 2022	U.S. employees 10% URM by 2025 (8% in 2020); URMs in Engineering and Product Support 8% (4%); URMs in Leadership (Bands L2-L4) 5% (2.7%); all by 2025.	ESG report, October 2023
Baxter International, Inc.	2022	Jul 2021	Increase representation of ethnic minorities in leadership roles (director and above) in United States to 25% by 2030.	Corporate Responsibility Report, 2022
Becton, Dickinson & Co.	2022	Oct 2021	Improve ethnic diversity, and separately gender diversity, by 1% year over year at management and executive levels through 2030.	Press release, Oct 18, 2021
Best Buy Co., Inc.	2021	Dec 2020	Fill one of three new non-hourly corporate positions with BIPOC (Black, Latinx, Indigenous) employees by 2025.	Press release, Dec 9, 2020
Biogen, Inc.	2021	Apr 2021	Increase diversity in U.S. manager positions and above by 30% by end of 2021 (share held by Black, African American and Latinx employees, plus Asian employees where underrepresented).	Proxy statement, April 2021
BlackRock, Inc.	2020	Jun 2020	Double representation of Black senior leaders (directors and above), from 3%, by 2024; increase overall U.S. Black workforce representation by 30%, from 5%, by 2024.	Company memo by Larry Fink and Rob Kapito, Jun 22, 2020
Brinker International, Inc.	2022	Sep 2021	Increase minority representation in restaurant operations leadership from 35% (2021) to 40% by fiscal 2025.	Sustainability report, Sep 2021
Bristol Myers Squibb Co.	2021	Aug 2020	Double U.S. Black/African American executives (VP+) from 3.0% to 6.0% and Hispanic/Latino executives from 3.7% to 7.4%, from June 2020 levels, by year-end 2022.	2020 ESG Report
Brown-Forman Corp.	2020	Jun 2020	Achieve 25% people of color in United States by 2030.	Form 10-K, June 2020

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Table A.6, continued

Firm	Cohort	Date	Race target(s)	Source
Cardinal Health, Inc.	2022	Sep 2021	Increase Black and African American representation at manager level and above in the U.S. to 11% (from 5%) by 2030; all other ethnically diverse groups to 23% (from 17%) by 2030.	Proxy statement, September 2021
Cisco Systems, Inc.	2021	Sep 2020	Achieve 25% increase in Black/African American employees entry-level through manager by 2023; 75% increase in Black/AA from Director through VP+ by 2023.	Company document 'Social Justice Beliefs and Actions', released Sep 2020
Citigroup, Inc.	2019	Aug 2018	Black AVP to MD share in North America: 8.1% to 11.5% by 2025. U.S. Hispanic and Latino AVP to MD share: 13.7% to 16% by 2025. 2018 goal: U.S. Black, 6% to 8% within three years.	Reuters/SRN News, Sep 2022; Bloomberg and internal memo, Aug 2018
Conagra Brands, Inc.	2021	Feb 2021	Double representation of people of color in management and middle-management roles within five years.	Press release, February 2021
Constellation Brands, Inc.	2022	Apr 2022	Achieve 30% ethnic diversity among U.S. salaried population by fiscal 2026.	Form 10-K, Apr 2022
Danaher Corp.	2021	Nov 2020	Raise people of color among U.S. associates to 35% (from 31% in 2019) by 2025; later raised goal to 38% by 2025.	Press releases, November 2020 and October 2021
Dave & Buster's Entertainment, Inc.	2022	Mar 2022	Improve representation of BIPOC team members in corporate and field leadership by 33% by end of FY2025.	Proxy statement, May 2022
Dell Technologies, Inc.	2020	Nov 2019	Black/African American and Hispanic/Latino U.S. workforce from 13% to 25% by 2030; U.S. people leaders (managers/executives) from 10% to 15% by 2030.	FY20 Progress Made Real Report (published Jul 21, 2020); announced Nov 12, 2019
Delta Air Lines, Inc.	2021	Jan 2021	Double representation of Black officers and director-level employees by 2025 compared to 2020 baseline.	ESG Report, 2022; CEO memo, Aug 2020
Dick's Sporting Goods, Inc.	2022	Aug 2021	Grow BIPOC representation in leadership roles by 30% by 2025.	Sustainability report (2020 Purpose Playbook), August 2021
DoorDash, Inc.	2022	Oct 2021	Raise underrepresented people of color from 32% to 40% of the U.S. corporate team by 2025; from 7% to 10% in technical roles; from 12% to 20% in director level and above.	Company newsroom, Oct 4, 2021
Eaton Corp. Plc	2022	Oct 2021	Increase salaried U.S. minorities to 34% by 2030.	Press release, Oct 28, 2022 (goals first set 2021)

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Table A.6, continued

Firm	Cohort	Date	Race target(s)	Source
Ecolab, Inc.	2021	2020	Increase management-level ethnic/racial diversity to 25% by 2030.	Form 10-K, fiscal 2020 (filed February 2021)
Emerson Electric Co.	2021	2021	Double representation of U.S. minorities at leadership level by 2030.	Proxy statement, Dec 2021
Expedia Group, Inc.	2021	Feb 2021	25% of U.S. external hires from racially and ethnically under-represented groups by end of 2021.	Inclusion and Diversity Report, 2020
FirstEnergy Corp.	2021	Jan 2021	Increase underrepresented racial/ethnic employees by 30% by 2025 from 2019 baseline: overall 10% to 13%, supervisor-and-above leadership 7.5% to 9.7%.	Press release, Jan 2021; restated in 2021 proxy
FMC Corp.	2021	Feb 2021	Reach 14% Black/African American representation in the U.S. workforce by 2027.	Form 10-K, fiscal 2020 (filed February 2021)
Fortive Corp.	2021	Jun 2021	BIPOC representation: 37% of U.S. workforce by 2025 (from 34% in 2020); 36% in professional and management roles by 2025 (from 26%).	Sustainability report, June 2021
FTI Consulting, Inc.	2021	Apr 2021	Reach 120 underrepresented minority Senior Managing Directors by 2025 (more than double the 2020 level).	Press release / 2020 Corporate Sustainability Report, Nov 2021
Gap, Inc.	2020	Jun 2020	Double Black and Latinx employees at all levels in U.S. HQ offices by 2025; increase Black Store Leader representation by 50% by 2025.	Company website, 'Equality & Belonging' page (corroborated by press release, Jun 2021)
Gilead Sciences, Inc.	2021	2020	Double the percentage of Black employees at Gilead by 2025; diverse hires at least 15% of all hiring by 2025.	Proxy statement, 2021
Global Payments, Inc.	2022	Aug 2021	Increase people of color from 17% to 29% of U.S. leadership roles and from 31% to 39% of the U.S. workforce by 2025.	Global responsibility report, Aug 2021
Granite Construction, Inc.	2021	Mar 2021	Increase persons of color from 14.7% to 20% workforce-wide by 2025; increase persons of color in leadership from 17% to 20% by 2025.	Form 10-Ks, fiscal 2020 and 2021
Hasbro, Inc.	2018	Apr 2018	Reach 25% minority representation in the U.S. workforce by 2025; raise U.S. annual hiring rate of minorities to 40% by 2025.	Proxy statements, DEF 14A April 2018 and DEFA14A May 2022
Hewlett Packard Enterprise Co.	2022	Feb 2022	Double U.S. Black and Hispanic executive headcounts by 2027, from 2020 levels.	Proxy statement, Feb 2022; Form 10-K, FY2022

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Table A.6, continued

Firm	Cohort	Date	Race target(s)	Source
Hilton Worldwide Holdings, Inc.	2021	Apr 2021	25% ethnic diversity at corporate leadership levels in the U.S. by 2027.	Newsroom Q&A, April 2021
HP, Inc.	2020	Jun 2020	Double the number of Black and African American executives by 2025.	Form 10-K, December 2020
Huntington Bancshares, Inc.	2022	Oct 2021	15% ethnically diverse in middle and executive management by end of 2021 (from 13.5% at end of 2020).	American Banker, October 2021; proxy statement, 2021
Hyatt Hotels Corp.	2022	Feb 2022	Double representation of Black leaders in key leadership roles by 2025 (Hyatt outward materials: U.S. vice president-plus and general manager levels).	Form 10-K, Feb 2022
Ingersoll Rand, Inc.	2021	Feb 2021	Increase underrepresented populations in the U.S. workforce from 25% to 30% by 2025.	Form 10-K, fiscal 2020 (filed February 2021)
Ingredion, Inc.	2022	May 2022	Black, Indigenous and People of Color (BIPOC) at 38% of management level in the U.S. by 2030.	Press release (GlobeNewswire), May 2022
Intercontinental Exchange, Inc.	2022	Mar 2022	Double representation of under-represented minorities in U.S. officer population from 6% to 12% by 2027.	Proxy statement, Mar 2022
International Flavors & Fragrances, Inc.	2022	Dec 2021	Reach 40% people of color in U.S. management roles (executive, upper, middle) by 2030, up from 27.2% in 2021.	Press release, Dec 2021; ESG+ Report, Jun 2022
International Paper Co.	2020	Feb 2020	30% minority representation in U.S. salaried positions by 2030.	Form 10-K, February 2021
Intuit, Inc.	2021	Feb 2021	Increase Black, Latino, and other underrepresented ethnic minorities from 12% to 16% of total workforce by 2023.	Company blog, 2021 (confirmed in Fast Company interview, Apr 2021, and DEI Report 2023)
Janus Henderson Group Plc	2022	Aug 2021	Increase racially and ethnically diverse employees in senior management positions from 11% to 16% by 2023.	DEF 14A proxy statement, Mar 24, 2022
JetBlue Airways Corp.	2021	Feb 2021	Double race and ethnic minority representation at officer and director level from 12.5% to 25% by 2025.	Press release, Feb 2021
Johnson & Johnson	2021	Nov 2020	Achieve 50% growth of African American talent at manager and above levels in United States by 2025.	Proxy statement, Mar 2021
Johnson Controls International Plc	2021	Jan 2021	Double the representation of minority leaders in the United States within five years (by 2026).	Press release, January 2021

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Table A.6, continued

Firm	Cohort	Date	Race target(s)	Source
Keurig Dr Pepper, Inc.	2021	Jun 2021	Increase by 25% the representation of people of color in Director and above roles (U.S.) by 2025 (from 17% in 2020).	Press release, June 2021
KeyCorp	2022	Feb 2022	Increase representation of people of color in senior leadership by 25% over five years (by 2026) and by 50% over ten years (by 2031), from 2021.	2022 ESG Report / investor relations press release, May 2023
Keysight Technologies, Inc.	2021	May 2021	45% of U.S. new hires underrepresented minorities by fiscal year 2021; increase to 47.4% of U.S. new hires by fiscal year 2022.	CSR press release, May 17, 2021 and Form 10-K (FY2021)
Lincoln National Corp.	2021	Sep 2020	Grow minority representation at Officer level (Assistant Vice President and above) by 50% by 2023, with focus on Black officers.	Press release, Sep 2020
Live Nation Entertainment, Inc.	2021	Jul 2020	Double Black leadership (Director and above) by 2025; 30% racially/ethnically diverse U.S. leadership (Director and above) by 2025.	Proxy statement, 2021
lululemon athletica, Inc.	2021	2021	40% racially diverse representation for stores and 30% racially diverse representation of directors and assistant store managers and above, by 2023.	Demographic survey findings report, 2020 (undated)
Macy's, Inc.	2020	Sep 2019	Achieve 30% ethnic diversity at senior director level and above by 2025.	Press release via BusinessWire, Sep 10, 2019
Mastercard, Inc.	2020	Jun 2020	Grow Black leadership at Vice President level and above globally by 50% by 2025.	Proxy statement, 2021
McDonald's Corp.	2021	Feb 2021	Increase representation of historically underrepresented groups in Senior Director and above roles in the U.S. from 29% to 35% by 2025.	Proxy statement, 2021
McKesson Corp.	2021	Jun 2021	Increase representation of people of color in leadership ranks (VP and above) by 20% relative to FY2021 (21% baseline) by 2025.	Proxy statement, 2021
Medtronic Plc	2019	Nov 2018	Maintain or exceed 20% ethnically diverse representation in U.S. management positions in fiscal 2020.	Inclusion, Diversity, and Equity Report, 2019
Merck & Co., Inc.	2022	Feb 2022	Raise U.S. Black/African American representation in senior management (VP or SVP) from 3% in 2020 to 10% by 2024; U.S. Hispanics/Latinos from 5% in 2020 to 10% by 2024.	Form 10-K (FY2021), filed Feb 25, 2022

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Table A.6, continued

Firm	Cohort	Date	Race target(s)	Source
Meta Platforms, Inc.	2020	Jun 2020	50% of workforce from underrepresented populations by 2024; increase people of color in U.S. leadership, including Black leadership, by 30% from 2020 to 2025; double Black and Hispanic employees within 5 years (by 2025).	Form 10-K, 2021 (filed 2021)
Microsoft Corp.	2020	Jun 2020	Double the number of Black and African American people managers, senior individual contributors, and senior leaders in the U.S. by 2025; also double Hispanic and Latinx in the same roles by 2025.	DEF 14A 2020-2021 (announcement Jun 2020)
Molson Coors Beverage Co.	2021	Jul 2020	Increase representation of people of color in U.S. operation by 25% by 2023 across salaried employees and leadership positions.	Earnings call, Jul 2020; restated in 2021 proxy
Mondelez International, Inc.	2021	Sep 2020	Double Black representation in U.S. management team by 2024.	Form 10-K, fiscal 2020 (filed February 2021)
NIKE, Inc.	2021	Mar 2021	35% racial and ethnic minorities in U.S. corporate workforce by 2025; 30% racial and ethnic minorities at Director level and above in the United States by 2025.	Proxy statement, March 2021 (quoting FY20 Impact Report)
NXP Semiconductors NV	2022	Feb 2022	Achieve 50% underrepresented minorities in U.S. workforce by 2025.	Form 10-K, FY2021, and Sustainability Report, Apr 2022
Palo Alto Networks, Inc.	2022	Oct 2021	Achieve 20% underrepresented minorities in U.S. workforce by 2025.	DEF 14A proxy statement, Oct 29, 2021
Paramount Skydance Corp.	2022	Oct 2021	A 40% U.S. hire and promotion rate for ethnically diverse vice presidents by end of 2021.	ViacomCBS CEO letter, Oct 1, 2021
PepsiCo, Inc.	2020	2020	Increase U.S. Black and Hispanic managerial population to 10% by 2025.	Proxy statements, 2024 and 2021
Pfizer Inc.	2020	Oct 2019	Raise U.S. minority representation at vice president level and above to 25%, from 19% in 2019, by 2025.	Press release, Oct 17, 2019
Pinterest, Inc.	2021	May 2021	Increase underrepresented races and ethnicities to 20% in technical roles and leadership by 2025.	Newsroom report, May 2021
PPG Industries, Inc.	2022	Mar 2022	Increase Black, Latinx and Asian employees by more than 45% in non-frontline, U.S.-based workforce by 2025, from a January 1, 2021 baseline.	Press release, 2022

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Table A.6, continued

Firm	Cohort	Date	Race target(s)	Source
PVH Corp.	2021	Jun 2021	Double Black and Hispanic/Latinx representation at Director and VP levels (U.S.) by 2026; increase total BIPOC at SVP+ by 50% by 2026.	Bloomberg sponsored article, Feb 2022; company website I&D page
QUALCOMM, Inc.	2021	Feb 2021	Increase underrepresented minorities in leadership (principal-and-above technical; director-and-above non-technical) by 15% by 2025; increase overall workforce by 20% by 2025.	Form 10-K, Nov 2021
Ralph Lauren Corp.	2021	Aug 2020	People of color at least 20% of the Global Leadership Team by 2023 (stated Aug 2020); at least 10% Black, African or African American added in May 2021.	Earnings call, Aug 4, 2020; Form 10-K, May 2021
Salesforce, Inc.	2020	Jul 2020	Double U.S. representation of Black employees in leadership (VP+) by end of 2023; increase U.S. Black representation 50% by end of 2023; increase underrepresented minorities (Black, Latinx, Indigenous, multiracial) employees and leaders 50% by 2023.	Press release, Jul 1, 2020; third target from update, Feb 11, 2021
Sealed Air Corp.	2021	May 2021	Increase representation of racial and ethnic minorities in the U.S. workforce to above 35% by 2025 (34% in 2021).	Form 10-K, 2022
Sensata Technologies Holding Plc	2022	Sep 2021	Reach 25% racial/ethnic diverse representation in manager and above roles in the U.S. by 2026.	Press release, 2021
Shake Shack, Inc.	2022	Sep 2021	Raise people of color to 50% (from 34.7%) of Shack leadership (general managers through regional vice presidents of operations) by 2025; to 30% (from 19.6%) of home office leadership (director through senior vice president) by 2025.	Proxy statement, April 2022; sustainability report, April 2021
Signet Jewelers Ltd.	2021	Jun 2021	Increase diversity in Director+ leadership by 25% by 2030; increase Black, Indigenous, and People of Color (BIPOC) in Director+ support center roles by 20% by 2030.	FY2022 Corporate Citizenship and Sustainability Report, Jun 2022 (goals announced Jun 30, 2021)
Southwest Airlines Co.	2021	Sep 2020	Double the percentage of racial diversity in the Senior Management Committee by 2025.	Press release, September 2020
State Street Corp.	2021	Jul 2020	Triple Black and Latinx leadership (senior vice presidents and above) by 2023; double percentage of Black and Latinx populations (all levels) by 2023.	Company web statement, July 2020
Sunrun, Inc.	2022	Apr 2022	Increase BIPOC leaders in manager roles by 25% by 2025 (from about 35% in 2021).	Impact report press release, Apr 22, 2022

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Table A.6, continued

Firm	Cohort	Date	Race target(s)	Source
Target Corp.	2021	Sep 2020	Increase representation of Black team members across the company by 20% over three years (by 2023).	Press release, Sep 10, 2020
TEGNA, Inc.	2021	Mar 2021	Increase BIPOC share of content teams (news, digital, marketing) to about 36% by 2025; increase BIPOC representation in content leadership roles by 50% by 2025.	Form 8-K, 2021
The Bank of New York Mellon Corp.	2022	Jul 2021	U.S., by year-end 2023: raise Black representation by 15%, to 12%, and Latinx by 15%, to almost 8%; raise Black senior leaders (Levels M/S) by 30%, to over 4%, and Latinx senior leaders by 30%, to over 5.5%.	2020 Enterprise ESG Report (published Q2 2021)
The Boeing Co.	2021	Aug 2020	Increase Black representation in the U.S. by 20% by 2025.	Proxy statements, 2023 and 2024
The Chemours Co.	2019	2018	20% of all U.S. positions filled with ethnically diverse employees by 2030 (2020 report); raised to 30% by 2030 (2023 proxy).	Corporate responsibility report, 2020; proxy statement, March 2023
The Coca-Cola Co.	2021	Mar 2021	Mirror U.S. census data at all job grade levels for U.S. roles by 2030: 13% Black, 18% Hispanic, 6% Asian.	Proxy statement, Mar 2021
The Goldman Sachs Group, Inc.	2021	Aug 2020	Reach 7% Black professionals in Americas and UK vice president population by 2025; reach 9% Hispanic/Latinx professionals in Americas vice president population by 2025; double U.S. campus analyst hires recruited from HBCUs by 2025.	Press release, 2020
The Kraft Heinz Co.	2022	Oct 2021	Reach 30% people of color (from 24%) in the U.S. salaried employee population by 2025.	ESG report and news release, October 2021
The New York Times Co.	2021	2021	Double share of Black/African American and Latino/Hispanic colleagues in leadership by 2025.	Diversity and Inclusion Report, 2021
The Sherwin-Williams Co.	2021	Jun 2021	Increase underrepresented racial/ethnic groups in U.S. management roles to 30% by 2025.	Sustainability report press release, June 2021
Tractor Supply Co.	2022	Sep 2021	Increase People of Color at manager level and above by 50% by 2026, from a 2021 baseline.	Press release, September 2021
Trane Technologies Plc	2021	Apr 2021	Increase racial and ethnic diversity of salaried U.S. workforce from 17% to 26% (a 50% increase) by 2030.	2020 ESG Report (published Apr 30, 2021)

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Table A.6, continued

Firm	Cohort	Date	Race target(s)	Source
Truist Financial Corp.	2021	Sep 2020	Racially and ethnically diverse employees in senior leadership from about 12% to at least 15% in three years (by 2023); hiring from HBCUs from 3% to 5% by 2023.	Press release, September 2020; press release, July 2021
Uber Technologies, Inc.	2021	Jul 2020	Double Black in Director titles and above (5 most senior levels at Uber) by 2025.	Proxy statement, 2021
Valmont Industries, Inc.	2021	May 2021	Increase representation of people of color by 50% by 2025 and double it by 2030, workforce-wide, from a 2018 baseline.	Form 8-K, May 2021
VF Corp.	2021	Feb 2021	BIPOC representation of 25% within Director and above population by 2030 (from 16% in 2021).	Press release, February 2021
Visa, Inc.	2021	Jul 2020	Increase underrepresented employee population by 50% at Vice President and above level in the U.S. by 2023 (three years from July 2020); increase overall U.S. underrepresented employee population by 50% by 2025 (five years).	Proxy statement, 2020
Walker & Dunlop, Inc.	2021	Sep 2020	Minorities in management (assistant VP/associate director and above) from 11% (2020) to 20% by 2025; underrepresented groups in management from 14% (2021) to 25% by 2025; minorities among top earners to 15% by 2025 (6% baseline).	Proxy statement, Mar 2021; ESG reports and EEO disclosure, 2021
Wells Fargo & Co.	2020	Jun 2020	Double Black leadership (senior managers and executive ranks) by 2025, baseline 5.8% in 2020.	CEO memo, June 2020; press coverage, September 2020
Whirlpool Corp.	2021	Jul 2020	Increase by 50% the number of Black employees across all levels and increase by 50% Black director-level and above employees by 2025.	Sustainability Report press release, Mar 2021
Workday, Inc.	2021	Sep 2020	Increase Black and Latinx employees in U.S. by 30% by 2023 (baseline 7.56% in 2021); double Black and Latinx leaders in U.S. by 2023.	Company blog post, Sep 2020
Zoetis, Inc.	2021	Feb 2021	Increase people of color from 21% to 25% in U.S. workforce by 2025; increase Black colleagues from 4% to 5%; increase Latinx colleagues from 5% to 6%.	Form 10-K, Feb 2021

B The Workforce Panel

B.1 Construction

The panel uses the November 2025 release of Revelio Labs' LinkedIn data. I keep positions in the United States that have a start date. For each worker and each July 1 from 2014 to 2024, the worker counts at the employer of the most recent position that started on or before that date and has not ended by then; if two positions started on the same day, the more senior one counts. Many people do not record end dates, so a later start at another employer ends a spell even when the profile never closed it. Positions still open when the data were extracted, on 1 October 2025, are right-censored, and I do not impute an end for them. Counting only the most recent position removes about 8 percent of the workers whose position at a firm is still open on the snapshot date, because their profiles show a later job elsewhere.

Revelio assigns each employer to its ultimate parent under current ownership, so a company acquired in 2022 counts as part of its acquirer in 2016 as well. For each parent I keep its largest affiliates up to 95 percent of its U.S. workers. This drops 4 percent of workers and removes small, often mismatched units. I drop positions whose title marks the holder as retired, former, a board member or a volunteer; they are 2.3 percent of the worker-years in the top band. I keep frontline positions.

The same release also gives a snapshot on 1 July 2025, which I build with the rules above and those of Appendices B.2 and B.5. Its 298,305 workers without BIFSG probabilities, all of them new to the panel in 2025, are scored with BIFSG without the education term (Section 3.4). I use the snapshot only in the checks on persistence. It adds the shaded event years 4 and 5 of Figure 6, where event year 4 covers the cohorts 2018 to 2021 (83 pledgers) and event year 5 the cohorts 2018 to 2020 (20 pledgers), and it gives the Black share at the first July 1 after a pledger's retreat in Panel B of Table E.1. The data were extracted on 1 October 2025, three months after this snapshot, so it likely records recent moves less completely than earlier snapshots do. The share of workers whose position started in the preceding twelve months falls from 19.3 percent in 2024 to 15.6 percent in 2025 at pledgers, and from 18.7 to 14.5 percent at controls, although part of this fall continues a decline in entries under way since 2022.

The hire-level designs use a panel of position starts instead of snapshots. It holds every U.S. position that started between 2016 and 2024 at the 115 pledgers and the 1,063 controls, 18.0 million starts in all, each assigned to its firm with the reassignments of

Appendix B.2. I date each start by the first July 1 on or after it, the clock of the cohorts, so a position that started in August 2024 counts in 2025. Its band is Revelio's level with the correction of Appendix B.5. A start is an external hire when the worker's previous position (the latest one at any employer and in any country that started before it) was at another employer, with or without a Revelio identifier, or when there is no previous position. When the previous position is at the same firm, the start is an internal move. I look up previous positions only for starts at Revelio levels 3 to 7, so entry-level starts cannot be classified, and the entry-level results of Appendix H.2 use all starts. The triple differences compare hires by their most likely label. At director and above, 702,265 of the 1,570,897 starts labeled Black, Hispanic or white male are external hires: 111,743 Black, 106,228 Hispanic and 484,294 white-male hires. In the middle band the count is 1,446,447 of 2,498,307 (290,978, 271,158 and 884,311). The flow decomposition of Appendix E.2 finds the workers who join a firm by comparing consecutive snapshots. Of those who join at director and above and match a start by the same worker at the same firm, dated to the same July 1, 94 percent are external hires by this rule.

B.2 Company Mapping

Because every position is assigned to a parent, a wrong parent moves whole workforces between firms. I screened the estimation firms for three signs of a wrong parent: a large unit, at least a tenth of the parent's U.S. workers or 3,000 workers, whose name shares no word with the parent's; a jump in the firm's headcount of more than 50 percent up or a third down from one year to the next; and a panel headcount far outside the firm's EEO-1 total. Independent readers checked each flagged unit against public records of ownership, and 47 units were reassigned. Eleven are The Huntington National Bank and ten units of its lineage, which Revelio had placed under Associated Banc-Corp, a control. The other 36 are mostly tenants' staff that Revelio had placed under their real-estate landlords, such as casino operators under the real-estate trusts that own their buildings. One further unit, the Four Points by Sheraton hotels, was moved from Blackstone to Marriott. When I rebuild the panel without the reassignments, the affected workers' histories reproduce more than 99.6 percent of their worker-years in Revelio's mapping, so the rebuild changes nothing beyond the reassigned units. Moving the Four Points hotels changes the main estimates by less than 0.002 percentage points.

The static mapping also breaks histories when ownership changes. When a firm sells or spins off a unit, workers whose profiles name the parent before the sale leave its his-

tory; when it acquires a firm, the acquired workers enter its earlier years. These breaks are unrelated to pledges.

B.3 Coverage

Not every worker has a profile. For the 461 panel firms whose names match a consolidated EEO-1 report for 2020 and whose ratio of LinkedIn to reported workers lies between 0 and 4, the median firm's LinkedIn headcount on 31 December 2020, inside the report's payroll period, is 74 percent of the U.S. headcount it reported to the government, with an interquartile range of 50 to 106 percent. Coverage is highest in information and finance and lowest in utilities and transportation, where it is about half. Because the outcomes are shares, low coverage matters only if it differs by race or between pledgers and controls. In North Carolina, 16.7 percent of Black and 15.5 percent of white registered voters have a matching profile. Table B.1 compares the panel's headcount on 1 July 2019 with the 2019 reports of the firms in the main comparison. The means are close, 86 percent at race pledgers and at general pledgers and 83 percent at controls, but the medians are 79, 77 and 66 percent. Part of the means comes from six firms whose LinkedIn headcount is more than three times the reported one, five of them controls. Without these six the means are 83, 86 and 73 percent.

TABLE B.1: LINKEDIN COVERAGE OF REPORTED U.S. EMPLOYMENT BY GROUP, 2019

			Percentile		Firms with
	Mean	Median	25th	75th	a 2019 report
Race pledgers (115)	86	79	51	112	78
General pledgers (193)	86	77	56	109	105
Never-pledged controls (1,062)	83	66	43	97	455
All three groups	84	69	45	100	638

Notes: Ratio, in percent, of a firm's LinkedIn U.S. workers on 1 July 2019 to the U.S. employees in its 2019 EEO-1 report, for the firms of each group with such a report (last column). Race pledgers are the main estimation sample, general pledgers the firms with general statements and no numerical race target, controls the never-pledged firms; the last row pools the three groups. EEO-1 reports are those of federal contractors released under the Freedom of Information Act, matched as in Appendix B.3. Workers and reports are the inputs of Table 4.

Profiles are updated with a delay, so the last years of the panel are recorded less completely. Headcounts in the panel fall slightly in 2023 and 2024, by about 5 percent at pledgers and 1 percent at controls from 2022 to 2024, which may reflect late updating as well as layoffs. The estimates hold when I drop the 2024 snapshot (Table 6).

B.4 Comparison with the Current Population Survey

Table B.2 compares the panel with the March 2019 Current Population Survey (CPS). The main benchmark is column 5: private wage and salary workers at firms with 1,000 or more employees. The panel is close to these workers on sex, region and the Black share (13.7 percent, against 14.4 percent in the CPS), but holds fewer Hispanic and more Asian workers. I impute race in the panel (Section 3.4), so the race rows compare an estimate with self-reported race. LinkedIn users are a selected group: the panel over-represents college graduates, younger workers and professional and managerial jobs. The industry mix reflects LinkedIn's uneven coverage across industries and the S&P 1500 frame, which includes no nonprofit hospitals or universities. Pledgers and controls are balanced when each firm counts once, as in the estimates (column 10: no normalized difference exceeds 0.16); pooled over workers, large firms widen the Asian, West, South and education and health gaps (column 9, at most 0.27).

B.5 Seniority Levels

Revelio assigns every title one of seven seniority levels. I group them into three bands: entry (levels 1 and 2), middle (3 and 4) and director and above (5 to 7). In 2019, 77 percent of the workers in the top band are at level 5, 21 percent at level 6 and 2 percent at level 7. The levels come from a model of job titles, so the same title can sit at different levels in different firms: for 91 of the 150 most common titles in the top band, fewer than 80 percent of firms place the title at its most common level. I correct one systematic error. When a worker's title at the same firm clearly goes up (manager to senior manager, associate director to director) but Revelio's level falls, I place the new job one level above the old one for as long as the worker holds that title. This recodes 43,506 promotions and moves no estimate by more than 0.013 percentage points. Each band has about 12,500 firm-year cells in the estimation panel; the median cell in the top band holds about 300 workers, and fewer than 0.3 percent of cells fall below the floor of five workers.

B.6 Index Membership over Time

The universe is the index as it stands today. Firms that were in the S&P 1500 during the pledge wave but have since been acquired, gone private or dropped out of the index are therefore missing, and firms that joined later are included. To measure how much this matters, I reconstruct the index on 30 June of each year from 2020 to 2024 from the

TABLE B.2: THE WORKFORCE PANEL AND CPS BENCHMARKS, 2019

	Panel, 1 July 2019			CPS ASEC 2019			Normalized difference			
				Employed	Private,	(5), BA	(1) vs		(2) vs (3)	
	All	Pledgers	Controls	in March	1,000+	or MBSA	(5)	(6)	Workers	Firms
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Demographics										
Female	43.7	42.6	44.5	47.5	48.2	50.3	−0.06	−0.09	−0.03	0.01
Age under 30	37.4	37.9	37.1	23.7	28.5	20.5	0.13	0.27	0.01	0.03
Age 30 to 44	42.6	43.7	41.8	32.1	31.8	36.9	0.16	0.08	0.03	0.01
Age 45 or older	20.0	18.4	21.2	44.2	39.7	42.5	−0.31	−0.35	−0.05	−0.04
Mean age (years)	35.2	34.8	35.5	42.4	40.6	42.5	−0.30	−0.42	−0.05	−0.05
Race and ethnicity										
Black	13.7	13.3	14.0	11.6	14.4	10.7	−0.01	0.07	−0.01	0.01
Hispanic	11.8	11.8	11.7	17.9	15.8	9.8	−0.08	0.05	0.00	0.00
Asian, Pacific Islander	13.0	17.3	10.2	6.8	7.9	11.4	0.12	0.03	0.15	0.07
White, non-Hispanic	61.9	57.8	64.5	63.7	61.9	68.2	0.00	−0.09	−0.10	−0.05
Education										
BA+, coded degree	80.0	82.2	78.5	38.8	39.2	76.6	0.65	0.06	0.07	0.03
BA+, lower bound	53.1	58.3	49.7	38.8	39.2	76.6	0.20	−0.36	0.12	0.08
Occupation										
MBSA occupations	59.5	61.4	58.4	41.1	40.3	78.8	0.28	−0.30	0.04	0.00
Frontline occupations	40.5	38.6	41.6	58.9	59.7	21.2	−0.28	0.30	−0.04	0.00
Industry										
Goods-producing	30.5	31.5	29.9	18.8	16.5	14.2	0.24	0.28	0.02	0.11
Trade, transport, util.	22.3	26.7	19.5	18.3	28.0	15.8	−0.09	0.12	0.12	0.04
Info., finance, prof.	37.7	37.3	37.9	21.0	22.7	33.9	0.23	0.06	−0.01	−0.15
Education and health	4.2	0.0	6.9	23.5	20.3	29.4	−0.36	−0.51	−0.27	−0.16
Other services, public	5.3	4.5	5.8	18.5	12.5	6.7	−0.18	−0.04	−0.04	0.11
Census region										
Northeast	17.6	19.4	16.4	17.5	17.5	19.8	0.00	−0.04	0.05	0.08
Midwest	19.3	16.8	20.8	21.6	23.0	21.8	−0.06	−0.04	−0.07	−0.01
South	35.2	29.2	39.1	37.2	36.8	35.1	−0.02	0.00	−0.15	−0.09
West	27.9	34.6	23.7	23.8	22.7	23.3	0.08	0.07	0.17	0.03
Obs. (thousands)	6,418	2,501	3,917	84.4	27.6	13.7				
Firms	1,177	115	1,062							

Notes: Percent unless stated. Panel: worker-years in estimation cells; sex is imputed from first names, race from names and locations (Black also from colleges), and age from education dates (42 percent of rows). Coded degrees (2025 profiles, 66 percent of rows) count unless the first bachelor's started or ended after 2019; the lower bound counts the rest as none. CPS: weighted persons 16 or older; job variables refer to the longest job of 2018. MBSA: SOC groups 11 to 29. Column 10 weights firms equally. Normalized difference: $(\bar{x}_1 - \bar{x}_2)/\sqrt{s_1^2 + s_2^2}$ (Imbens and Wooldridge, 2009).

holdings reported to the SEC by an exchange-traded fund that replicates the S&P 1500. Of the 1,454 companies in the index on 30 June 2020, 1,075 are in the universe. Most of the others left the index after 2020, and a few are current members that the name matching had missed. About 370 firms of the universe joined the index after June 2020.

C Measuring Race

C.1 The BIFSG Classifier

On the statistical properties of BIFSG and its refinements, see Derby et al. (2025) and Imai et al. (2022). The name terms use the first-name and surname tables of Rosenman et al. (2023), which I downloaded in August 2026 as distributed with the `wru` package for R: 1,044,047 first names and 1,484,515 surnames. For each group, the tables give the probability of a name, the terms $P(f \mid r)$ and $P(s \mid r)$ of Section 3.4. When a surname is missing from the surname table, I look it up in the Census Bureau's list of the 162,253 surnames that occur at least 100 times in the 2010 Census, which adds almost no matches among the panel's workers. Since the tables report no counts, I do not smooth the probabilities of rare names. I also leave out the authors' middle-name table, because LinkedIn profiles record no middle name. Ninety-one percent of all workers are scored based on all three of first name, surname and location. Since some first names or surnames are not in the name tables, 6 percent of workers are scored on first name and place, 2 percent on surname and place, and 1 percent on place alone.

The place term comes from the Census Bureau's Vintage 2023 population estimates by county, race and Hispanic origin, for 1 July 2023 and all ages. The Census asks about Hispanic origin and about race in two separate questions, so Hispanic Americans can be of any race. I treat Hispanic as an absorbing category: it includes white Hispanics, Black Hispanics and Hispanics of any other race. A county's prior is its shares of non-Hispanic white, non-Hispanic Black, non-Hispanic Asian and Hispanic residents, with everyone else in a residual group, and state and national priors add up the counties. The classifier computes the posterior over these five groups and then rescales the paper's four groups to sum to one.

Each worker gets one geography for her whole career. I resolve the city of each of her U.S. positions to a county with the Census Bureau's 2020 list of places by county, searching only within the position's state; a place that spans several counties gets the average of their priors, weighted by population. Her county is the one that occurs most often among her positions. Each position counts once, whatever its length, and a tie goes to the county of the position she started most recently. If none of her positions resolves to a county, she gets the most frequent metropolitan area among her positions, then the most frequent state, then the national mix. Revelio's metropolitan labels match the Office of Management and Budget's 2009 delineation, and a metropolitan area's prior averages its

counties by population. When the positions do not give a county, I also parse the location in the profile and keep the finer of the two answers. Of the 25,426,806 workers scored, 73.2 percent get a county from their positions and 11.3 percent from their profiles. If the mapping is not clear (15 percent of the workers in the panel), I use either the metropolitan area (1 percent), their state (6 percent) or the national mix (8 percent). Puerto Rico and the other territories have no county prior, so positions there fall through to the national mix.

The prior is fixed over time. I use the 2023 estimates for every year of the panel, and the county comes from all of a worker's positions, whenever she held them, so she keeps a single score throughout. Among the North Carolina voters matched to the panel (Appendix C.3), this rule places 73.5 percent in the county where they are registered to vote. Scoring them with their registered county instead lowers the factor for the Black share from 1.62 to 1.60 (from 1.43 to 1.41 with most likely labels).

The education-augmented BIFSG of the main text adds one term. If a worker's college has a Black share of at least one half among its graduates, I multiply the Black-versus-rest odds of her posterior by that college's likelihood ratio: the odds that its graduates are Black over the national odds, from the IPEDS counts of graduates by race. The rule covers 138 colleges, 87 of them historically Black, with the campuses of a college system pooled. The term enters with a weight of about 0.66, fitted on the North Carolina voters (Appendix C.3). A vision model labels the proxy photographs of 4,450 directors and officers, and 343 of them appear Black. Among these Black directors and officers, the share whose most likely label is Black rises from 41 to 46 percent, while the false-positive rate stays at 4 percent.

C.2 Measurement Model and Correction Factors

Consider a cell of N workers, a share s of whom belong to the group, and let p_i be the score the classifier gives worker i: her probability of belonging to the group for the share outcomes, or an indicator that the group is her most likely label for the hire-level designs. The measured share is the average score,

$$m = \frac{1}{N}\sum_{i=1}^{N} p_i = s\,\pi_1 + (1-s)\,\pi_0 = \pi_0 + (\pi_1 - \pi_0)\,s,$$

where π_1 is the average score of the members of the cell and π_0 that of the non-members. With most likely labels, π_1 is the classifier's sensitivity and π_0 its false-positive rate. For the Black and Hispanic share, a member belongs to either group, and the score is the sum

of the two probabilities or an indicator that the most likely label is one of the two.

The event studies compare changes in m at pledgers and controls. Within a band, write the measured share of firm f in year t as $m_{ft} = \pi_{0,ft} + (\pi_1 - \pi_0)\, s_{ft}$. If $\pi_{0,ft}$ is the sum of a firm term and a year term, the firm and year fixed effects of Section 4.3 absorb it. If in addition $\pi_1 - \pi_0$ is the same in every cell, each coefficient of the event study on m equals $\pi_1 - \pi_0$ times the corresponding coefficient on s, and the true effect is the measured one times $1/(\pi_1 - \pi_0)$, the correction factor. The classifier misses some members ($\pi_1 < 1$) and gives some weight to non-members ($\pi_0 > 0$), so $\pi_1 - \pi_0$ is below one and the factor exceeds one.

The voter file records each voter's race (Appendix C.3), and a vision model labels the apparent race of each photographed director or officer (Appendix C.5). For these two sources, π_1 and π_0 are the average scores of the members and of the non-members, and the factor is $1/(\pi_1 - \pi_0)$. The EEO-1 reports record no individual's race, only each firm's official share s_{ft} (Appendix C.4). Across firms the model gives $m_{ft} = \pi_0 + (\pi_1 - \pi_0)\, s_{ft}$, so a regression of the measured share on the official one has intercept π_0 and slope $\pi_1 - \pi_0$, and the factor is the inverse of the slope. That slope also picks up coverage and the mapping of Revelio's levels into EEO-1 job categories, and, since it compares firms, where each firm's workers live.

The correction rests on two assumptions: the classifier errs at the same rates at pledgers and controls, and its rates did not change with the pledges. Since each worker carries one score (Appendix C.1), the rates in a cell change only when the names and places of its workers change. Appendix C.7 tests both assumptions among the North Carolina voters. The sources also cover different people. The voters are a broad population, the EEO-1 categories only approximate the top band, and the photographed directors and officers sit higher than most workers in that band. Table C.1 reports the factors from each source for education-augmented BIFSG, BIFSG and Revelio's classifier, scored by probabilities and by most likely labels.

C.3 Voter Files

North Carolina's voter file records the race and Hispanic ethnicity that each voter reports. I match the panel's workers whose LinkedIn location is in North Carolina to the 8,275,932 voters with a recorded race, on first name, last name and city, and keep the 239,207 workers who match exactly one voter. The sample is the 230,940 of them recorded as white, Black, Asian or Hispanic (Hispanic ethnicity takes precedence over race), of whom 53,323

TABLE C.1: CLASSIFIER CORRECTION FACTORS BY VALIDATION SOURCE

		Education-augmented BIFSG		BIFSG		Revelio	
Source	Population	Black	Black and Hispanic	Black	Black and Hispanic	Black	Black and Hispanic
Panel A: Posterior probabilities, used for the share outcomes							
North Carolina voter file	230,940 voters in the panel	1.62	1.65	1.75	1.76	2.48	2.41
EEO-1 reports	Managers at 190 firms	1.30	2.23	1.37	2.31	1.59	2.68
Proxy photographs	1,351 executives and directors	2.72	2.94	3.00	3.20	4.86	4.77
Panel B: Most likely label, used for the hire-level designs							
North Carolina voter file	230,940 voters in the panel	1.43	1.45	1.54	1.54	2.10	2.05
EEO-1 reports	Managers at 190 firms	1.44	2.33	1.53	2.43	1.59	2.55
Proxy photographs	1,351 executives and directors	2.48	2.67	2.80	2.97	4.75	4.40

Notes: Each entry is the factor that scales a measured change in the share to the true change. Voter file and photographs: $1/(\pi_1 - \pi_0)$, with π_1 and π_0 the average scores of group members and non-members (probabilities in Panel A, most likely labels in Panel B). EEO-1: the inverse slope of the measured share on the official share across 739 firm-years (Revelio levels 6 and 7 against officials and managers, 2016 to 2020), which also absorbs coverage and the level-category mismatch. BIFSG as in Section 3.4; education-augmented BIFSG raises the Black probability of graduates of majority-Black colleges and rescales the other groups in proportion; Revelio is the vendor's classifier, which uses names and location.

are Black and 8,738 Hispanic. The voter-file factors in Table C.1 use these workers because the paper's education-augmented BIFSG probabilities, which place each worker in the county of her positions, exist only for workers in the panel.

With most likely labels, education-augmented BIFSG labels 76.0 percent of the Black voters Black (BIFSG: 70.8) and 6.1 percent of the others, so its factor for the Black share is $1/(0.760 - 0.061) = 1.43$. Revelio's classifier labels 55.3 and 7.7 percent, a factor of 2.10. With probabilities, education-augmented BIFSG gives the Black voters an average probability of 73.3 percent of being Black and the others 11.7 percent, a factor of 1.62 (bootstrap 95 percent interval 1.62 to 1.63). North Carolina is one of the six states whose voter files Rosenman et al. (2023) use to estimate the name tables, so here the names are tested on a population like the one they were estimated from.

Florida's voter file also records the race each voter reports. In a random sample of 500,000 Florida voters recorded as white, Black, Hispanic or Asian, scored with first, middle and last names and the registered county, BIFSG's sensitivity is 60 to 64 percent among

Black voters born in the 1940s to 1960s and 78 to 82 percent among those born in the 1980s to 2000s. The check favors the classifier, since Florida is one of the six states behind the name tables and LinkedIn profiles do not record middle names.

C.4 EEO-1 Reports

The consolidated EEO-1 reports that federal contractors filed for 2016 to 2020 count each firm's U.S. employees by race, sex and job category. I match them to firms by company name, after removing punctuation and legal suffixes such as Inc., Corp. and Holdings. The comparison uses an earlier build of the panel that covers 395 S&P 1500 firms, of which 190 match a report exactly, for 739 firm-years. In each firm-year, the measured share is the average score the classifier gives the firm's LinkedIn workers at Revelio levels 6 and 7. The official share is the share of the firm's officials and managers (EEO-1 categories 1.1 and 1.2) who are Black, or Black or Hispanic.

Across the 739 firm-years, a regression of the measured Black share on the official one has a slope of 0.77 with education-augmented BIFSG's probabilities, so the factor is 1.30 (95 percent interval 1.03 to 1.65, bootstrap over firms). For the Black and Hispanic share the slope is 0.45 and the factor 2.23 (1.80 to 2.92), because the measured Hispanic share follows the official one less closely (slope 0.36). With most likely labels the factors are 1.44 and 2.33, and with Revelio's probabilities 1.59 and 2.68 (Table C.1).

Because it compares firms, the slope picks up the classifier's use of place, in addition to coverage and the mapping of levels into job categories. BIFSG gives every worker a higher score in a place with many Black residents, so across places the measured share follows the true share more closely than it does within a place. I gauge the size of this effect with the North Carolina voters matched to the panel. Across the 361 employers with at least 50 matched workers in 2019, a regression of each employer's measured Black share on its voter-file Black share gives a factor of 1.41 for education-augmented BIFSG's probabilities and 1.22 for its most likely labels. Within employers (all 1,309 employers of these workers), the gap between Black and other workers gives 1.68 and 1.48, close to the 1.66 and 1.45 for the same workers pooled. Across the state's counties the factor is about one. The event study measures a change within a firm, and a comparison across firms understates the correction for such a change. For the Black share, the EEO-1 factor is the lowest of the three sources.

C.5 Proxy Photographs

Proxy statements filed on EDGAR carry photographs of executive officers and directors. I use a vision model to label the apparent race of each person in them. The model never sees the person's name, and no image is kept. After the match to the panel, the labels cover 1,351 people, most of whom sit on corporate boards. Education-augmented BIFSG calls 44 percent of the people who appear Black in their photographs Black, well below its 76 percent among voters. It gives them an average probability of 45.0 percent of being Black, and everyone else 8.1 percent. These averages imply factors of 2.72 for the Black share and 2.94 for the Black and Hispanic share, or 2.48 and 2.67 with most likely labels (Table C.1). The factors are imprecise, because only 108 of the 1,351 appear Black: the bootstrap 95 percent interval for the Black-share factor runs from 2.3 to 3.3. Revelio's classifier calls 27 percent of them Black, with factors of 4.86 and 4.77 (4.75 and 4.40 with most likely labels). A study that takes this vendor field as given, as Cai et al. (2024) do, therefore understates a race effect at the top by a factor of almost five. The classifier misses Black workers most often at the top, so the estimates for the top band are the most understated. The labels are a model's reading of a photograph, and the main text uses these factors only as the upper end of the correction range (Section 5.1).

The photographs also give the precision of the most likely label, which the hire-level designs use. Of the 100 photographed people whose most likely education-augmented BIFSG label is Black, 48 appear Black in their photographs, a precision of 48 percent (bootstrap 95 percent interval 38 to 58 percent). Of the other 52, 46 appear white. Errors in the match to LinkedIn lower this number. Among the 1,009 people matched on their exact names, 43 of the 76 labeled Black appear Black, or 57 percent. Appendix H.3 applies the 48 percent to the hires labeled Black at director and above. Precision also depends on how many people in a population are Black, and the photographs cannot measure that for these hires. If I carry over to the hires the share that appears Black within each range of BIFSG probability, precision is 49 percent. Carrying over the label's sensitivity and false-positive rate instead gives 52 percent. If the Black share of these hires were 5.9 percent, precision would be 31 percent. That is the share among officials and managers in the 2016 to 2020 EEO-1 reports of the 451 sample firms that match a report.

C.6 Historically Black Universities

The Integrated Postsecondary Education Data System (IPEDS) records the race of each school's students. In the fall enrollment surveys of 2010 to 2023, Black students made up

93 to 99 percent of the undergraduates at six historically Black universities. These shares count only the four groups Black, Hispanic, Asian or Pacific Islander and white, and so leave out nonresident aliens, students of two or more races and students of unknown race (Table C.2). Among the panel's workers who completed a bachelor's degree at one of these schools in 2014 to 2024, Revelio's classifier labels 48 to 60 percent Black and BIFSG 64 to 74 percent. The share labeled Black rises with the year of graduation. With the six schools pooled, BIFSG labels 57 percent of those who graduated in 1999 or earlier Black, 62 percent of the 2000 to 2013 graduates and 68 percent of the 2014 to 2024 graduates (Revelio: 44, 46 and 51 percent). The Florida voters show the same gradient by year of birth (Appendix C.3). Distinctively Black first names are less common among older Black Americans (Fryer and Levitt, 2004). Since first names carry much of the classifier's information, part of the gap at the top likely comes from naming cohorts (Appendix C.5).

TABLE C.2: BLACK SHARE OF GRADUATES OF SIX HISTORICALLY BLACK UNIVERSITIES

			Revelio		BIFSG	
School	IPEDS	Graduates	Probability	Label	Probability	Label
Howard University	93.1	5,798	46.9	49.2	62.6	65.9
Spelman College	98.3	1,917	49.1	51.6	68.1	72.2
Morehouse College	99.4	2,054	46.0	48.5	61.9	64.1
Hampton University	96.0	2,445	46.9	50.2	63.2	66.6
Tuskegee University	98.0	1,505	48.0	50.9	65.8	69.1
Jackson State University	95.2	2,517	56.0	59.6	69.7	74.2

Notes: IPEDS is the Black share of the undergraduates in four groups (Black, Hispanic, Asian or Pacific Islander, white), excluding nonresident aliens and students of two or more races or of unknown race, in each school's fall enrollment surveys of 2010 to 2023. Graduates are the workers scored by the classifiers whose LinkedIn profiles list a U.S. bachelor's degree from the school completed in 2014 to 2024. Probability is their average probability of being Black; label is the share whose most likely group is Black. Entries in percent except graduates.

C.7 Error Rates at Pledgers and Controls

The correction factors rescale the estimates only if the classifier's error rates are the same at pledgers and controls and did not change with the pledges. I check this on the North Carolina voters matched to the panel. At the level of worker-years, I compare the gap between the education-augmented BIFSG scores of Black and other voters at the 115 pledgers and at the 1,004 controls that employ matched voters (113 and 806 at director and above), in July 2014 to 2019 and July 2021 to 2024. Before 2020 the gaps imply factors of 1.64 at pledgers and 1.72 at controls. After 2020 the gap moved by 0.002 more at

pledgers than at controls, and by 0.011 more at director and above (firm-cluster bootstrap SEs 0.009 and 0.021). If I read this as a difference in differences of the measured Black share net of the true change scaled by the classifier's pre-2020 gap between Black and other voters, the classifier's own contribution is -0.04 percentage points (SE 0.27), and -0.15 at director and above (SE 0.36). A change in the names or places of the workers that pledgers hire could still weaken the assumption, since the classifier reads both.

D Estimation and Inference

D.1 The Event-Study Estimator

I estimate Equation (6) separately for each band and for the whole company, on cells of firm, band and July 1 year from 2014 to 2024, with the bands of Appendix B.5. The outcome is 100 times the mean education-augmented BIFSG probability of being Black among a cell's workers. I drop cells with fewer than five workers, at pledgers and controls alike. A pledger enters only from event year -4 to event year 3; the controls enter in every year. The pledgers' windows are balanced. Each of the 115 pledgers has a cell in every band at every event year from -4 to 2, and so do the 83 pledgers of the 2018 to 2021 cohorts at event year 3. This gives 888 pledger cells in each band. The 1,062 controls add between 11,609 cells (middle band) and 11,670 (whole company), of the 11,682 possible.

Each regression is ordinary least squares, and every cell counts once. Besides firm and year effects, it has one indicator for each cohort and event year other than -1, 34 in all, since the 2022 cohort is not observed at event year 3. Every event year of a pledger except -1 has its own indicator, so the pledgers' cells carry no information about the year effects, which come from the controls alone. Each $\beta^b_{g,e}$ is then the change in cohort g's mean Black share between years $g-1$ and $g+e$, minus the change in the year effects between the same two years. This is the cohort-specific effect of Sun and Abraham (2021), with the never-pledged firms as the comparison group. The leads are also differences against year -1, so they trace each cohort's path before its pledge relative to the controls.

Equation (7) averages the cohort coefficients at each event year with weights $\omega_{g,e}$, each cohort's share of the pledgers observed at that event year. With balanced windows these weights are the same in every band. They are the cohort shares of all 115 pledgers at event years -4 to 2, and of the 83 observed at event year 3 (Table D.1). The 2021 cohort, whose earliest numerical race pledges fall between 2 July 2020 and 1 July 2021, carries 55 percent of the weight at event years -4 to 2, 76 percent at event year 3 and 60 percent of τ_b. The 2018 and 2019 cohorts together carry 4 percent of τ_b.

I cluster the standard errors by firm (1,177 firms). They come from the cluster-robust covariance matrix $\hat{V}$ of all the regression's coefficients, with the finite-sample factor $G(N-1)/[(G-1)(N-K)]$ for G firms, N cells and K coefficients. Since K includes the 1,176 firm effects, the standard errors are about 5 percent larger than with a factor that leaves them out. Each event-year estimate is a linear combination $c_e'\hat{b}$ of the coefficient vector $\hat{b}$, with the weights $\omega_{g,e}$ in c_e. The average effect is then $\hat{\tau}_b = \bar{c}\,'\hat{b}$ with $\bar{c} = \frac{1}{4}\sum_{e=0}^{3} c_e$,

TABLE D.1: PLEDGE COHORTS AND THEIR WEIGHTS IN THE EVENT STUDY

Cohort g	Pledgers	Years in the window	Weight $\omega_{g,e}$ (percent) $e = -4$ to 2	Weight $\omega_{g,e}$ (percent) $e = 3$	Weight in τ_b (percent)
2018	1	2014 to 2021	0.9	1.2	1.0
2019	3	2015 to 2022	2.6	3.6	2.9
2020	16	2016 to 2023	13.9	19.3	15.3
2021	63	2017 to 2024	54.8	75.9	60.1
2022	32	2018 to 2024	27.8	0.0	20.9
All	115		100.0	100.0	100.0

Notes: Cohort g is the year of the first July 1 on or after a firm's earliest numerical race pledge. The window runs from event year -4 to event year 3, or to 2024. $\omega_{g,e}$ is the cohort's weight in the event-year average of Equation (7); the last column is its weight in τ_b, the mean over event years 0 to 3. 115 pledgers of the main estimation sample; the weights are the same in every band. Source: pledge dataset (Section 3.1).

and its standard error is $(\bar{c}'\hat{V}\bar{c})^{1/2}$. This standard error uses the covariances between the four post-pledge estimates. They are large and positive (correlations of 0.42 to 0.92 across pairs and bands), because all four are measured against the same year -1 cells and the same year effects. Without these covariances, the standard error at director and above would be 0.042 instead of 0.072. I treat the weights as known, so the standard error conditions on the cohort sizes. For the joint test of the leads, I compare the Wald statistic $(R\hat{b})'(R\hat{V}R')^{-1}(R\hat{b})$, where R stacks c'_{-4}, c'_{-3} and c'_{-2}, with a χ^2 distribution with three degrees of freedom. The p-values of τ_b use the normal distribution.

D.2 Sensitivity to Pre-Trends

Parallel trends after the pledge cannot be tested directly. Rambachan and Roth (2023) replace the assumption with a restriction on the bias δ_e, the gap between the pledgers' and the controls' paths that would have opened without the pledges, and give confidence sets for τ_b that are valid under that restriction. If firms did not anticipate their pledges, the pre-pledge coefficients contain no effect, so they measure the bias before the pledge. I compute the sets with the authors' HonestDiD package. The inputs are the seven event-year coefficients of Appendix D.1 ($e = -4, -3, -2$ and 0 to 3) and their firm-clustered covariance, and the target is τ_b, the mean of the four post-pledge coefficients.

Under the smoothness restriction the bias can bend, but its slope may change by at most M percentage points from one year to the next. With $M = 0$ the bias is a straight line through the pre-pledge coefficients, extended past the pledge. The interval is then that of an estimate net of a linear pre-pledge trend, close to the trend-adjusted estimates

in Panel B of Table 7. A positive M lets the slope drift. By event year e the bias can depart from the line that continues its last pre-pledge slope by up to $M(e+1)(e+2)/2$, that is M, $3M$, $6M$ and $10M$ at event years 0 to 3, or $5M$ on average. With $M = 0.01$, for instance, the average bias over event years 0 to 3 can sit 0.05 points off the line, about a quarter of the estimate at director and above. The relative-magnitudes restriction bounds every year-to-year change in the bias from event year -1 onward by $\bar{M}$ times the largest year-to-year change up to event year -1. With $\bar{M} = 1$, each year after the pledge can see a change as large as the largest pre-pledge one. For each restriction I use the confidence sets that Rambachan and Roth (2023) recommend: fixed-length intervals for smoothness and conditional least-favorable hybrid sets for relative magnitudes.

Under a linear trend the intervals exclude zero at director and above (0.05 to 0.43), at entry (0.08 to 0.55) and across the firm (0.10 to 0.42). They include zero in the middle band (-0.09 to 0.35), whose coefficient three years before the pledge dips to -0.18 (Table D.2). At director and above they still exclude zero with $M = 0.01$ and first include zero at $M = 0.02$; at entry and across the firm they still exclude zero with $M = 0.01$ and first include zero at $M = 0.02$. The pre-pledge coefficients cannot tell these values of M apart: at director and above their slope changes by 0.037 and then by -0.019 points from one pre-pledge year to the next, with standard errors of 0.085 and 0.062. Under relative magnitudes with $\bar{M} = 0.5$, the intervals exclude zero at entry (0.08 to 0.54) and across the firm (0.04 to 0.37) and include it at director and above (-0.01 to 0.40) and in the middle band. With $\bar{M} = 1$ every interval includes zero. The package also has a third restriction, which bounds each change in the slope of the bias from event year -1 onward by $\bar{M}$ times the largest such change up to event year -1. At $\bar{M} = 0.5$ its intervals include zero in every band.

D.3 Trend Adjustment

I report several estimates net of a linear pre-pledge trend. Those listed here all use the same line: Panel B and the slope rows of Table 7, column (4) of Table H.2, the columns beyond the trend in Table H.3, and the added share net of its pre-trend in Appendix H.3. Let $\hat{\beta}_e$ be the event-time coefficients of an event study or a triple difference with reference year e_0. The reference year is -1, except for the 24-month outcomes, where it is -3. I fit the line by generalized least squares through zero at e_0 to the three coefficients before e_0,

TABLE D.2: CONFIDENCE SETS ALLOWING FOR VIOLATIONS OF PARALLEL TRENDS

	Director and above	Middle	Entry	Company
ATT (percentage points)	0.199	0.226	0.302	0.209
Conventional 95% interval	[0.058, 0.340]	[0.076, 0.376]	[0.146, 0.459]	[0.099, 0.320]
Panel A: Smoothness, slope of the bias changes by at most M points a year				
$M = 0$ (linear trend)	[0.050, 0.425]	[−0.093, 0.354]	[0.084, 0.554]	[0.099, 0.423]
$M = 0.005$	[0.036, 0.438]	[−0.101, 0.367]	[0.071, 0.567]	[0.071, 0.427]
$M = 0.01$	[0.005, 0.467]	[−0.122, 0.395]	[0.032, 0.591]	[0.017, 0.438]
$M = 0.02$	[−0.064, 0.532]	[−0.175, 0.463]	[−0.050, 0.655]	[−0.091, 0.451]
Panel B: Relative magnitudes, at most $\bar{M}$ times the largest pre-pledge change				
$\bar{M} = 0.5$	[−0.013, 0.396]	[−0.164, 0.590]	[0.085, 0.543]	[0.044, 0.374]
$\bar{M} = 1$	[−0.140, 0.520]	[−0.474, 0.894]	[−0.056, 0.687]	[−0.044, 0.464]

Notes: 95 percent confidence sets of Rambachan and Roth (2023) for the ATT of Table 5 (Black share, mean of event years 0 to 3, percentage points), from the seven event-time coefficients and their firm-clustered covariance. Panel A bounds the yearly change in the slope of the bias by M, and $M = 0$ extrapolates a linear pre-pledge trend (fixed-length intervals). Panel B bounds each yearly change in the bias from event year -1 on by $\bar{M}$ times the largest one up to year -1 (conditional least-favorable hybrid sets). Source: Revelio Labs.

$\hat{\beta}_{\text{pre}} = (\hat{\beta}_{e_0-3}, \hat{\beta}_{e_0-2}, \hat{\beta}_{e_0-1})'$, with their estimated covariance S:

$$\hat{d} = g'\hat{\beta}_{\text{pre}}, \qquad g = \frac{S^{-1}x}{x'S^{-1}x}, \qquad x = (-3, -2, -1)', \qquad \text{SE}(\hat{d}) = \left(x'S^{-1}x\right)^{-1/2}. \quad \text{(D.1)}$$

Extrapolating the line to event years 0 to 3 and subtracting it gives the adjusted estimate

$$\hat{\tau}^{\text{adj}} = \frac{1}{4}\sum_{e=0}^{3}\left[\hat{\beta}_e - \hat{d}\,(e - e_0)\right] = \hat{\tau} - \bar{k}\,\hat{d}, \qquad \bar{k} = \frac{1}{4}\sum_{e=0}^{3}(e - e_0), \quad \text{(D.2)}$$

so $\bar{k} = 2.5$ when $e_0 = -1$ and $\bar{k} = 4.5$ when $e_0 = -3$. The adjusted estimate is a linear combination of the event-time coefficients, with weight $1/4$ on each post-pledge coefficient and $-\bar{k}\,g$ on the three leads. Its standard error comes from their firm-clustered covariance with g held fixed. In the 24-month triple differences, the partly exposed years -2 and -1 are neither leads nor post-pledge years, so the line rests on years -6 to -4. The linear pre-trend tests I report are the t-test of $\hat{d}$.

The Black share at director and above (Table 7) has a slope of -0.014 points a year (SE 0.025). The adjustment moves the estimate from 0.199 to 0.233 and raises its standard error by a third, from 0.072 to 0.096. Promotion within 24 months at entry (Table H.3) has a slope of 0.23 (SE 0.12), and subtracting 4.5 times it turns the estimate of 0.56 (SE 0.44) into -0.47 (SE 0.68). For the added share a of Appendix H.3, I fit the line to the Poisson

coefficients b_{-4}, b_{-3} and b_{-2} of Equation (H.1), on the log scale, and adjust each post-pledge coefficient before the transformation: $a^{\text{adj}} = \frac{1}{4}\sum_{e=0}^{3}\left\{1-\exp\left[-\left(b_e-\hat{d}\,(e+1)\right)\right]\right\}$. The slope of b_e is 0.020 a year ($p = 0.07$), and a falls from 0.068 to 0.020 (SE 0.043).

D.4 Alternative Estimators

Table D.3 reports the main estimate under eight estimators for staggered adoption. At director and above the effect lies between 0.186 and 0.199 points, and at the company level between 0.209 and 0.229. The two imputation estimators give a larger effect in the middle band (0.313 against 0.226), and no estimator rejects the joint lead test at the 5 percent level in any band.

TABLE D.3: BLACK SHARE ATT BY ESTIMATOR

	Director and above	Middle	Entry	Company
Panel A: ATT, mean of event years 0 to 3 (percentage points)				
Sun-Abraham (main)	0.199***	0.226***	0.302***	0.209***
	(0.072)	(0.077)	(0.080)	(0.056)
Two-way fixed effects, dynamic	0.195***	0.232***	0.306***	0.214***
	(0.071)	(0.075)	(0.081)	(0.056)
Callaway-Sant'Anna, never-treated	0.199***	0.226***	0.302***	0.209***
	(0.070)	(0.075)	(0.079)	(0.055)
Callaway-Sant'Anna, not-yet-treated	0.199***	0.226***	0.303***	0.210***
	(0.070)	(0.075)	(0.079)	(0.055)
Borusyak-Jaravel-Spiess imputation	0.186**	0.313***	0.310***	0.229***
	(0.079)	(0.082)	(0.096)	(0.067)
Stacked regression	0.193***	0.231***	0.303***	0.212***
	(0.073)	(0.077)	(0.081)	(0.057)
Gardner two-stage	0.186**	0.313***	0.310***	0.229***
	(0.078)	(0.082)	(0.096)	(0.067)
Local-projection DiD	0.199***	0.225***	0.302***	0.209***
	(0.069)	(0.073)	(0.076)	(0.054)
Panel B: Joint test of the three leads (p-value)				
Sun-Abraham (main)	0.92	0.08	0.86	0.35
Two-way fixed effects, dynamic	0.90	0.09	0.84	0.29
Callaway-Sant'Anna, never-treated	0.90	0.06	0.85	0.33
Callaway-Sant'Anna, not-yet-treated	0.90	0.07	0.83	0.27
Borusyak-Jaravel-Spiess imputation	0.90	0.07	0.82	0.26
Stacked regression	0.93	0.10	0.87	0.41
Gardner two-stage	0.90	0.07	0.81	0.25
Local-projection DiD	0.90	0.06	0.82	0.28
Numerical race pledgers	115	115	115	115
Never-pledged controls	1,062	1,062	1,062	1,062

Notes: Black share measured as in Section 3.4, Sun-Abraham row as in the main table. Panel A: mean of event years 0 to 3 of each estimator's event-study coefficients, in percentage points, standard errors in parentheses clustered by firm (bootstrapped for Callaway-Sant'Anna and Borusyak-Jaravel-Spiess). Panel B: joint test of event years -4 to -2. Unweighted firm-band-year cells; the Gardner two-stage estimate drops one control in the middle band. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Source: Revelio Labs.

E Additional Workforce Results

E.1 Persistence through the Backlash

By their 2025 filings, 114 of the 115 pledgers had softened or withdrawn their race targets, 106 of them by the second quarter of 2025, when the hire data end. Firms chose to retreat, and nearly all of them did, so the designs below compare firms that retreated at different times. They check a mechanism and give no causal estimates. Table E.1 aligns each pledger on the quarter of its first retreat filing. Panel A compares the quarters after it with those before, once against pledgers that had not yet retreated and once against the controls. The Black hire rate did not fall in any comparison. At director level the comparison with pledgers that had not yet retreated fails its lead test, so for that band the result rests on the comparison with the controls (+0.015 per 100 positions a quarter, SE 0.012). The Black share of hires rose in two of the four comparisons, but the rise disappears when the data end in the first quarter of 2025. Panel B shows the Black share of the workforce at the first July 1 after the retreat. No band falls, and the firm-wide interval excludes falls larger than about two fifths of the pledge effect. The data end in mid-2025, so half of the retreating firms are observed for only one or two quarters after their retreat.

E.2 Flow Decomposition

The decomposition of Section 5.4 follows each band's Black share from one July 1 snapshot to the next and assigns every change to four flows: hires from outside the firm, promotions into the band from a lower band of the same firm, separations, and promotions out of the band. A separation is dated to the first snapshot at which the worker is gone from the firm. Moves down to a lower band are left out, and a clear title promotion counts as a promotion (Section 3.3). For each pledge cohort, I compare the pledgers' yearly flow contributions after the pledge with those before it, relative to the same difference at the controls and with the timing of the event study, and then stack the cohorts. The pooled version sums workers across firms, so large firms weigh more. The firm-average version computes the contributions firm by firm and averages them, so each firm counts once, as in the event study. The intervals come from a bootstrap over firms, and Table E.2 reports both versions.

Table 8 splits the hiring channel at director and above by the hire's previous employer. Each row is the origin's contribution to the yearly change in the Black share, pledgers

TABLE E.1: BLACK HIRING AND THE BLACK SHARE AFTER THE FIRST RETREAT

	Director and above		Middle	
	Black hire rate	Black share of hires (pp)	Black hire rate	Black share of hires (pp)
Panel A: Quarters 0 to 4 minus quarters −4 to −1 around the first retreat				
Against not-yet-retreated pledgers	−0.009	0.469	0.022	1.604**
	(0.023)	(0.859)	(0.034)	(0.666)
Joint lead test, p	<0.001	0.025	0.250	0.891
Against never-pledged controls	0.015	1.169**	0.023	0.556
	(0.012)	(0.503)	(0.018)	(0.370)
Joint lead test, p	0.164	0.114	0.683	0.137
Panel B: Black share of the workforce at the first July 1 after the retreat (pp)				
	Director and above	Middle	Entry	Company
Against not-yet-retreated pledgers	0.082	−0.025	−0.045	−0.007
	(0.059)	(0.063)	(0.047)	(0.037)
Joint lead test, p	0.017	0.784	0.458	0.513

Notes: First retreat: the first filing that softens or drops the race target. Panel A: hires dated by start month, against pledgers not yet retreated (Callaway-Sant'Anna) or the controls in the same quarter; Black hire rate per 100 positions a quarter. Data end in 2025 Q2, so for 53 of the 106 retreating firms the post window is one or two quarters. At director level the comparisons with pledgers not yet retreated fail their lead tests. Panel B: event year 0, July 2024 or July 2025. Bootstrap standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

relative to controls, so the rows add up to the hiring channel of Table E.2. The Black hire rate rose significantly from only two small origins: education, nonprofits and health (at the 5 percent level), and a residual group of moves from the firm's own units abroad, other index firms and unidentified employers. From the other origins, the contributions come from a lower volume of hiring at an unchanged or higher Black share. Never-pledged index firms account for 5 percent of the increase, which is the basis of the spillover bound in Section 5.4.

E.3 Headcount on LinkedIn and in the EEO-1 Reports

Table E.4 reports the estimates behind Figure E.1.

The headcount result of Section 5.3 rests on LinkedIn, so it depends on who keeps a profile. The firms' EEO-1 reports give a second count. I use the reports of federal contractors for 2016 to 2020 (Appendix C.4) and, from 2021 on, the reports that firms published themselves or filed in Illinois, leaving out the few whose composition jumps implausibly from one year to the next. As in Section 5.1, each report is dated to the first

TABLE E.2: FLOW DECOMPOSITION OF THE CHANGE IN THE BLACK SHARE: POOLED AND FIRM-AVERAGE VERSIONS

	Director and above		Middle	
	Pooled	Firm average	Pooled	Firm average
Panel A: Contribution to the yearly change in the Black share (percentage points)				
External hiring	0.079	0.076	0.079	0.093
	[0.031, 0.122]	[0.028, 0.128]	[0.026, 0.131]	[0.026, 0.165]
Promotion into the band	0.021	0.011	0.027	0.002
	[−0.003, 0.044]	[−0.016, 0.039]	[0.007, 0.046]	[−0.032, 0.034]
Separations	−0.005	−0.005	−0.037	−0.031
	[−0.030, 0.023]	[−0.040, 0.032]	[−0.070, 0.008]	[−0.071, 0.009]
Promotion out of the band	0	0	0.001	−0.001
			[−0.015, 0.015]	[−0.019, 0.018]
Net: sum of the four flows	0.095	0.083	0.071	0.063
	[0.034, 0.161]	[0.015, 0.155]	[0.004, 0.141]	[−0.019, 0.148]
Panel B: Share of the net (percent)				
External hiring	83	92	112	146
Promotion into the band	22	14	38	4
Separations	−5	−6	−52	−49
Promotion out of the band	0	0	1	−1

Notes: Contribution of each flow to the yearly change in the band's Black share: change after the pledge (event years −1 to 3) minus change before (−4 to −1), pledgers minus controls, by cohort, with cohorts weighted by their number of pledgers. Pooled sums flows and stocks over firms; firm average weights firms equally. Net sums the four flows; moves down are left out, and nobody is promoted out of the top band. Brackets: 95 percent firm-cluster bootstrap intervals (1,000 draws). 115 numerical race pledgers, 1,063 never-pledged controls; LinkedIn panel.

July 1 after its count. Of the main sample, 105 pledgers and 708 controls have at least one report, and 80 of the pledgers have reports from both before and after their pledge.

An EEO-1 report counts the employees of the legal entities that file it, so a firm's official count jumps when it buys or sells a business. LinkedIn assigns past jobs to today's parent company and has no such jumps. I therefore cut a firm's series wherever the log of its official count moves by more than 0.3 from one report to the next while the log of its LinkedIn count moves by less than 0.1, and at known mergers, and give each stretch its own fixed effect. Table E.3 applies the estimator of Figure E.1 to the log of the official count. With the series cut, relative headcount in the four years after the pledge is 2.2 log points higher than in the year before, with a 95 percent interval from −4.0 to 8.3. Keeping every firm-year gives 2.0, with an upper end of 7.9. The reports cannot confirm the decline that LinkedIn shows, which is small on the same firm-years (−1.2) and once the pre-pledge decline is removed (−1.6). They rule out a relative expansion of more than about 9 percent.

TABLE E.3: RELATIVE HEADCOUNT IN THE EEO-1 REPORTS AND ON LINKEDIN

			95% upper end			
	Estimate	SE	Normal	Bootstrap	Pledgers	Controls
EEO-1, cut at mergers	2.19	3.45	8.96	8.25	105	708
EEO-1, all firm-years	2.02	3.29	8.48	7.93	105	708
LinkedIn, same firm-years	−1.17	1.90	2.55		105	706
LinkedIn, all firms	−4.77	1.31	−2.22		115	1,062
LinkedIn, net of pre-trend	−1.63	2.18	2.64		115	1,062

Notes: Estimates of Equation (6) for 100 times the log of U.S. headcount, averaged over event years 0 to 3, with standard errors clustered by firm. EEO-1 counts are the federal contractor reports to 2020 and the reports that firms published or filed in Illinois after; a report of data year t is dated July 1 of $t + 1$, as in Table 5. In the first row each stretch of a firm's series between mergers or divestitures has its own fixed effect. LinkedIn counts in the third row are those of July 1 of $t + 1$ for the same firm-years. Upper ends use the normal approximation or a wild cluster bootstrap with 9,999 draws. The last row removes a linear trend fitted to event years -4 to -1.

TABLE E.4: HEADCOUNT BY BAND AFTER A NUMERICAL RACE PLEDGE

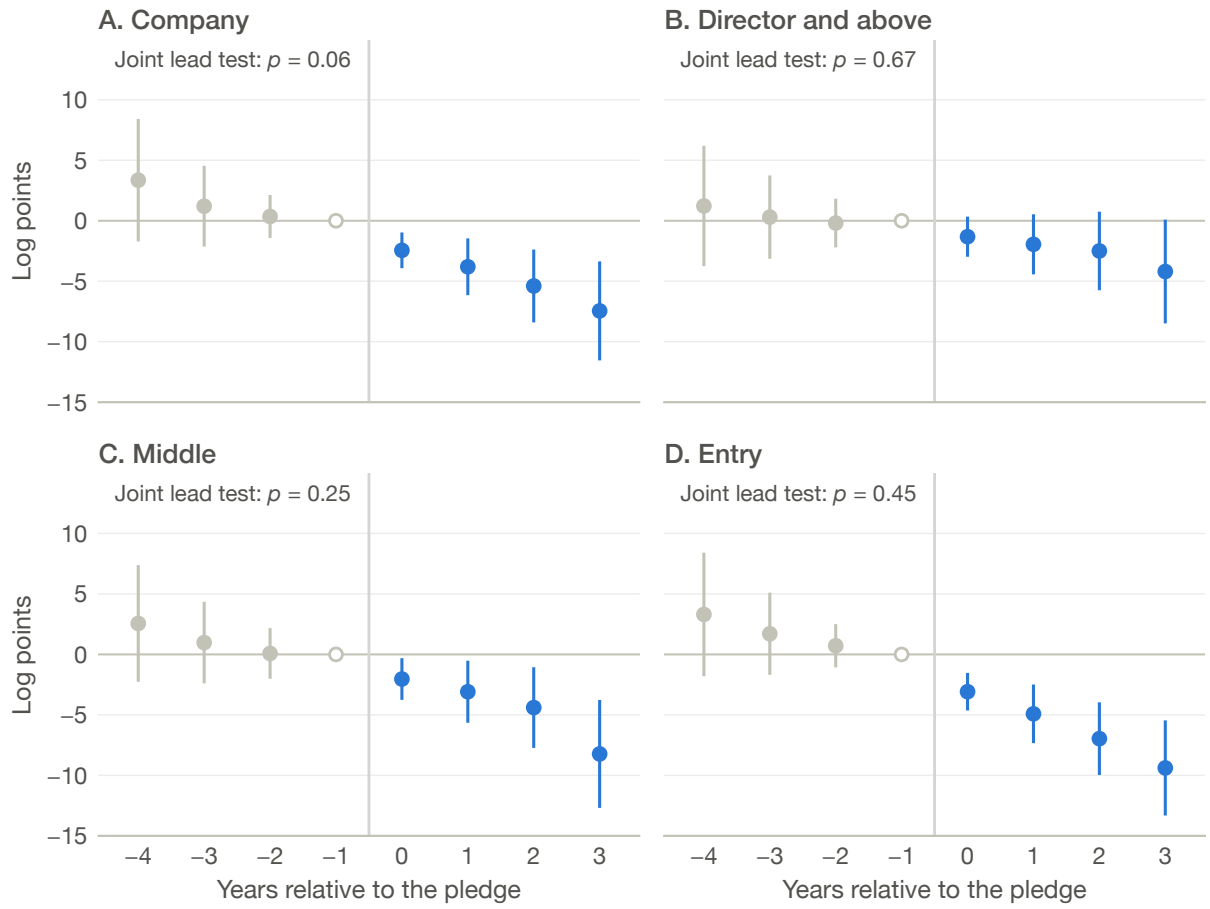
	Director and above	Middle	Entry	Company
Headcount, 100 × log (about percent)	−2.49*	−4.43***	−6.09***	−4.77***
	(1.40)	(1.43)	(1.28)	(1.30)
Headcount without the 2024 snapshot	−2.31	−4.69**	−6.18***	−4.46**
	(1.77)	(1.86)	(1.68)	(1.77)
Band's share of the firm's workforce (pp)	0.38***	0.09	−0.47***	
	(0.12)	(0.13)	(0.18)	
Joint lead p, headcount	0.67	0.25	0.45	0.06

Notes: Estimator: interaction-weighted event study of Sun and Abraham (2021), Equation (6), with firm and year fixed effects, event years −4 to 3, reference year −1. Estimates of τ_b (Equation (7)), the mean of the coefficients for event years 0 to 3, with firm-clustered standard errors in parentheses; 115 numerical race pledgers against 1,062 never-pledged controls. Headcount is 100 times the log of the band's LinkedIn workers on 1 July, or of the whole firm's in the Company column. The band's share is in percentage points (pp). The last row tests jointly that the three pre-pledge coefficients of the first row are zero. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Source: LinkedIn profiles via Revelio Labs (November 2025 release).

E.4 Other Groups' Shares

At the pledgers, the white-male share fell and the Asian share rose relative to the controls (Panel A of Table 7 in Section 5.3), and both movements had begun before the pledges and kept their pace afterwards. The white-male and Asian shares fail the lead test in most bands, and Figure E.2 and the slope rows of Table 7 show why. Relative to the controls, the pledgers' white-male share was already falling and their Asian share already rising, significantly so in every band except entry. Panel B of Table 7 removes each group's linear trend through its own pre-pledge coefficients, fitted by generalized least squares and extrapolated to event years 0 to 3. Against its own trend, no group's share other than the Black one moves significantly in any band. The Black share, in contrast, breaks from its trend. Its pre-pledge coefficients have almost no slope, and the trend-adjusted increase is 0.23 points at director and above, significant at the 5 percent level, and stays significant at entry and across the firm. Only in the middle band, with its early dip, does it lose significance. The sensitivity analysis of Rambachan and Roth (2023) gives the same answer at director and above, at entry and across the firm. The answer holds as long as the pre-pledge paths are close to straight (Appendix D.2). White men lost the most in raw terms, but their share was already falling at pledgers.

FIGURE E.1: EFFECT OF NUMERICAL RACE PLEDGES ON FIRM HEADCOUNT BY SENIORITY LEVEL



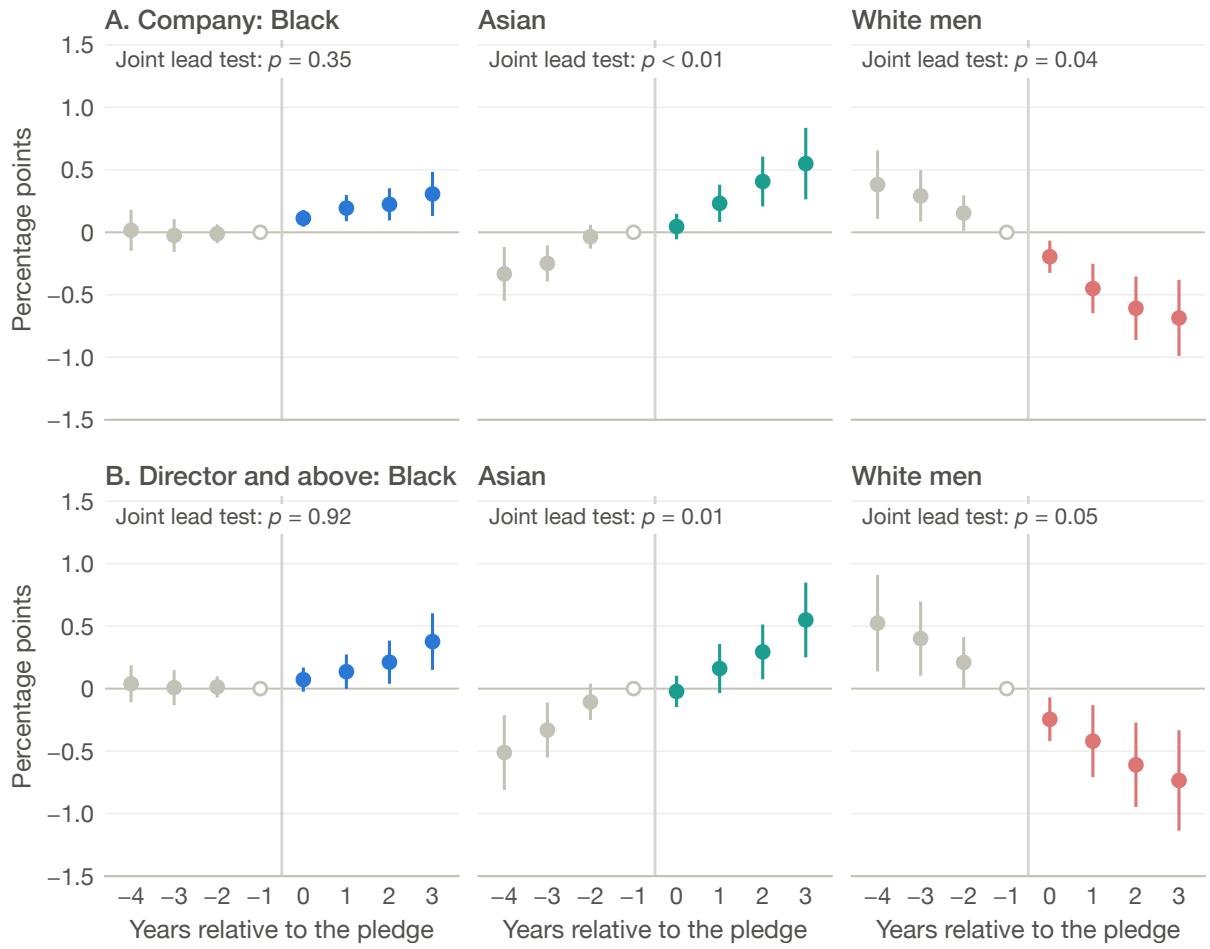
Notes: The figure plots event-study coefficients for 100 times the log of LinkedIn headcount, approximately the percent difference, in the company and in each band. The estimator is the interaction-weighted event study of Sun and Abraham (2021), Equation (6), with firm and year fixed effects and 95 percent confidence intervals based on firm-clustered standard errors. Firm fixed effects absorb differences in LinkedIn coverage, so the coefficients compare growth and say nothing about levels. Timing is as in Figure 6, and the sample is 115 pledgers and 1,062 controls. Source: Revelio Labs.

F Economy-Wide Data and New Seats

F.1 Refills of the Movers' Seats

A move is an outside hire into director and above at a pledger, in event years 0 to 3, of a worker labeled Black whose last employee position before the entry was at director level or above at another U.S. employer, the origin. A seat counts as refilled when a different person starts at the origin within 12 months in the same role, one of Revelio's 150 role groups, and in the same band; the first such start is the refill. The benchmark

FIGURE E.2: EFFECT OF NUMERICAL RACE PLEDGES ON WORKFORCE SHARES BY DEMOGRAPHIC GROUP



Notes: The figure plots event-study coefficients for the Black, Asian and white-male shares of the whole firm (Panel A) and of the director-and-above band (Panel B), in percentage points. The estimator is the interaction-weighted event study of Sun and Abraham (2021), Equation (6), with firm and year fixed effects; bars are 95 percent confidence intervals based on firm-clustered standard errors. The white probability is split between white women and white men according to the first-name probability of being female. Sample and timing follow Figure 6. Sources: LinkedIn profiles via Revelio Labs; pledge dataset.

is the origin's Black share among its other director-level starts in the 24 months before the departure: it leaves out the mover's own start, which falls in that window for 57.5 percent of the moves. Of the 9,348 moves, 77.9 percent were refilled, 43.0 percent by an internal promotion and 34.8 percent by an outside hire. With race scored by the posterior, the Black refill rate is 9.7 percent instead of 7.4, with ties between same-day starts broken at random 7.8 percent, and counting a start in any role 9.2 percent. A refill is the origin's next hire into the role, not a person-for-person replacement. The count of new seats in Table 10 is $(1 - L) + L\,r$ times the added seats, where L is the share of the added entries

TABLE F.1: REFILL OF SEATS VACATED BY DEPARTING DIRECTORS

		Percent of seats or refills				Difference (4) − (5)		
	Moves	Refilled within 12 months	Refilled by a Black worker	Black share of refills	Origin's usual Black share	Points	95% interval	p vs. row 1
	(1)	(2)	(3)	(4)	(5)			
Panel A. Black directors, by the employer they moved to								
Pledger, after its pledge	9,348	77.9	7.4	9.5	9.8	−0.2	[−0.9, 0.5]	
Pledger, before its pledge	5,806	78.0	7.4	9.5	8.5	1.0	[0.1, 1.9]	0.02
General pledger	12,568	74.5	8.4	11.3	10.4	0.9	[0.2, 1.7]	0.03
Never-pledged firm	16,972	72.7	7.9	10.8	10.2	0.6	[−0.1, 1.2]	0.10
Panel B. White men								
Pledger, after its pledge	34,528	76.4	6.0	7.8	8.1	−0.3	[−0.6, 0.1]	0.89
Panel C. Black directors added back per 100 seats left for a pledger after its pledge								
Every Black refill as new			7.4 [6.8, 8.0]					
Refills from below, first link			3.8 [3.3, 4.2]					
Following the vacancy chain			6.4 [5.6, 7.1]					
Panel D. Row 1 by origin								
Over 82 director jobs	7,252	91.8	8.8	9.5	9.5	0.0	[−0.7, 0.7]	
82 or fewer	2,096	29.9	2.8	9.4	12.1	−2.7	[−5.1, −0.2]	0.04
Black share over 11.5%	3,685	73.2	9.3	12.6	13.4	−0.8	[−2.0, 0.4]	
Black share 11.5% or less	5,260	86.6	6.6	7.6	7.5	0.1	[−0.7, 0.8]	0.23
One of the 115 pledgers	1,590	98.6	9.0	9.1	8.9	0.2	[−1.1, 1.5]	0.26
Another race pledger	415	98.3	11.6	11.8	8.5	3.3	[0.0, 5.6]	0.03
Not a race pledger	7,343	72.2	6.9	9.5	10.1	−0.6	[−1.4, 0.2]	

Notes: Moves from a director-level job at another U.S. employer into a director-level job, 2015 to September 2024; pledger moves in event years 0 to 3 or −4 to −1. A seat is refilled when someone else starts in that role and band within 12 months. The origin's usual Black share is that of its director-level starts in the previous 24 months, leaving out the mover's own start. Rows 3 to 5 are reweighted to row 1's calendar years. In Panel C, refills whose last job was at director level pass the vacancy on. Panel D splits row 1 at the median origin (July 2019 director jobs; Black share at that level) and by the origin's own pledge; p-values compare each pair, or a pledger row with the last. Most likely race label; 1,000 bootstrap draws over origins. Sources: Revelio Labs, pledge dataset.

that came from directors who moved (45 in 100) and r the Black refill rate; it ignores later steps of the chain, in which refilled seats open further vacancies, which Panel C of Table F.1 follows. With the posterior refill rate instead of the label rate, 708 of the 1,200 added seats are new. Origins are observed for 91.9 percent of the Black outside hires at pledgers after their pledges; the rest came from abroad (3.8 percent), had no earlier job (3.1 percent) or came from an unresolved employer (0.6 percent).

F.2 Origin Firms and Rivals

For the 38,444 U.S. employers with at least 20 director-level workers in 2019, the pledgers excluded, exposure is the share of their director-level leavers in 2016 to 2019 who moved to a pledger (mean 2.4 percent, standard deviation 5.0 points). In an event study with employer fixed effects and year effects by size and industry, one standard deviation of exposure changes the Black director share by 0.006 points on average over 2020 to 2024 (95 percent interval -0.022 to 0.033), with flat leads ($p = 0.57$). Employers that lost a Black director to a pledger in 2020 to 2022 changed by -0.04 points (-0.13 to 0.07) relative to matched employers from 2019 to 2023. Their Black director share was already rising faster before 2020, and net of that pre-trend the change is -0.27 (-0.42 to -0.10), about a fifth of the mechanical loss of 1.5 points (28 percent at the end of the interval). The pledgers' rivals show no consistent change. Non-pledging employers in 4-digit industries where the pledgers held more director jobs in 2019 changed by -0.028 points (standard error 0.016) per standard deviation of exposure, and those in metropolitan areas where the pledgers were larger by $+0.054$ (0.023). Only 59 industries and 39 metropolitan areas contain a pledger, and permutation p-values are 0.17 and 0.21, so neither result is reliable.

F.3 Routes into the Added Seats

Table 10 splits the hiring line of Table E.2 by the job each entrant held just before: the last employee position that ended before the entry, leaving out concurrent roles. The routes add up exactly to the hiring line. Table F.2 repeats the split under two other definitions. With the most recently started earlier job, as in the hire design of Section 6, outside hires who were already directors account for 52 percent of the increase. Counting anyone who had ever held a director-level job, they account for 74 percent, with a 95 percent interval from 44 to 158 percent. These shares are of the net increase, which also subtracts extra separations; per 100 entries through inside promotions and outside hires, as in Table 10, the three definitions give 45, 49 and 70. When each pledger counts equally, the intervals for inside promotions and for hires who were already directors include zero. The shares charge each route for the dilution its hires cause in the band. Counted by number of hires instead, the added outside hires split 43 percent already directors, 46 percent from below and 10 percent with no earlier job that had ended.

The count model of Appendix H.3 points the same way. Estimated separately by the hire's most recent earlier level, the mean coefficient over event years 0 to 3 is 0.056 (SE 0.029) for Black hires from below director level, 0.105 (SE 0.036) from director level and

0.146 (SE 0.047) from higher levels. Seventy percent of the implied added hires come from director level or above. The gap between hires from director level or above and hires from below is 0.062 (SE 0.032), and 0.032 (SE 0.065) once the linear pre-trend is removed, so the model cannot separate the routes net of that trend.

I also estimate the share of sitting directors on the entries themselves. For every entry into director and above, by promotion or outside hire, I regress the entrant's probability of being Black on an indicator for a pledger in the years after its pledge, with firm and year effects, in the stacked design of Section 4.5. A second regression uses that probability times an indicator for coming straight from a director-level job at another firm, and the ratio of the two coefficients is the share. The Black share of entrants rose by 0.77 points (SE 0.13, first-stage F of 33) and the ratio is 43 percent, with a 95 percent interval of 28 to 65 from 1,000 firm-bootstrap draws and of 27 to 64 from a test that stays valid when the first stage is weak. The 2022 cohort has 12 percent of the weight in this regression, against 28 percent in Table 10. Its Black share of entrants rose by 0.04 points (SE 0.23), against 0.64 (SE 0.14) for the 2021 cohort and 1.19 (SE 0.21) for the 2020 cohort. With the cohort weights of Table 10, the share is 41 percent (18 to 76). The Black share of entrants was already rising at the pledgers: its coefficients at event years -3 and -2 are -0.36 and -0.21 (SE 0.16 each, joint $p = 0.08$). With each firm-year of at least five entries counted equally, the share is 29 percent (1 to 60).

Table F.3 compares the moves of Black and white-male hires into director and above with the triple difference of Table 9. Relative to white men, the Black hires did not climb more levels on the way in, and the predicted pay of the new position did not rise more relative to the previous one, either overall or within cells of earlier and new level. The same holds for the hires who were already directors. With all pre-pledge years as the reference, the level change is -0.055 (SE 0.023), significant at the 5 percent level, consistent with more Black hires moving at the same level.

F.4 The Pledgers in the U.S. Labor Market

Across all U.S. employers on LinkedIn, which held about 10.6 million director-level jobs in 2023, the Black share of director-level jobs rose by 0.22 points from 2019 to 2023, from 11.41 to 11.63 percent. At the pledgers it rose by 0.63 points, and at the never-pledged index firms by 0.21 points (Figure F.1). The pledgers held 5 percent of director-level jobs and accounted for about 15 percent of the economy-wide rise, and their share of all Black

TABLE F.2: ROUTES INTO THE ADDED SEATS UNDER THREE DEFINITIONS OF THE EARLIER JOB

	Pooled over firms			Firm average
	Just before	Latest started	Ever held	Just before
Promoted inside the firm	22 [−4, 49]	22 [−4, 49]	22 [−4, 49]	14 [−37, 54]
Outside hire, already a director	48 [26, 102]	52 [29, 112]	74 [44, 158]	36 [−4, 115]
Outside hire, from below	39 [18, 71]	35 [7, 64]	13 [−28, 34]	61 [19, 173]
Outside hire, no earlier job that had ended	−4 [−33, 8]	−4 [−28, 6]	−4 [−28, 6]	−5 [−44, 15]
Separations	−5 [−58, 20]	−5 [−58, 20]	−5 [−58, 20]	−6 [−122, 35]

Notes: Shares of the net change in the Black share at director and above, in percent, with 95 percent intervals from 1,000 firm-bootstrap draws. Just before: the last employee position that ended before the entry. Latest started: the most recently started earlier position, as in the hire design. Ever held: any earlier employee position at director level or above. The last column weights each pledger equally.

TABLE F.3: MOVES OF BLACK AND WHITE-MALE HIRES INTO DIRECTOR AND ABOVE

	Reference: year before		All pre-pledge years
	Estimate	Lead p	Estimate
Level change (levels)	−0.043 (0.030)	0.69	−0.055 (0.023)
Moved up a level (pp)	−1.43 (0.94)	0.79	−1.48 (0.70)
Change in log predicted pay (×100)	−1.16 (1.76)	0.91	−0.62 (0.98)
within cells of earlier and new level	−0.29 (1.50)	0.34	0.45 (0.91)
Already directors: moved up a level (pp)	−0.86 (1.31)	0.76	−0.08 (0.93)
Already directors: log pay change (×100)	−0.51 (2.23)	0.45	1.33 (1.32)

Notes: Triple difference of Table 9: Black against white-male external hires into director and above, pledgers against controls, mean of event years 0 to 3, standard errors clustered by firm in parentheses. The earlier job is the most recently started one. Predicted pay is Revelio's estimate for the position, which depends on title, level, place and employer. The last column uses all pre-pledge years as the reference.

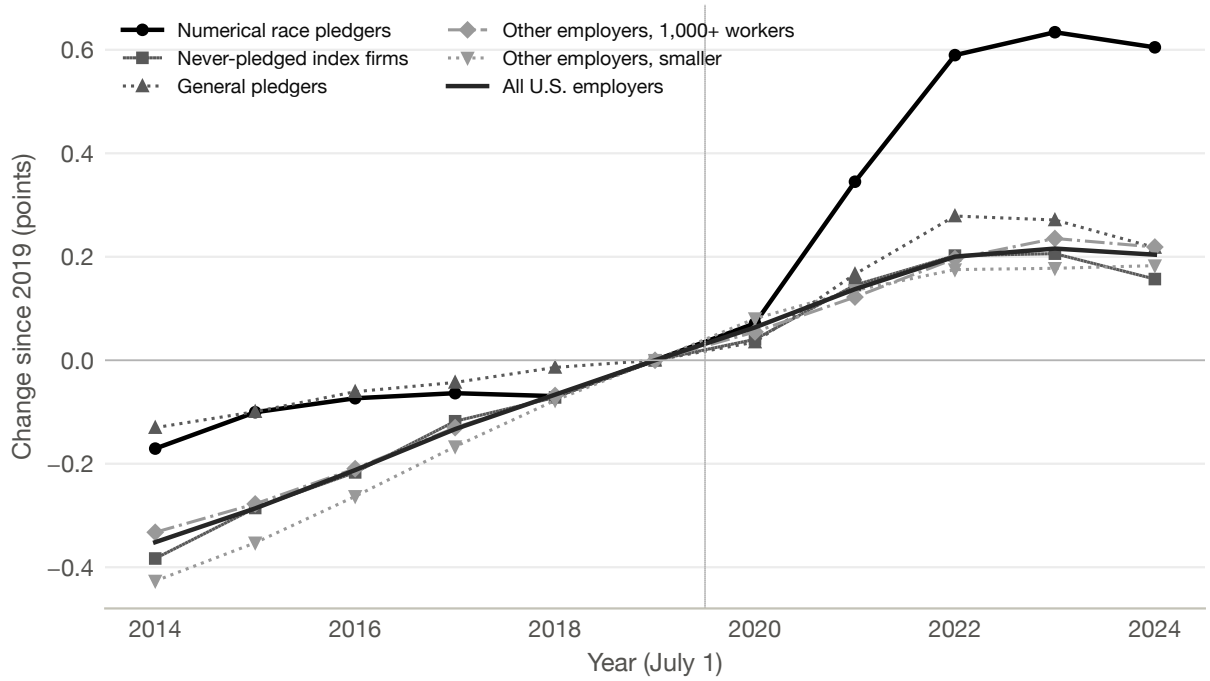
directors grew from 4.75 percent in 2019 to 5.13 percent in 2024.[28] The part of that rise which the pledges caused and which was new to the economy is smaller. On average over the four years after the pledges, the 700 net new Black directors raise the economy-wide share by about 0.007 points, 3 percent of the 2019 to 2023 rise.[29] Net of the seats that Black workers refilled and of the 17 percent of vacated seats that were at other pledgers,[30] the pledges took about 420 Black directors from other employers. This lowered the other employers' Black share by about 0.004 points, 2 to 4 percent of the 0.20-point rise they recorded.

[28] Non-pledging employers in the industries and metropolitan areas where the pledgers held more director jobs did not move in a consistent direction (Appendix F.2).

[29] The 700 count each pledger equally, like the effects of Section 5. If I weight pledgers by size, as in Figure F.1, the 0.43-point change at the pledgers relative to the never-pledged index firms implies about 2,600 added and 1,500 net new Black directors, about 7 percent of the rise.

[30] Of the seats that Black directors left for pledgers after their pledges (Figure 9), 17 percent were at another of the 115 pledgers.

FIGURE F.1: EVOLUTION OF THE BLACK SHARE AT DIRECTOR AND ABOVE ACROSS U.S. EMPLOYERS ON LINKEDIN (2014 TO 2024)



Notes: The figure plots the change since July 1, 2019, in the share of Black workers at director and above, by group of employers, in percentage points. Shares are weighted by workers and use the education-augmented BIFSG posterior (Section 3.4). Pledgers are the 115 numerical race pledgers. General pledgers made public diversity commitments without a numerical race target. Other employers (those outside the S&P 1500) are split at 1,000 U.S. workers on LinkedIn, using each employer's largest count from 2014 to 2024. Pseudo-employers such as "Retired" are excluded. Sources: Revelio Labs, pledge dataset.

G Robustness of the Workforce Estimates

In Table G.1 and Figure G.1 I weight each firm-band-year cell by its number of workers, which moves the estimand toward the effect on the average worker at a pledging firm. The weighted estimates are larger at director and above (0.278 against 0.199 percentage points) and in the middle band (0.386 against 0.226), smaller at entry (0.254 against 0.302) and close at the company level (0.223 against 0.209). Weighting also moves the leads, which fall below zero in every band: the joint lead test gives $p = 0.05$ in the middle band and $p = 0.06$ at the company level, and when cohorts are also weighted by worker-years, or firms carry fixed pre-2020 weights, it also rejects at the 5 percent level at director and above ($p = 0.03$). Amazon accounts for part of the gap at director and above and in the middle band: without it the weighted estimates are 0.249 and 0.351, and the middle-band lead test gives $p = 0.16$. I keep the firm average as the main estimand because each pledge sets a target for that firm's own workforce, so the firm is the unit whose commitment I

evaluate. Figure G.2 shows the estimates against the controls reweighted to the pledgers' size, sector and 2019 Black share at director and above (Section 5.2).

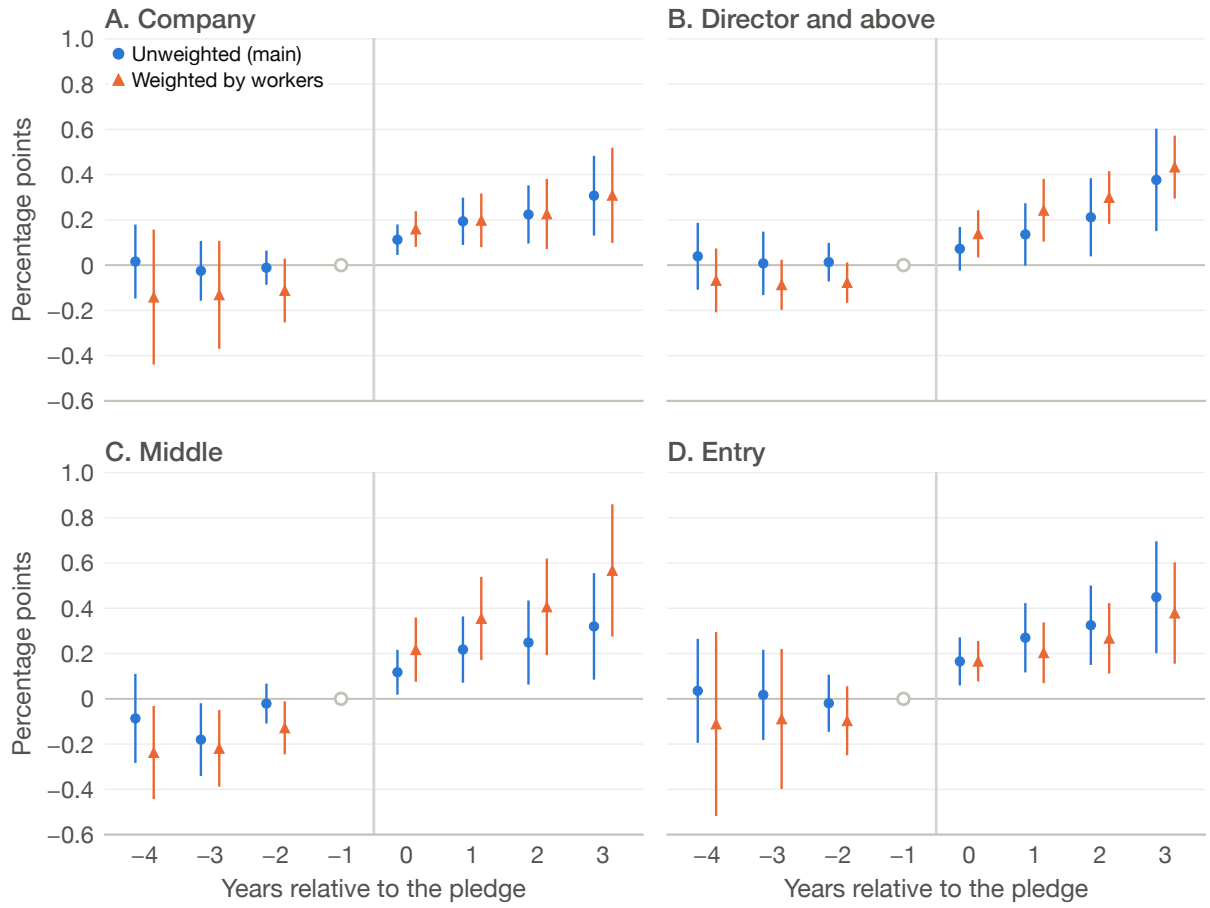
TABLE G.1: BLACK SHARE ATT WITH CELLS WEIGHTED BY WORKERS

	Director and above	Middle	Entry	Company
Panel A: ATT, mean of event years 0 to 3 (percentage points)				
Unweighted (main)	0.199***	0.226***	0.302***	0.209***
	(0.072)	(0.077)	(0.080)	(0.056)
Weighted by workers	0.278***	0.386***	0.254***	0.223***
	(0.053)	(0.098)	(0.068)	(0.064)
Weighted, without Amazon	0.249***	0.351***	0.230***	0.251***
	(0.054)	(0.120)	(0.076)	(0.071)
Weighted, cohorts by worker-years	0.359***	0.381***	0.254***	0.245***
	(0.062)	(0.089)	(0.073)	(0.068)
Fixed pre-2020 firm weights	0.309***	0.320***	0.199***	0.208***
	(0.062)	(0.094)	(0.076)	(0.069)
Panel B: Joint test of the three leads (p-value)				
Unweighted (main)	0.92	0.08	0.86	0.35
Weighted by workers	0.24	0.05	0.25	0.06
Weighted, without Amazon	0.53	0.16	0.34	0.22
Weighted, cohorts by worker-years	0.03	0.01	0.31	0.09
Fixed pre-2020 firm weights	0.03	0.01	0.52	0.09
Numerical race pledgers	115	115	115	115
Never-pledged controls	1,062	1,062	1,062	1,062

Notes: Black share measured as in Section 3.4, Sun-Abraham, mean of event years 0 to 3 in percentage points, standard errors clustered by firm. The main row counts each firm-band-year cell once. Weighted rows weight cells by workers and cohorts by firms, or by treated worker-years where stated. Fixed weights use each firm's mean pre-2020 cell size for cells and cohorts, so they cannot respond to the pledge (one middle-band control drops). Without Amazon, 114 pledgers. Panel B: event years -4 to -2. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Source: Revelio Labs.

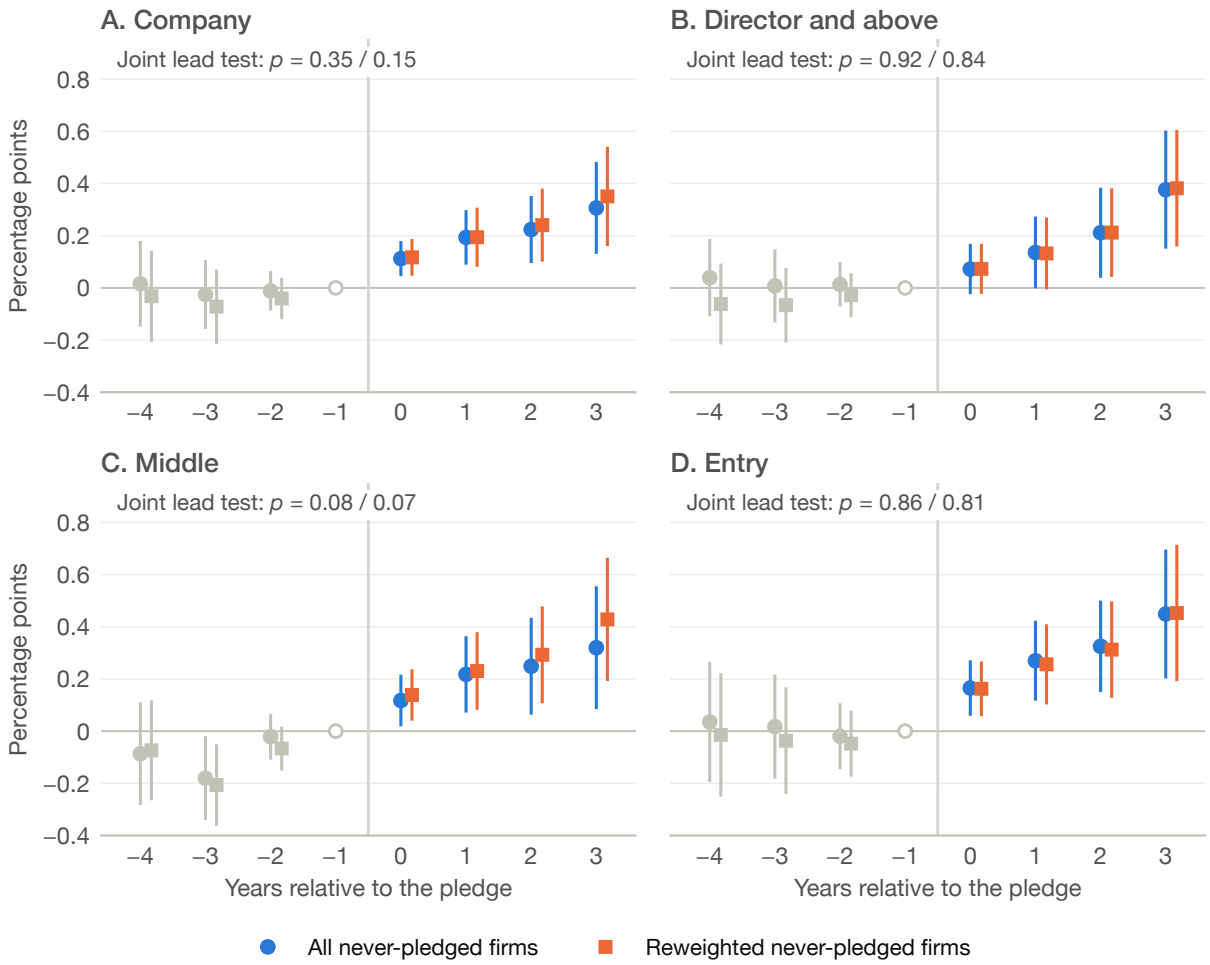
Earlier cohorts. Four firms made a numerical race pledge before the window of the main sample: Intel and Starbucks in the 2015 cohort, and Airbnb and Iron Mountain in the 2017 cohort. I leave them out because the EEO-1 reports and the hire data begin in 2016, which leaves little or no period before their pledges. The LinkedIn panel and the photographs start earlier. With these four firms, the effect is 0.19 points at director and above (SE 0.07), 0.20 across the firm (SE 0.05) and 0.39 in the photographs (SE 0.10). On LinkedIn, the 2015 cohort has no lead year, and the 2017 cohort has two.

FIGURE G.1: EVENT-STUDY ESTIMATES FOR THE BLACK SHARE WITH CELLS WEIGHTED BY WORKERS



Notes: Sun-Abraham event-study coefficients for the Black share of each band (measured as in Section 3.4), in percentage points, with 95 percent confidence intervals clustered by firm. Unweighted firm-band-year cells (main) and the same cells weighted by their workers, offset within each year; the weighted estimates combine cohorts by their number of firms. Year −1 (open marker) is the reference. 115 numerical race pledgers and 1,062 never-pledged controls; cells with at least five workers. Source: Revelio Labs.

FIGURE G.2: EFFECT OF NUMERICAL RACE PLEDGES ON THE BLACK SHARE AGAINST NEVER-PLEDGED FIRMS BALANCED ON THREE CHARACTERISTICS



Notes: The figure plots event-study coefficients, in percentage points, for the Black share of the company and of each band, against all never-pledged controls (circles) and against the same controls reweighted by entropy balancing to the pledgers' means of log 2019 headcount, the 2019 Black share at director and above and eight sector groups (squares; effective number of controls 99). The estimator is that of Equation (6), with 95 percent confidence intervals based on firm-clustered standard errors; year −1 is the reference. The sample is 115 numerical race pledgers and 1,062 never-pledged controls. Sources: LinkedIn profiles via Revelio Labs; pledge dataset.

H Hire-Level Results

Table H.1 defines the hire-level outcomes of Section 6.

TABLE H.1: DEFINITIONS OF HIRE-LEVEL VARIABLES AND OUTCOMES

Outcome	Definition	Sample
Previous employer's rank	Where the hire's previous employer ranks among U.S. employers, in percentiles. The ranking uses the moves workers chose to make in 2014 to 2019 (Sorkin, 2018).	Hires who came from another employer
Staff function	Equals 1 if the job is in administration, marketing, sustainability or reception (32 of Revelio's 150 functions), and 0 if it is in a line function such as engineering, finance, operations, sales or science.	Placement sample, job recorded under the parent company
Path-to-power score	How often jobs in the function start at Revelio level 6 or 7, standardized across the 150 functions. It does not vary across firms.	Same as above
New role	Equals 1 if nobody at the firm started a job in the same role (one of Revelio's 1,500 roles) in the three years before the hire.	Same as above
Manages people	Equals 1 if the title says manager, director, head of or supervisor, names a team, group, squad or shift lead, or is at director level or above outside the individual-contributor track. Product, program, project, account, category, relationship, portfolio and vendor managers get a 0.	Placement sample
Bottom fifth of pay	Equals 1 if the predicted salary is below the 20th percentile among the firm's 2016 to 2019 outside hires into director and above who are labeled Black, Hispanic or white male (Table H.4).	External hires into director and above
Bottom fifth of previous employers	Equals 1 if the previous employer's rank is below the 20th percentile among the same hires, keeping only those who came from another employer (Table H.4).	Hires who came from another employer
Lower-tail cuts for the bounds	Two cutoffs: the 10th percentile of predicted salary among the white men the firm hired into director and above in 2016 to 2019, moves within the firm included, and the 25th percentile of the previous employer's rank among the white men it hired from other employers (Table H.5).	White men hired in 2016 to 2019

Notes: The table defines each hire-level variable and outcome and its sample. The previous employer is the employer of the latest position that started before the hire. Ranks come from the revealed-preference method of Sorkin (2018), estimated on the 32,787 employers with at least 100 moves in or out. The path-to-power score uses positions started in 2015 to 2019 at panel firms that did not pledge. The manager rule uses the title rules of Appendix B.5. The placement sample leaves out titles that mark board members, advisers, contractors, consultants to the company, former or retired employees, and volunteers. Predicted salary comes from Revelio. Source: LinkedIn profiles via Revelio Labs (November 2025 release).

H.1 Alternative Designs

Table H.2 sets the main design of Section 6 against four alternatives. The main design compares Black and white-male hires within the same firm and hire year, and dates each firm's hires relative to its own pledge. The alternatives are the same comparison pooled across firms; a common cutoff in 2020 for every pledger, pooled across firms; the main design net of a linear pre-pledge trend; and, in the pooled version, the general pledgers in place of the never-pledged controls (the within-firm version gives the same picture at director and above). At director and above the designs agree on the outcomes shown, although the human-resources title of Section 6.1 falls only in the aligned designs. Only two gaps move. The gap in the rank of the previous employer narrows in every design, by 0.9 to 2.2 percentile points, and in every design the DEI-title share of Black hires falls relative to white men's, by 0.21 to 0.45 points. Below the top the designs disagree. The pooled designs find entry-level differences in pay, promotion, retention and placement in staff functions; these shrink when hires are compared within firms, and for promotion and retention they disappear once the pre-pledge trend is taken out.

H.2 Hires below Director Level

Table H.3 reports the main design of Section 6 for the middle and entry bands. Below the top, three gaps changed within firms (Tables H.3 and H.2, which use all position starts). In the middle band, the predicted pay of Black hires fell relative to white men (5 percent level, with flat leads), as did their chance of a top-100 degree (1 percent level, although it fails its lead test). At entry, Black hires shifted into staff functions (5 percent level), and relative to white men their chance of promotion within 24 months rose by 0.56 points (SE 0.44) and retention by 0.65 points (SE 0.46). The promotion gap had been closing before the pledges, however, by 0.23 points a year (SE 0.12), and net of that trend the estimate is -0.47 (SE 0.68). Pooled across firms, the same comparisons give entry-level advantages of about 1.5 points in promotion and 1.7 in retention, as well as higher pay and more placement in staff functions. Comparing hires within the same firm and year removes most of them: 19 percent of the pooled pay estimate remains, and 39 to 44 percent of the others. In the middle band, the comparison of previous employers also fails the pre-trend checks. I therefore read the results below the top with caution and do not report bounds on the added hires there.

TABLE H.2: BLACK HIRES UNDER ALTERNATIVE DESIGNS

	(1) Aligned, within firm (all starts)	(2) Aligned, pooled	(3) Calendar, pooled	(4) Beyond a linear pre-trend	(5) General pledgers as controls
Panel A: Director and above (levels 5 to 7)					
Top-100 university (pp)	−0.33	−0.53	−0.20	−0.88	−0.45
	(0.61)	(0.64)	(0.58)	(1.22)	(0.64)
Years since bachelor's degree	0.04	−0.05	−0.21	0.25	−0.18
	(0.18)	(0.17)	(0.15)	(0.32)	(0.18)
Previous employer's rank (percentile, movers only)	1.82***	1.51**	0.87*	2.21**	1.17*
	(0.56)	(0.65)	(0.51)	(1.02)	(0.62)
Predicted salary (dollars)	−845	−1,102	−491	−602	−1,764
	(1,092)	(1,217)	(1,162)	(1,891)	(1,259)
Still at the firm after 24 months (pp)	−0.30	−0.07	−0.38	0.40	0.01
	(0.73)	(0.74)	(0.52)	(2.58)	(0.71)
Promoted within 24 months (pp)	−0.28	−0.18	−0.09	0.00	−0.13
	(0.31)	(0.33)	(0.24)	(0.99)	(0.34)
DEI title (pp)	−0.35***	−0.31**	−0.45***	−0.37*	−0.21*
	(0.13)	(0.13)	(0.10)	(0.21)	(0.12)
Panel B: Middle (levels 3 and 4)					
Promoted within 24 months (pp)	0.44	0.93*	0.81**	0.08	0.43
	(0.53)	(0.50)	(0.40)	(1.21)	(0.53)
Panel C: Entry (levels 1 and 2)					
Promoted within 24 months (pp)	0.56	1.45***	1.44***	−0.47	1.06*
	(0.44)	(0.55)	(0.41)	(0.68)	(0.59)
Still at the firm after 24 months (pp)	0.65	1.65*	2.32***	0.10	1.11
	(0.46)	(0.86)	(0.81)	(1.17)	(0.91)
Staff rather than line function (pp)	0.59**	1.34***	1.68**	0.47	1.22**
	(0.29)	(0.44)	(0.67)	(0.47)	(0.54)
Predicted salary (dollars)	277	1,461*	3,419***	134	1,613**
	(361)	(746)	(1,219)	(572)	(774)

Notes: Black minus white-male hires (U.S. position starts, 2016 to 2024), 115 numerical race pledgers minus 1,063 never-pledged controls; firm-clustered standard errors. (1) The design of Table 9 on all position starts, moves inside the firm included (Table 9 keeps outside hires): within destination firm and July 1 hire year, aligned on the pledge: mean of event years 0 to 3 against year −1 (24-month outcomes: hires to 2022, year −3). (2) Pooled across firms. (3) Pooled, hires from 2020 on against earlier ones. (4) Column (1) net of its linear pre-trend. (5) Column (2) against the 193 general pledgers. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Source: Revelio Labs.

TABLE H.3: BLACK HIRES BELOW THE DIRECTOR LEVEL

	Middle (levels 3 and 4)			Entry (levels 1 and 2)		
	θ	Lead p-value	Beyond pre-trend	θ	Lead p-value	Beyond pre-trend
Top-100 university (pp)	-1.30^{***}	<0.01	-1.84^{**}	0.46	0.53	0.68
	(0.49)		(0.82)	(0.34)		(0.48)
Years since bachelor's degree	-0.18	0.70	-0.09	0.08	0.42	0.21
	(0.11)		(0.22)	(0.12)		(0.20)
Previous employer's rank (percentile, movers only)	-0.12	0.02	-1.29^{**}	not measured		
	(0.36)		(0.65)			
Predicted salary (dollars)	$-1{,}184^{**}$	0.67	$-1{,}274$	277	0.49	134
	(573)		(962)	(361)		(572)
Still at the firm after 24 months (pp)	0.06	0.64	0.77	0.65	0.14	0.10
	(0.57)		(1.97)	(0.46)		(1.17)
Promoted within 24 months (pp)	0.44	0.94	0.08	0.56	0.28	-0.47
	(0.53)		(1.21)	(0.44)		(0.68)
Staff rather than line function (pp)	0.28	0.11	0.66	0.59^{**}	0.24	0.47
	(0.45)		(0.88)	(0.29)		(0.47)

Notes: θ: mean of the event-year coefficients 0 to 3 of the Black minus white-male gap within destination firm and July 1 hire year, aligned on the pledge, pledgers relative to controls; firm-clustered standard errors. Lead p-value: joint test of the three coefficients before the reference year (-1; -3 for 24-month outcomes). Beyond pre-trend: θ net of a line fitted through them. U.S. position starts, 2016 to 2024 (24-month outcomes to 2022), at the 115 numerical race pledgers and 1,063 never-pledged controls. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Source: Revelio Labs.

H.3 Bounds on the Added Hires

Take the external Black hires of a pledger into director and above in a post-pledge event year e. Write their number as $N_e = N_e^0 + A_e$, where N_e^0 is the number the firm would have hired without the pledge and A_e the number the pledge added, and let $a_e = A_e / N_e$ be the added share. Suppose the pledge changed neither the outcomes of the Black workers the firm would have hired anyway nor which white men it hired. The coefficient θ_e of Equation (8) then measures $a_e\,(Q_e^A - Q_e^0)$, where Q_e^0 and Q_e^A are the mean outcomes of the two groups. If this gap Δ is the same in each post-pledge year, the averages over event years 0 to 3 satisfy $\theta = a\,\Delta$, so θ / a measures Δ. If the pledge also moved the mean outcome of the white-male hires, the ratio compares the added hires with the other Black hires, net of the change among the white men hired alongside them.

I estimate a_e from counts of hires. On a balanced panel of firms and July 1 years, stacked as in Equation (8), a Poisson pseudo-maximum-likelihood regression specifies

$$\mathrm{E}\big[n^B_{c,f,t}\big] = T_{f,t}\,\exp\Big(\alpha_{c,f} + \delta_{c,t} + \sum_{e \neq -1} b_e\,\mathbf{1}\{f \in \mathcal{P}_c\}\,\mathbf{1}\{t - c = e\}\Big), \tag{H.1}$$

where $n^B_{c,f,t}$ is the number of external Black hires into director and above, the exposure $T_{f,t}$ is the number of external hires of every label into the band, and $\alpha_{c,f}$ and $\delta_{c,t}$ are stack-by-firm and stack-by-year effects. At given total hires, the added share is then $a_e = 1 - \exp(-b_e)$. For each outcome, I estimate a on the Black hires whose outcome is observed. For the two 24-month outcomes, θ uses the reference year -3 and a keeps -1: a count of hires is not exposed to an outcome measured later.

I report a Fieller set, which allows for the sampling error in a and its covariance with θ, both computed from firm-level influence functions clustered by firm. The set holds the values q at which a test of $\theta - q\,a = 0$ does not reject at 5 percent. The instrumental-variables version replaces a with $\tilde{a}$, the mean over event years 0 to 3 of the coefficients of Equation (8) estimated on the log Black share of the firm's external hires into the band, which is close to a when a is small. Its Anderson-Rubin set stays valid when the first stage is weak.

The added share a of the external Black hires into director and above, averaged over event years 0 to 3, is 0.068 (SE 0.024). Table H.5 reports θ, the ratio θ/a with its 95 percent Fieller set, and the within-firm instrumental-variables version with Anderson-Rubin sets (first-stage F of 9.4 for pay).

The sets are wide. Compared with the other Black hires, the added hires entered positions predicted to pay between \$45,400 less and \$63,300 more (Fieller), or between \$36,200 less and \$51,000 more (instrumental variables). For the rank of the previous employer both sets exclude zero, at 6.6 to 65.6 and 5.7 to 52.5 percentile points. Relative to the white men hired alongside them, the added hires therefore came from employers at least as well ranked (a measured among hires from other employers is 0.085, SE 0.027). In the lower tail of pay, below the pre-2020 10th percentile of the firm's white-male hires, the share of added hires is at most about 21 points higher than among the other Black hires. Table H.4 adds retention and promotion within 36 months for hires made by June 2022, and the share of Black hires in the bottom fifth of the firm's pre-2020 hires by pay and previous employer. Section 6.2 discusses them.

The ratio depends on a, and a is fragile. The Black share of the pledgers' external hires was already drifting up before the pledges, by 0.020 a year ($p = 0.07$). Net of a linear pre-trend (Appendix D.3), a is 0.020 (SE 0.043), and at that value the set for pay, like most of the sets, covers the whole line. The interval for θ itself does not need a. It bounds how much the pledges moved the average: $-$\$2,260 to \$3,040 for predicted pay.

Mislabeling adds a mechanical shift. If the pledge adds only Black hires, every added hire with the label is Black, so the purity of the label, the share of labeled-Black hires who

are Black, rises from p_0 to $p_0 + (1 - p_0)\,a$. This alone moves the labeled-Black mean by $-(1 - p_0)\,a\,G$, where G is the mean outcome of the hires who carry the label without being Black minus that of the Black hires. The photograph validation of Appendix C.5 implies $p_0 = 0.47$ at the pledgers, close to its precision of 48 percent among photographed directors and officers. I do not observe G, so I use two values: the white-male minus labeled-Black gap at the pledgers in event years -4 to -1, and that gap divided by p_0, which equals the white-male minus Black gap if the mislabeled hires look like white men. On all position starts, the added share for predicted pay is $a = 0.068$, so purity rises by about 4 points. That moves θ for pay by \$470 to \$1,010 toward a lowered bar, between a fifth and nearly half of its 95 percent half-width. For predicted pay, G is positive, so the bounds without the shift are conservative.

TABLE H.4: RETENTION, PROMOTION AND THE LOWER TAIL OF EXTERNAL BLACK HIRES

	θ	95% interval	Lead p-value	Post years	Firms, last year	Black hires
A. Within 24 months, hires to 2022 (reference year −3)						
Still at the firm	−0.36	[−2.32, 1.59]	0.22	0 to 3	19	21,225
	(1.00)					
Promoted	−0.66	[−1.47, 0.15]	0.22	0 to 3	3	17,073
	(0.41)					
B. Within 24 months, hires to June 2022 (reference year −4)						
Still at the firm	0.00	[−2.29, 2.29]	0.11	0 to 2	16	16,557
	(1.17)					
Promoted	−0.16	[−0.87, 0.54]	0.11	0 to 2	16	15,156
	(0.36)					
C. Within 36 months, hires to June 2022 (reference year −4)						
Still at the firm	0.18	[−2.42, 2.79]	0.05	0 to 2	16	16,557
	(1.33)					
Promoted	0.08	[−0.70, 0.86]	0.25	0 to 2	16	16,557
	(0.40)					
Promoted, July 1 levels, to 2021	−0.11	[−1.37, 1.15]	0.10	0 to 1	16	7,572
	(0.64)					
D. At hire, bottom quintile of the firm's 2016 to 2019 hires (reference year −1)						
Predicted salary	−0.54	[−1.99, 0.92]	0.79	0 to 3	108	26,558
	(0.74)					
Previous employer's rank	0.04	[−1.71, 1.78]	0.06	0 to 3	102	19,472
	(0.89)					

Notes: External hires at director and above, 115 race pledgers and 1,063 controls. Percentage points. θ: mean of the post-pledge event-year coefficients on the Black minus white-male gap, within firm and July 1 hire year; firm-clustered standard errors. Reference years leave out hires whose follow-up overlaps the pledge. Lead p-value: joint test of the earlier years. Firms, last year: pledgers behind the final post year; Black hires: at pledgers, post years. Panel D: below the 20th percentile of the firm's 2016 to 2019 external hires (previous employer: movers). † lead test rejects at 5 percent. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Source: Revelio Labs.

TABLE H.5: BOUNDS ON THE ADDED BLACK HIRES

	Triple difference θ (SE)	Per added hire θ/a [95% set]	Within-firm IV [95% set]
Predicted salary (dollars)	391	5,775	4,933
	(1,352)	[−45,377, 63,272]	[−36,225, 51,033]
Top-100 university (pp)	−0.88	−13.0	−11.1
	(1.09)	[−50.6, 30.6]	[−41.1, 24.4]
Previous employer's rank	1.79***	21.0	17.9
(percentile, movers only)	(0.57)	[6.6, 65.6]	[5.7, 52.5]
Still at the firm after 24 months (pp)	−0.36	−4.5	−2.8
	(1.00)	[−34.5, 25.6]	[−20.3, 13.6]
Promoted within 24 months (pp)	−0.66	−5.9	−3.4
	(0.41)	[−15.4, 1.4]	[−9.2, 0.7]
Lower tails (pp)			
Pay below the firm's pre-2020	−0.10	−1.4	−1.3
white-male 10th percentile	(0.62)	[−27.5, 20.9]	[−22.7, 17.5]
Previous employer below the firm's	0.02	0.2	0.2
pre-2020 white-male 25th percentile	(0.96)	[−32.6, 21.5]	[−29.7, 19.0]

Notes: Black minus white-male gaps among external hires at director and above, 2016 to 2024, within firm and July 1 hire year; 115 numerical race pledgers against 1,063 never-pledged controls, stacked by pledge cohort. θ averages event years 0 to 3 against year -1 (-3 for 24-month outcomes); firm-clustered standard errors. θ/a divides θ by the added share of post-pledge Black hires, $a = 0.068$ (0.024), estimated on each row's sample; Fieller sets. The IV divides θ by the pledges' effect on the log Black share of hires; Anderson-Rubin sets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Source: Revelio Labs.

I The Framework: Derivations and Extensions

I.1 Derivations

The first-order condition for a counted channel is $v_f + \lambda_f = x_m/(\phi_m Z_m)$ and for an uncounted one $v_f = x_m/(\phi_m Z_m)$, which gives (2). The pre-pledge solution is $\bar{x}_m = v_f \phi_m Z_m$, so $C_f = \lambda_f \Sigma_M$ when the bar is not used. Substituting into $\lambda_f = \eta_f(\Gamma_f - C_f)$ gives (3). The share of the increase through channel m is $\omega_m = \phi_m Z_m / \Sigma_M$, independent of η_f, Γ_f and λ_f.

For condition (5), write $a = \Sigma_{PU}$ and $b = \phi_L Z_L$, so $\Sigma = a + b$ and $\sigma_L = b/(a+b)$. Net new Black directors are $\lambda^S(a + b\rho r)$ under a stock target, with $\lambda^S = \eta\Gamma/(1 + \eta(a+b))$, and $\lambda^F a$ under a flow target, with $\lambda^F = \eta\Gamma/(1+\eta a)$. Their difference has the sign of $a(1+\eta a+\eta b) - (a+b\rho r)(1+\eta a) = b\,[\eta a(1-\rho r) - \rho r]$, which is positive if and only if $\eta a > \rho r/(1-\rho r)$. The ratio of the two is $\frac{(1-\sigma_L)(1+\eta\Sigma)}{(1-\sigma_L+\sigma_L\rho r)(1+\eta\Sigma_{PU})}$; at $\eta\Sigma$ between 0.11 and 0.14, the values implied by a delivered fraction of a tenth to an eighth, and $\sigma_L = 0.45$, it lies between 0.99 and 1.00.

I.2 Channel-Specific Curvature

Let the marginal cost on channel m be $(x_m/Z_m)^{\gamma_m}/\phi_m$, so the cost is $Z_m(x_m/Z_m)^{1+\gamma_m}/(\phi_m(1+\gamma_m))$ and $\gamma_m = 1$ is the quadratic case. Then

$$x_m = Z_m\left[\phi_m(v_f + \lambda_f \mathbf{1}\{m \in M_f\})\right]^{1/\gamma_m}, \tag{I.1}$$

and the elasticity of channel m with respect to the shadow value is $1/\gamma_m$. A pledge raises every counted channel, and raises the one with the smallest γ_m by the largest proportion: the added hires tilt toward the channel whose marginal cost rises most slowly, and keep the firm's usual mix only when the γ_m are equal. A firm in a hurry, with a short horizon and a thin pipeline, is one for which $\gamma_L < \gamma_P$. Section 7 puts the lateral share of the added outside hires at 45 of 79 in the probability-weighted accounting and near four in ten by heads, against about 40 percent of all Black outside hires at the pledgers, and Section 6.2 finds the Black hires shifted toward people who had already held a director-level job: a tilt toward the fast channel in the first and third measures, none in the second. The data do not separate $\gamma_L < \gamma_P$ from $\gamma_L = \gamma_P$ sharply, and the shadow value and the delivered fraction no longer have the closed forms of (3); the logic of condition (5) is unchanged, its threshold is not.

I.3 Chains

If a refill is itself a sitting director, which holds for a share $\bar{\sigma}$ of refills, the vacancy moves on (White, 1970; Mercan and Schoefer, 2020). Followed to its end, where a seat is filled from outside the director level or left empty, a vacated seat yields

$$g = \frac{\rho\,(1-\bar{\sigma})\,r_E}{1-\rho\bar{\sigma}} \tag{I.2}$$

net new Black directors, with r_E the Black share among refills from outside the director level. The one-link rate ρr in (4) credits a Black lateral refill in full and ignores later links; the two errors offset, and at plausible values of $\bar{\sigma}$ and r_E replacing ρr by g moves $\mathcal{I}$ by less than a point.

I.4 Implications Not Tested in the Paper

Two implications are testable with the pledge data and not pursued here: the effect should be larger at pledgers with larger targets (larger Γ_f), and the lateral share of the added hires should be larger at pledgers whose 2019 middle-band Black pipeline is thin and whose deadline is short. Appendix A.7 compares the pledgers whose targets are tied to executive pay (larger η_f) with the others.

I.5 What the Framework Leaves Out

Wages: the inducement on the ladder is a pay premium that the data cannot see within a title. Worker responses: Coate and Loury's concern is that a lower bar weakens the targeted group's incentive to invest; the horizon here is too short for investment to respond, and the pools are taken as given. Dynamics: the horizon enters only through ϕ_P. The pledge itself: η_f and Γ_f are given, and selection into pledging is a matter for the design (Section 4.4). Lower bands: the same problem one level down, where an upward hire vacates a middle-band seat; Section 7 counts director seats only.

TABLE I.1: OBJECTS IN THE FRAMEWORK AND WHERE THEY ARE ESTIMATED

Object	Framework	Estimate	Section
Firm-level effect	Δ_f / N_f, Eq. (3)	ATT τ^b	5.1
Fraction delivered	$\eta\Sigma/(1+\eta\Sigma)$	A tenth to an eighth of the promised pace	4.1, A.6
Bite	η_f: number $>$ statement $>$ none	Numerical vs. general pledgers	5.2
Composition	ω_m; tilt, Eq. (I.1)	Flow decomposition; Table 10, Panel A	5.4, 7
Headcount	N_f fixed while induced hires $<$ turnover	Log headcount	5.3
Bar	$x_Q > 0$ iff $v_f + \lambda_f > \kappa_f$	Triple difference θ; bounds	6.2
Placement	x_Q without κ_f	DEI, HR, staff roles	6.1
Ladder	Origin rank rises with x_L	Previous employer's rank	6.2
Refill	$\rho = 0.78$; $\rho r = 0.074$	Seats refilled; refilled by a Black worker	7
Incidence	$\mathcal{I}$, Eq. (4)	58 per 100	7
Depletion	r_o at its usual rate when μ is small	9.5 vs. 9.8 percent	7
Flow vs. stock target	Condition (5)	Implied by the delivered fraction	4.1